

Forder Book IV — Circle Geometry

Formal Reconstruction — Technical Documentation

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Document Status

This volume documents the production formalization of the Forder-labelled circle sequence IV.12–IV.20 in CGJteamLab. The numbering is Forder numbering, not Euclid Book IV numbering.

The documentation accompanies the formal development. The current Lean source is authoritative for theorem statements, hypotheses, dependencies, helper ownership, and proof status.

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Introduction

This volume documents the CGJteamLab formal reconstruction of the Forder-labelled circle sequence IV.12–IV.20.

The numbering is Forder numbering and must not be confused with Euclid Book IV numbering. The mathematical block begins with the inscribed-angle theorem, passes through right angles in a semicircle, cyclic quadrilateral supplements and same/opposite chord cases, develops the tangent–chord equivalence, and finishes with the two circle converses.

The current production architecture has two layers:

```
CGJteamLab/HilbertInterfaceIV.lean
CGJteamLab/Proposition4_12.lean
...
CGJteamLab/Proposition4_20.lean
```

The interface owns proposition-independent circle, order, ray, tangent, and angle infrastructure. Each numbered module owns its proposition-specific proof. This replaces the former monolithic organization and gives a directed dependency graph matching the mathematics.

The Lean source is authoritative for theorem statements, hypotheses, dependencies, helper ownership, and proof status.

Chapter 12

Proposition IV.12

12.1 Introduction

Forder IV.12 is the central inscribed-angle theorem in the circle block used by the present development. In Forder’s notation, if A, B, C lie on a circle with centre O , and O, C lie on the same side of the chord AB , then the angle ACB is one half of the central angle AOB [1].

The theorem looks elementary when stated in the language of angle measure. The formal reconstruction shows why the synthetic statement is substantially richer. There is no numerical angle measure and there is no operation “divide an angle by two”. Instead, “half” is represented by an actual interior bisector ray. The production theorem therefore constructs a point T such that

$$KT \text{ is interior to } \angle AKB, \quad \angle AKT \cong \angle BKT, \quad \angle ACB \cong \angle AKT.$$

The proof also has to make explicit the order distinctions that Forder compresses into a conventional diagram. After extending CK through the centre to an antipodal point X , the line KX becomes the reference axis. The proof then separates the following configurations:

A, K, C collinear,	diameter endpoint case,
B, K, C collinear,	symmetric diameter endpoint case,
A, B on opposite sides of KX ,	addition case,
A, B on the same side of KX ,	subtraction case.

This is the first proposition in the present Forder Book IV documentation where the difference between a short source proof and a complete synthetic formal reconstruction becomes especially visible.

The production module is

```
CGJteamLab/Proposition4_12.lean
```

and it exports both IV.12 and the corollary IV.12.1.

12.2 Classical Configuration

Let K be the centre of a circle and let A, B, C be points on the circle. Let AB be the chord under consideration and assume that C and K lie in the same half-plane determined by the line

C and K on the same side of chord AB

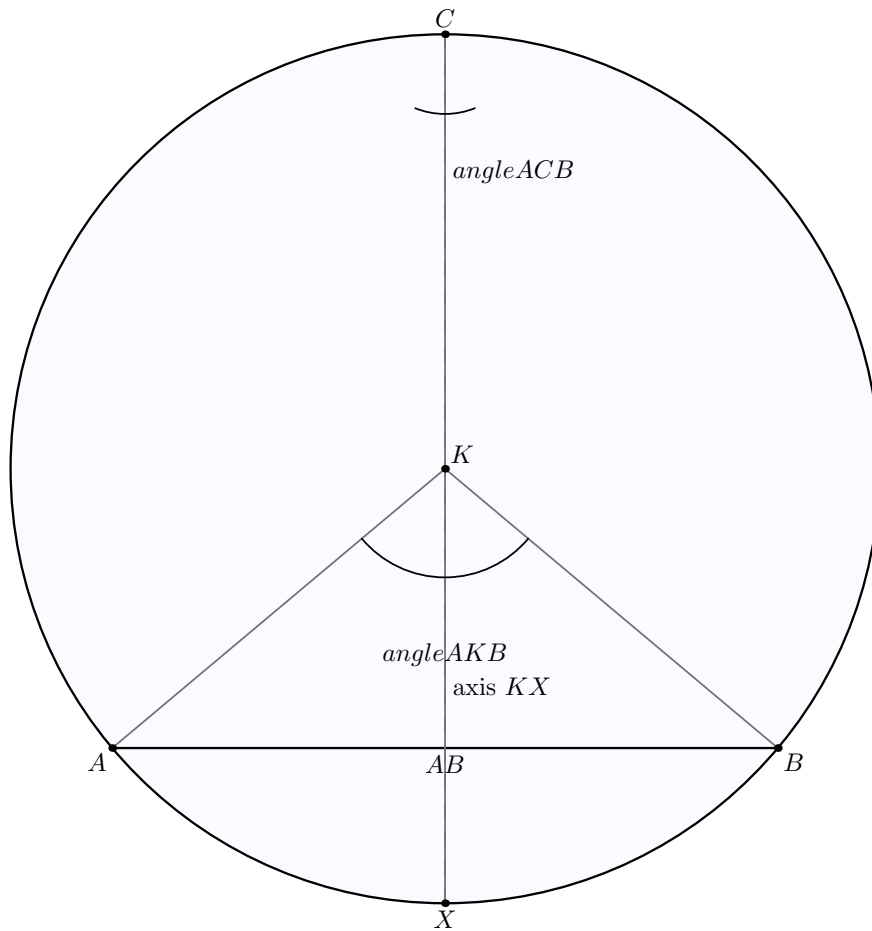


Figure 12.1: Global source configuration for Forder IV.12. The chord AB lies below the centre, so C and K are on the same side of the chord. The auxiliary antipodal point X determines the axis KX used in the formal case split.

AB .

The geometric content of IV.12 is

$$\angle ACB = \frac{1}{2}\angle AKB.$$

For the formal reconstruction this formula is only mnemonic. The actual synthetic target is the existence of an interior ray KT such that

$$\angle AKT \cong \angle TKB$$

and

$$\angle ACB \cong \angle AKT.$$

Extend CK through K to the antipodal point X , so that

$$C - K - X$$

and X lies on the same circle. The line KX is the axis used to organize the proof. Relative to this axis there are two generic configurations and two exceptional ones.

Addition configuration. If A and B lie on opposite sides of the axis KX , the axis passes through the central angle $\angle AKB$. The two exterior-central half-angle constructions attached to A, C, X and B, C, X contribute two pairs of equal angular components. The formal proof rearranges these four components to obtain a bisector of the whole central angle.

Subtraction configuration. If A and B lie on the same side of KX , one of the radius rays is nested inside the larger exterior-central angle determined by the other. Forder's subtraction of angle magnitudes must then be replaced by a synthetic comparison of two decompositions.

Diameter endpoint configurations. If A or B lies on the axis KX , then the corresponding chord through C passes through the centre and is therefore a diameter. The formal proof treats these configurations separately rather than forcing them through a generic angle-decomposition lemma.

12.3 Forder's Statement and Proof

Forder introduces IV.12 after explicitly saying that he wishes to treat angles in the same segment without invoking the full theory of angular measure, in particular avoiding reflex angles [1]. His statement is:

Let A, B, C be on a circle centre O , and let O, C be on the same side of AB ; then ACB is half AOB .

The proof extends CO through O to a point X on the circle and then separates the order configurations.

12.3.1 The diameter-aligned case

When X lies on OA (and symmetrically when it lies on OB), the line through C, O, X is also the diameter through A . Equal radii give an isosceles triangle on the remaining circle point. The central angle is then decomposed into two equal base-angle contributions, yielding the doubling relation required for IV.12.

This is the source ancestor of the two explicit production lemmas

```
hilbert_forder_IV12_diameter_A
hilbert_forder_IV12_diameter_B
```

12.3.2 The addition case

In the generic case Forder first obtains

$$\angle AOX = 2\angle OCA, \quad \angle BOX = 2\angle OCB.$$

If A and B lie on opposite sides of OX , then the ray OX lies inside the central angle and the corresponding ray through C lies inside the inscribed angle. Consequently the two decompositions add:

$$\begin{aligned} \angle AOB &= \angle AOX + \angle XOB, \\ \angle ACB &= \angle ACX + \angle XCB. \end{aligned}$$

Substituting the two exterior-central doubling relations gives

$$\angle AOB = 2\angle ACB.$$

In Lean this numerical-looking addition is replaced by an explicit rearrangement of interior rays and congruent angle components.

12.3.3 The subtraction case

If A and B lie on the same side of OX , one of the rays OA, OB lies inside the angle formed by the other ray and OX . Forder then writes the central and inscribed angles as differences:

$$\begin{aligned} \angle AOB &= \angle BOX - \angle AOX, \\ \angle ACB &= \angle OCB - \angle OCA \end{aligned}$$

up to the symmetric orientation. From the two doubled exterior-central relations he concludes again

$$\angle AOB = 2\angle ACB.$$

The formal proof cannot simply subtract angle measures. It reconstructs this step from nested rays, ordinary angle bisectors, uniqueness of halves, and the project's synthetic angle-decomposition theorems.

12.4 Formal Statement

The proposition interface is defined in `HilbertInterfaceIV.lean`:

```

def HilbertForderIV12CentralHalf
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
      H.OnLine A chord ->
      H.OnLine B chord ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R B ->
      HilbertCircle Geo K R C ->
      HilbertSameSide Geo C K chord ->
      exists T : Geo.Point,
        HilbertRayMeetsSegment Geo K T A B /\
        Geo.AngleCongruent A K T B K T /\
        Geo.AngleCongruent A C B A K T

```

The public production theorem is

```

theorem hilbert_IV12_central_half
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV12CentralHalf (Geo := Geo)

```

The distinction between the interface predicate and the theorem is useful. The predicate records the mathematical contract of IV.12 using only the order vocabulary needed to state it. The theorem then proves that contract under the project’s Euclidean-plane structure.

12.5 Why “Half an Angle” Is a Witnessed Construction

The conclusion

$$\angle ACB = \frac{1}{2} \angle AKB$$

cannot be copied literally into the synthetic language without introducing an arithmetic of angle magnitudes. The reconstruction instead asks for a point T satisfying three independent geometric obligations.

First,

```
HilbertRayMeetsSegment Geo K T A B
```

asserts that the ray KT genuinely lies inside the angle AKB . This is stronger than merely saying that T is noncollinear with the sides: the order witness records the interior position.

Second,

```
Geo.AngleCongruent A K T B K T
```

says that KT bisects the central angle.

Third,

```
Geo.AngleCongruent A C B A K T
```

identifies the inscribed angle with one of those two equal halves.

Thus IV.12 is formalized not as an equation of numbers but as the existence of a geometric configuration realizing the equation.

12.6 The Antipodal Reduction

The complete production proof begins by constructing the point X opposite C on the same diameter:

$$C - K - X.$$

The reusable theorem responsible for this construction is now

```
hilbert_circle_antipode
```

in `HilbertInterfaceIV`.

The proof then constructs the carrier line of KX and proves that C lies on it. This line is the axis relative to which all later order cases are classified.

Structurally the top-level proof has the form

```
by_cases hAaxis : H.OnLine A axis
...
by_cases hBaxis : H.OnLine B axis
...
by_cases hSameAB : HilbertSameSide Geo A B axis
...
```

The four outcomes are therefore exhaustive by incidence and plane separation; the final theorem introduces no additional geometric idea after the branch lemmas have been proved.

12.7 The Diameter Endpoint Branches

Suppose first that A, K, C are collinear. Because A and C lie on the circle and K is the centre, the reusable theorem

```
hilbert_circle_center_midpoint_of_chord
```

shows that K is the midpoint of AC . Hence

$$A - K - C.$$

Now consider triangle BCK . The radii KB and KC are congruent, so the triangle is isosceles

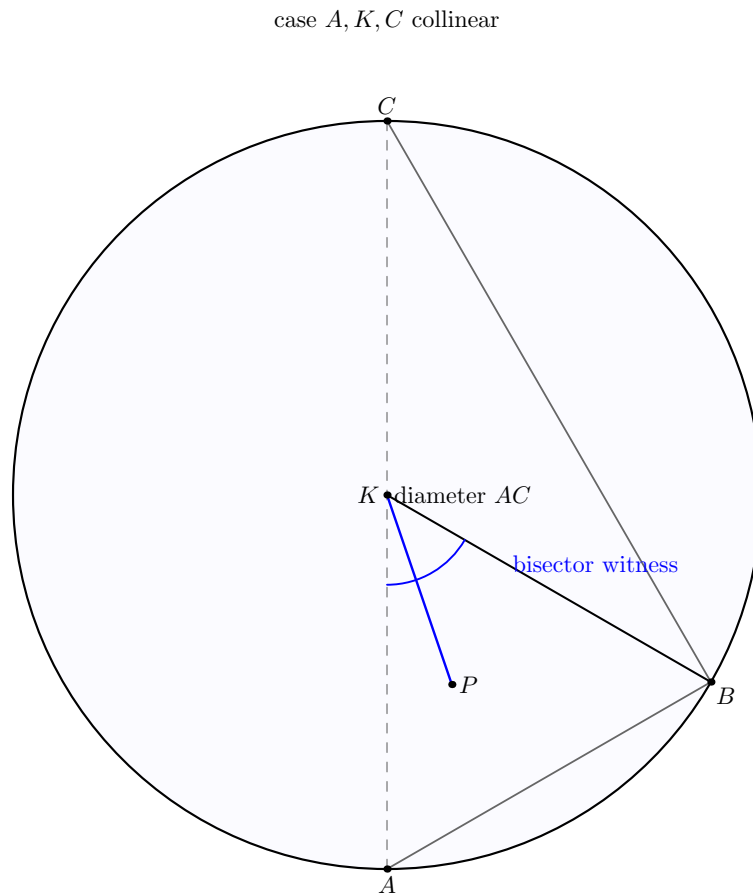


Figure 12.2: One of the two exceptional diameter branches: A, K, C are collinear, so AC is a diameter. The proof then uses the midpoint property of the centre together with Euclid I.32 to construct the required bisector witness.

and its base angles are congruent. Euclid I.32 is applied with CK produced through K to A . The exterior angle at K is thereby decomposed into two angles congruent to the equal base angles. This produces an actual bisector ray KP of the required central angle.

The final task is order-theoretic: the point P supplied by I.32 lies between B and A , so the ray KP is an interior ray of $\angle AKB$. Since C, K, A lie on one ray from C , angle transport identifies the original inscribed angle ACB with the appropriate base angle. The desired half-angle witness follows.

The B, K, C case is obtained by the symmetric production helper

```
hilbert_forder_IV12_diameter_B
```

The diameter branch is a useful indication of the proof's foundational profile: a source statement about circles reaches Euclid I.32 once the exceptional order configuration is made explicit.

12.8 The Exterior-Central Half Construction

Both generic branches use a reusable theorem extracted from the IV.12 development:

```
hilbert_circle_exterior_central_half
```

For a circle point A , the point C , and the antipodal continuation $C - K - X$, it produces an interior ray KT in the exterior-central angle AKX such that

$$\angle AKT \cong \angle XKT$$

and

$$\angle ACK \cong \angle AKT.$$

The same construction applied to B produces a second half-angle ray KU . These two witnessed half-angle configurations are the synthetic replacements for Forder's formulas

$$\angle AKX = 2\angle ACK, \quad \angle BKY = 2\angle BCK.$$

This reusable theorem belongs in `HilbertInterfaceIV`, not in `Proposition4_12.lean`, because later circle arguments can use the same exterior-central geometry independently of IV.12.

12.9 The Addition Branch

Assume now that A and B are on opposite sides of the axis KX . The axis therefore crosses the interior of the central angle. The proof obtains two local bisectors:

$$\begin{aligned} KT &\text{ bisects } \angle AKX, \\ KU &\text{ bisects } \angle XKB. \end{aligned}$$

The four angular components occur in the order

$$\alpha, \alpha, \beta, \beta.$$

A, B on opposite sides of axis KX

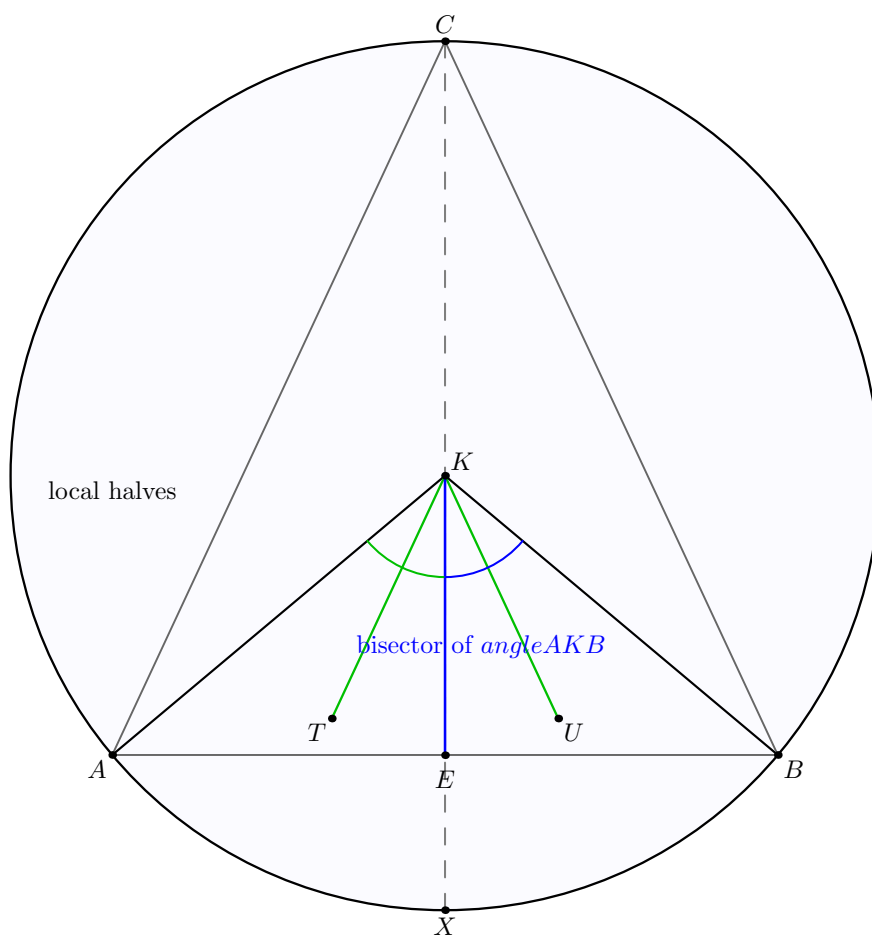


Figure 12.3: Addition branch of IV.12. The points A and B lie on opposite sides of the axis KX . Two local half-angle constructions are combined into a bisector of the whole central angle.

The central combinatorial theorem of the branch is the synthetic rearrangement

$$\alpha, \alpha, \beta, \beta \longrightarrow \alpha, \beta \mid \alpha, \beta.$$

In production this is supplied by the reusable theorem

```
hilbert_angleDecomposition_double_sum_bisector
```

It constructs a new ray KE inside $\angle AKB$ and proves

$$\angle AKE \cong \angle BKE.$$

At the same time the nested-ray data identify the relevant component of this new half with the corresponding component of the inscribed angle at C . Angle addition then gives

$$\angle ACB \cong \angle AKE.$$

The proposition-specific result is

```
hilbert_forder_IV12_addition_case
```

This branch is a good example of why the formal proof should not be described as “angle arithmetic”. No numerical sum of angle values exists in the proof object. The whole calculation is expressed by interior-ray witnesses and congruence transport.

12.10 The Subtraction Branch

The same-side branch is structurally more delicate. Assume that A and B are both off the axis KX and lie on the same side of it.

At K , one of the two radius rays lies inside the angle formed by the other radius ray and KX . At C , an analogous nesting occurs among the chord rays and CK . A priori the two choices could be crossed: the “larger” orientation at the centre need not automatically be synchronized with the “larger” orientation at the circumference.

The proof therefore first establishes a ray-order classification and then a synchronization theorem:

```
hilbert_forder_IV12_sameSide_ray_orders
hilbert_forder_IV12_sameSide_orders_synchronized
```

The synchronized alternatives are exactly:

$$\begin{aligned} &KA \text{ is interior to } \angle B K X \quad \text{and} \quad CA \text{ is interior to } \angle B C K, \\ &\text{or} \\ &KB \text{ is interior to } \angle A K X \quad \text{and} \quad CB \text{ is interior to } \angle A C K. \end{aligned}$$

The synchronization step uses strict monotonicity of angle halves: crossed orientations would force incompatible strict comparisons between congruent half-angles.

After synchronization, the proof chooses an ordinary bisector KE of the target central angle $\angle AKB$. It also has the two exterior-central half constructions described above. In either

Figure 12.4: Subtraction branch of IV.12. The points A and B lie on the same side of the axis KX . The proof requires an additional synchronization theorem for the ray order before the final synthetic subtraction step can be carried out.

orientation, the already proved addition theorem is used to reconstruct a bisector KW of the larger exterior-central angle from

$$\text{half}(\text{target}) + \text{half}(\text{smaller exterior angle}).$$

Uniqueness of angle halves identifies this reconstructed bisector with the previously constructed exterior-central bisector. The two complete angle decompositions now have matching larger parts and matching smaller parts. A single synthetic subtraction principle compares the remaining components, giving

$$\angle ACB \cong \angle AKE.$$

The production helper closing this branch is

```
hilbert_forder_IV12_subtraction_case
```

This is the formal counterpart of Forder’s line

$$2\beta - 2\alpha = 2(\beta - \alpha),$$

but no subtraction operation on numerical angle measures is introduced.

12.11 Completion of IV.12

Once the four branches have been isolated, the public proof is conceptually short.

1. Construct the antipodal point X with $C - K - X$.
2. Let KX be the reference axis.
3. If A lies on the axis, use the first diameter branch.
4. Else, if B lies on the axis, use the symmetric diameter branch.
5. Otherwise both points are off the axis.
6. If they lie on the same side, use the subtraction branch.
7. If they do not lie on the same side, plane separation gives opposite sides, and the addition branch applies.

This final case split is implemented by the public theorem

```
hilbert_IV12_central_half
```

after the production refactor. The earlier development-stage theorem performed exactly the same exhaustive classification; the production file retains the mathematics but removes stage-labelled public API.

12.12 Forder IV.12.1: Equal Angles in the Same Segment

Forder immediately records the same-segment corollary: if A, B, C, D lie on one circle and C, D, K lie in the same half-plane determined by chord AB , then

$$\angle ACB \cong \angle ADB.$$

The formal proposition interface is

```
def HilbertForderIV12_1SameSegment
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C D : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
      H.OnLine A chord ->
      H.OnLine B chord ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R B ->
      HilbertCircle Geo K R C ->
      HilbertCircle Geo K R D ->
      HilbertSameSide Geo C K chord ->
      HilbertSameSide Geo D K chord ->
      Geo.AngleCongruent A C B A D B
```

The proof is intentionally short once IV.12 is available. Apply IV.12 to C and to D . Both inscribed angles are represented as halves of the same central angle AKB . The synthetic analogue of Forder IV.46.2 then identifies those halves.

In production this reduction is isolated as

```
hilbert_forder_IV12_1_of_IV12_aux
```

and the closed public theorem is

```
theorem hilbert_IV12_same_segment
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV12_1SameSegment (Geo := Geo)
```

Thus IV.12.1 is not reproved by reopening the full circle configuration. It is a genuine corollary of the half-central-angle theorem.

12.13 Lean Formalization

At the highest level the production proof has a small public surface and a substantial proposition-specific interior.

The public endpoints are:

```

hilbert_IV12_central_half
hilbert_IV12_same_segment

```

The proposition-specific proof architecture is represented by:

```

hilbert_forder_IV12_1_of_IV12_aux
hilbert_forder_IV12_addition_configuration
hilbert_forder_IV12_addition_case
hilbert_forder_IV12_sameSide_ray_orders
hilbert_forder_IV12_sameSide_orders_synchronized
hilbert_forder_IV12_subtraction_case
hilbert_forder_IV12_diameter_A
hilbert_forder_IV12_diameter_B

```

These declarations remain local to the numbered proposition because their purpose is to organize the proof of IV.12 itself.

By contrast, the following kinds of lemmas were promoted to the reusable interface:

```

hilbert_circle_antipode
hilbert_circle_center_midpoint_of_chord
hilbert_circle_exterior_central_half
hilbert_circle_chord_inner_point_inside
hilbert_circle_chord_axis_order
hilbert_angle_bisector_with_interior
hilbert_angleDecomposition_swap_components
hilbert_angleDecomposition_nested_inside_left
hilbert_angleDecomposition_nested_inside_right
hilbert_angleDecomposition_middle_divider
hilbert_angleDecomposition_double_sum_bisector
hilbert_angleDecomposition_half_less_of_whole_less

```

The distinction is semantic rather than historical. A lemma belongs to `HilbertInterfaceIV` if it states a general circle/order/angle fact that a later proposition may reuse. It belongs to `Proposition4_12.lean` if it exists only to organize the IV.12 case analysis.

12.14 Hilbert, Book Zero, and Earlier Euclid Components

12.14.1 Incidence and plane separation

The theorem uses explicit carrier lines, same-side and opposite-side relations, and line uniqueness. These replace the informal visual judgment that a point or ray lies “inside” one of Forder’s diagrams.

The top-level addition/subtraction split is ultimately a plane-separation statement: once A, B are known to be off the axis, failure of same-side implies opposite-side.

12.14.2 Order and betweenness

The antipodal configuration $C - K - X$, midpoint configurations on diameters, intersection points on chords, and all interior-angle witnesses are encoded by betweenness or by `HilbertRayMeetsSegment`. This order information is not auxiliary bookkeeping: it determines whether angle components are to be added or subtracted.

12.14.3 Congruence

Equal radii produce isosceles triangles; angle congruence transports the equal base-angle information into the decompositions used by the proof. Congruence also supplies the synthetic notion of equal angle halves.

12.14.4 Euclid I.32

The exceptional diameter branch invokes the exterior-angle theorem I.32. The current theorem signature therefore uses

```
[HE : @HilbertEuclideanPlane Geo H]
```

rather than only the neutral incidence/order/congruence layers.

12.14.5 Angle-decomposition infrastructure

The generic addition and subtraction branches depend on the project's synthetic angle-decomposition library. Its role is not to assign magnitudes to angles but to compare and rearrange witnessed interior decompositions. This is the formal replacement for the algebraic appearance of Forder's angle calculations.

12.15 Foundational Boundary

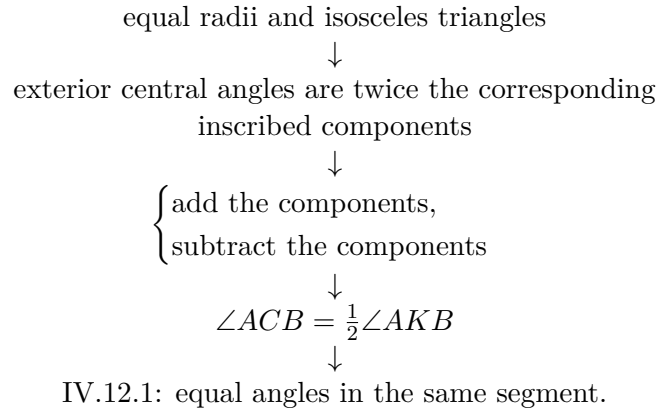
IV.12 introduces no proposition-local geometric axiom. Its public theorem is proved from the existing `HilbertEuclideanPlane` infrastructure and from reusable theorems already available in the production library.

The mathematically important boundary exposed by the proof is not a new continuity principle but the transition from neutral circle/order/congruence reasoning to the Euclidean angle machinery used in the diameter branch and the general angle constructions.

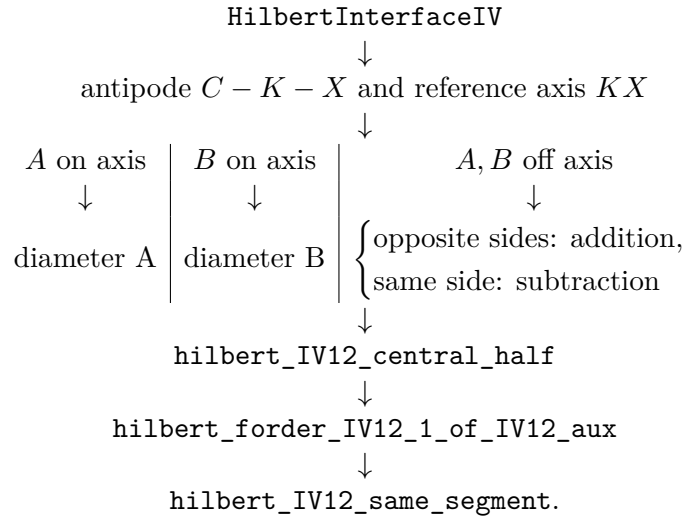
A separate raw `#print axioms` table is intentionally not reproduced here. As in the documentation of the other books, the purpose of this section is to explain the geometric role of the foundational layers rather than to copy implementation output.

12.16 Classical and Formal Dependency Structures

Forder's conceptual dependency is:



The current production architecture is:



At module level the direct dependency is simply

```
HilbertInterfaceIV
-> Proposition4_12
```

because the reusable support extracted during the proof has been moved out of the numbered module.

12.17 Diagrammatic Audit

IV.12 is one of the propositions for which figures are mathematically useful, not decorative. A complete documentation pass should contain three Asymptote figures.

1. **Forder source configuration.** Circle with centre K , chord AB , point C , antipode X , and the axis KX . The figure should show the role of the same-side hypothesis for C, K relative to AB .

2. **Addition branch.** A, B on opposite sides of KX , together with the two local half-angle rays and the reconstructed bisector of $\angle AKB$. The purpose is to make the $\alpha, \alpha, \beta, \beta$ rearrangement visible.
3. **Subtraction branch.** A, B on the same side of KX , showing the two synchronized nesting alternatives and the target bisector. This figure should explain why a genuine order theorem is needed before synthetic subtraction can be applied.

The two diameter endpoint cases can be shown as small panels in the first figure rather than as separate full figures.

12.18 Conclusion

Forder IV.12 is not merely an elementary statement about an inscribed angle being half a central angle. In a strictly synthetic reconstruction it is a theorem about the existence and position of a bisecting ray.

The formal proof exposes exactly the order distinctions suppressed by the classical diagram. The antipodal axis produces two exceptional diameter cases and two generic cases. The opposite-side case is solved by synthetic angle addition; the same-side case requires synchronized ray order and a synthetic subtraction argument. IV.12.1 then becomes a clean corollary: two inscribed angles in the same segment are two halves of the same central angle.

The proposition also fixes an important architectural pattern for the rest of Forder Book IV. General circle and angle-decomposition facts belong to `HilbertInterfaceIV`; only the proposition-specific branching remains inside `Proposition4_12.lean`. The resulting production module is therefore both mathematically transparent and reusable downstream.

Chapter 13

Proposition IV.13

13.1 Introduction

Forder IV.13 is the theorem that an angle in a semicircle is right. In Forder’s formulation, if AC is a diameter of a circle and B is a third point of that circle, then

$$\angle ABC$$

is a right angle [1].

The statement is familiar, but its formal reconstruction is instructive. The proof does not appeal to a numerical statement saying that the angle is 90° , nor does it identify a semicircle with an angular measure of π . The right-angle conclusion is expressed synthetically: after extending one side of the angle, the two adjacent angles are proved congruent.

The production file is

```
CGJteamLab/Proposition4_13.lean
```

and its public endpoint is

```
hilbert_IV13_diameter_right_angle
```

Unlike IV.12, IV.13 has no large order case split. Its difficulty is different: the formal proof must make explicit the midpoint property of the centre, the two isosceles triangles determined by the radii, the transport of their base angles along the diameter, and the final assembly of two angle decompositions.

13.2 Classical Configuration

Let K be the centre of the circle and let A, B, C lie on the circle. Assume that A, K, C lie on the diameter carrier, with B off that line.

The source statement uses AC as diameter and concludes that

$$\angle ABC$$

is right. In the production interface the letters are permuted: the endpoints of the diameter are A, B , the third circle point is C , and the target is

$$\angle ACB.$$

This is only a change of names. The geometric content is identical.

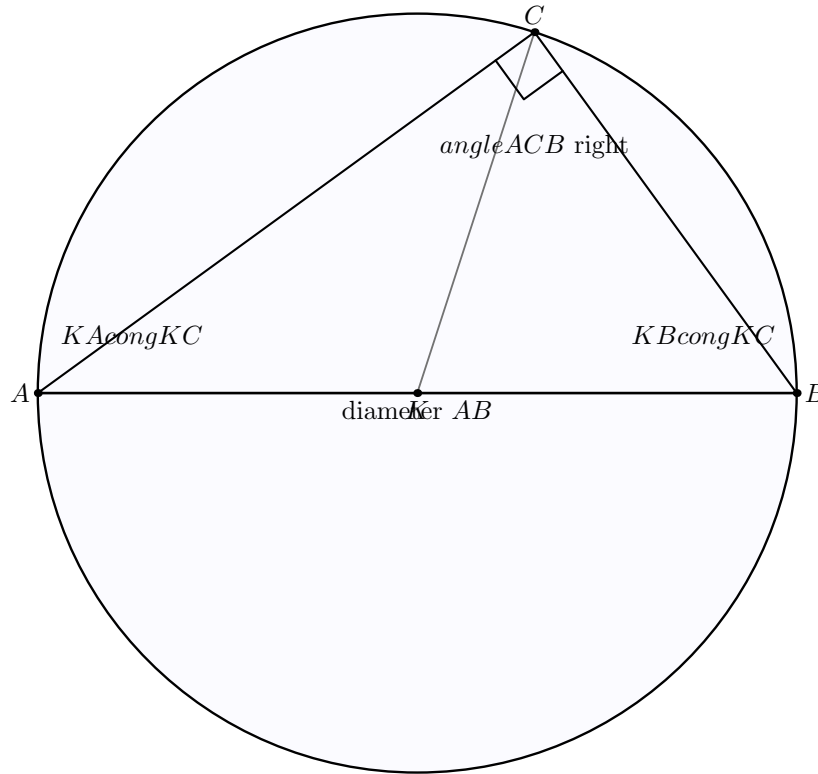


Figure 13.1: Forder IV.13 in the production naming convention. The centre K lies on the diameter AB , while C is a third point of the circle. The target angle ACB is right.

The central fact hidden in the picture is not merely that K lies on AB . Since A and B are distinct points of one circle centered at K , the centre is the strict midpoint:

$$A - K - B, \quad KA \cong KB.$$

That strict order is used repeatedly to transport angle arms along the same rays.

13.3 Forder's Statement and Proof

Forder states:

If AC is a diameter of a circle through B , then ABC is a right angle.

His proof begins with the centre O . The three radii are congruent:

$$OA \cong OB \cong OC.$$

Hence the two triangles formed by the centre and the point B are isosceles. Their base angles are equal.

Because the centre lies on the diameter, the ray from A through O is the same ray as the ray from A through the other endpoint of the diameter, and similarly from the opposite endpoint. Thus the equal base-angle information can be read as information about the two pieces of the angle at the circumference.

Forder then extends one side at the circumference and compares the angle inside the triangle with its adjacent exterior angle. The two are shown congruent, which is precisely the synthetic criterion for a right angle [1].

The formal reconstruction follows the same geometry, but expands every order and angle-composition step that the printed proof leaves in the diagram.

13.4 Formal Statement

The proposition interface lives in `HilbertInterfaceIV.lean`:

```
def HilbertForderIV13DiameterRightAngle
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H] : Prop :=
  forall K R A B C : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
      H.OnLine A chord ->
      H.OnLine B chord ->
      H.OnLine K chord ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R B ->
      HilbertCircle Geo K R C ->
      Not (H.OnLine C chord) ->
      HilbertRightAngle Geo A C B
```

The assumptions say:

- A and B are distinct;
- A, B, K lie on one carrier line;
- A, B, C lie on the same circle centered at K ;
- C does not lie on the diameter carrier.

The conclusion

```
HilbertRightAngle Geo A C B
```

is the project's synthetic right-angle predicate.

The public theorem is:

```
theorem hilbert_IV13_diameter_right_angle
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
```

```
HilbertForderIV13DiameterRightAngle (Geo := Geo)
```

The statement itself needs only incidence and congruence in its interface. The proof is carried out under the full Euclidean-plane structure because it uses the order and angle machinery, including Euclid I.32.

13.5 The Centre Is the Midpoint of the Diameter

The first substantial step is to upgrade

$$A, K, B \text{ collinear}$$

to the strict order

$$A - K - B.$$

The reusable theorem is

```
hilbert_circle_center_midpoint_of_chord
```

Applied to two distinct circle points A, B whose chord carrier contains the centre, it gives

```
HilbertIsMidpoint Geo K A B
```

and hence in particular

$$\text{Between}(A, K, B).$$

This is more than a convenience. It provides both nondegeneracy and the same-ray information required later:

$$K, B \text{ lie on the same ray from } A,$$

and

$$K, A \text{ lie on the same ray from } B.$$

In the development-stage proof this is immediately unpacked into Ne facts, collinearity data, and the reversed betweenness $B - K - A$. The production proof retains the same mathematics even though the reusable midpoint theorem now lives in the interface layer.

13.6 The Two Isosceles Radius Triangles

Since A, B, C lie on the circle centered at K ,

$$KA \cong KC, \quad KB \cong KC.$$

Thus the triangles KAC and KBC are isosceles.

The proof obtains

$$\angle KAC \cong \angle KCA$$

and

$$\angle KBC \cong \angle KCB$$

from the general isosceles-base-angle theorem.

The diameter order now allows the first arms to be transported.

Because K and B lie on the same ray from A ,

$$\angle KAC = \angle BAC.$$

Because K and A lie on the same ray from B ,

$$\angle KBC = \angle ABC.$$

Therefore the isosceles information becomes

$$\angle BAC \cong \angle KCA, \quad \angle ABC \cong \angle KCB.$$

This is the central comparison inherited directly from Forder's proof.

13.7 Extending the Side AC

The formal proof next extends AC through C to a point X :

$$A - C - X.$$

This is the line along which the adjacent exterior angle will be formed. The target for rightness is therefore to prove

$$\angle ACB \cong \angle BCX.$$

The production proof does not jump directly to that statement. Instead it applies Euclid I.32 to triangle BAC with AC produced through C to X .

The result supplies a point P between B and X such that

$$\angle ABC \cong \angle BCP$$

and

$$\angle BAC \cong \angle PCX.$$

Geometrically, CP is the interior splitter of the exterior angle $\angle BCX$.

This use of I.32 is one of the main differences between the polished source proof and the formal proof object: the angle-sum relation is represented by an explicit splitter point and two angle-congruence statements.

13.8 Two Parallel Angle Decompositions

At this stage the proof has two decompositions.

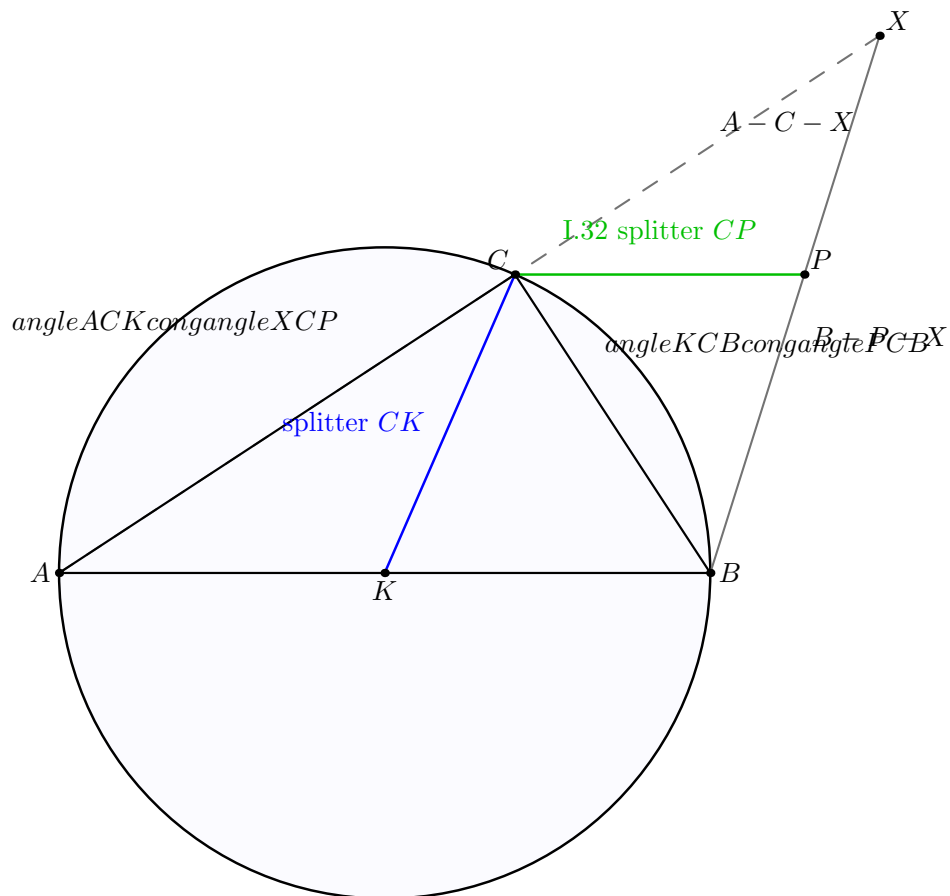


Figure 13.2: The formal I.32 configuration. The side AC is extended through C to X . The point P lies on BX , and the ray CP splits the exterior angle BCX into the two components corresponding to the remote interior angles of triangle BAC . The ray CK simultaneously splits the original angle ACB .

The first is the original angle at C , split by CK :

$$\angle ACB = \angle ACK + \angle KCB$$

in synthetic decomposition form.

The second is the adjacent exterior angle, split by CP :

$$\angle XCB = \angle XCP + \angle PCB.$$

The already established isosceles and I.32 congruences provide the componentwise matches

$$\angle ACK \cong \angle XCP$$

and

$$\angle KCB \cong \angle PCB.$$

The proof then invokes the general angle-addition theorem to conclude

$$\angle ACB \cong \angle XCB.$$

After reversing the order of the arms in the second angle, this becomes

$$\angle ACB \cong \angle BCX.$$

This is exactly the adjacent-angle equality required for a right angle when $A - C - X$.

13.9 Why the Side-of-Line Work Appears

A conspicuous part of the Lean proof is devoted to carrier lines and same-side/opposite-side conditions for the two decompositions. This is not an accidental formalization overhead.

The general angle-addition theorem does not infer from a diagram that the splitter ray CK lies inside $\angle ACB$, nor that CP lies inside $\angle XCB$. Those facts are certified through plane-separation data.

For the first decomposition, the diameter order $A - K - B$ shows that A and B lie on opposite sides of the carrier CK .

For the second, the I.32 point satisfies $B - P - X$, so B and X lie on opposite sides of the carrier CP .

The proof packages these two facts into the compatibility condition needed by the synthetic angle-addition theorem. Only then may the two component congruences be assembled.

Thus the formal proof makes explicit a fact the source diagram silently supplies: both splitters cut the corresponding angles in the same combinatorial manner.

13.10 The Final Right-Angle Step

The final result of the angle-addition stage is

$$\angle ACB \cong \angle BCX,$$

with

$$A - C - X.$$

The rays CA and CX are opposite. Hence the angles ACB and BCX are adjacent angles on a straight line. Their congruence is the synthetic definition of rightness used by the project.

The theorem therefore returns

```
HilbertRightAngle Geo A C B
```

without introducing degrees or a numerical angle measure.

This point is worth emphasizing: “right angle” is not proved by evaluating $\angle ACB$ as 90° . It is witnessed geometrically by an extension point X and equality of the two adjacent angles.

13.11 Lean Formalization

The public API is deliberately small:

```
HilbertForderIV13DiameterRightAngle
hilbert_IV13_diameter_right_angle
```

The body of the proof depends on reusable geometric results rather than exporting proposition-specific helper lemmas as new API.

Representative reusable ingredients are:

```
hilbert_circle_center_midpoint_of_chord
hilbert_circle_center_congruent
hilbert_isosceles_base_angles
hilbert_sameRay_of_between
hilbert_angle_eq_of_sameRay_first
euclid_proposition_32_exterior
hilbert_angle_addition
```

The proof pattern can be summarized as:

```
have hMidK :
  HilbertIsMidpoint Geo K A B := ...

have hIsoA :
  Geo.AngleCongruent K A C K C A := ...

have hIsoB :
  Geo.AngleCongruent K B C K C B := ...

rcases HilbertOrder.between_extension A C hAC with
  <X, hACX>

rcases euclid_proposition_32_exterior
  (Geo := Geo) B A C X hBAC hACX
```

```

with <P, hBPX, hABC_BCP, hBAC_PCX>

have hACB_XCB :
  Geo.AngleCongruent A C B X C B :=
  hilbert_angle_addition ...

exact <X, hACX, ...>

```

The production split places the general midpoint and circle lemmas in `HilbertInterfaceIV`. The numbered module therefore owns the mathematical argument of IV.13 without becoming the owner of generic circle infrastructure.

13.12 Relation to IV.12

Classically, IV.13 can look like an immediate specialization of the inscribed-angle theorem: an angle standing on a diameter intercepts a central straight angle and is therefore half of it.

That is not the dependency chosen by the production formalization. `Proposition4_13.lean` does not import `Proposition4_12.lean`. Instead it proves the right-angle theorem directly from:

- the midpoint order of the centre on a diameter,
- equal radii and isosceles triangles,
- Euclid I.32,
- synthetic angle addition.

This is a mathematically useful independence. IV.13 does not inherit the large addition/subtraction case analysis of IV.12, and IV.14 can subsequently depend on IV.13 through a much narrower theorem.

13.13 Hilbert and Earlier Euclid Components Used

13.13.1 Incidence

The proof repeatedly creates carrier lines through two distinct points and uses uniqueness of a line through two points to transfer collinearity and off-line information.

13.13.2 Order

Betweenness is essential in two places:

$$A - K - B$$

for the diameter midpoint, and

$$A - C - X$$

for the adjacent-angle construction. The I.32 point also gives

$$B - P - X.$$

These order facts generate the same-ray and opposite-side relations needed by angle transport and addition.

13.13.3 Congruence

Circle membership yields congruent radii. The congruent radii yield isosceles-base-angle congruences. The entire proof remains within segment and angle congruence; no metric coordinate is introduced.

13.13.4 Euclid I.32

The exterior-angle theorem is the main Euclidean proposition used in the formal assembly. It replaces a numerical equation of angle sums by an explicit point P and two angle-congruence relations.

13.14 Foundational Boundary

No new axiom is introduced for IV.13. The theorem is proved under

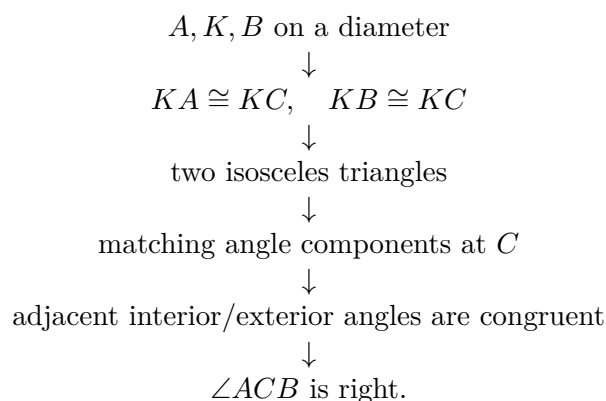
[HE : @HilbertEuclideanPlane Geo H]

using the project's existing incidence, order, congruence, extension, and Euclidean angle infrastructure.

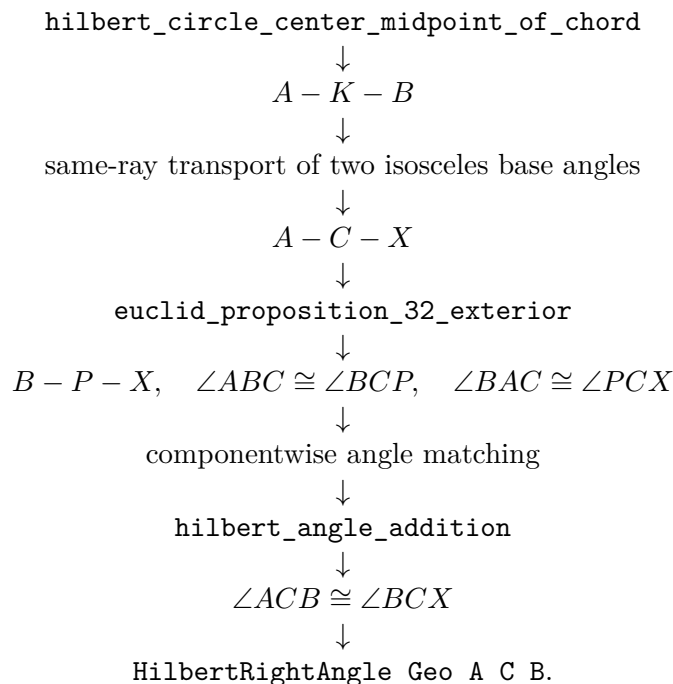
The proof is notably independent of IV.12 as a numbered theorem. Its foundational burden is therefore more transparent than the classical slogan “an angle in a semicircle is half a straight angle” might suggest.

13.15 Classical and Formal Dependency Structures

The classical argument can be summarized as



The formal proof expands the middle of that chain:



At module level the production dependency is simply

```
HilbertInterfaceIV
-> Proposition4_13
```

13.16 Diagrammatic Audit

IV.13 needs fewer figures than IV.12, but two figures are justified.

1. The semicircle configuration records the actual theorem and makes the midpoint role of the centre visible.
2. The I.32 configuration explains the formal proof architecture: the splitter CK of the original angle and the splitter CP of the exterior angle are compared component by component.

A separate dependency diagram would add little; the algebraic display above already captures the proof structure.

13.17 Conclusion

For IV.13 is short in the source because the diagram carries much of its order information. The formal reconstruction exposes that information without changing the mathematical idea.

The centre of the diameter is first made into a strict midpoint. Equal radii give two isosceles triangles, and the midpoint order transports their base angles to the two components of the target angle. After extending one side, Euclid I.32 constructs a matching decomposition of the

adjacent exterior angle. Synthetic angle addition then shows the two adjacent angles congruent, which is exactly the project's definition of a right angle.

The proposition is also architecturally clean: it depends only on the reusable Book IV interface and does not import IV.12. This makes IV.13 a narrow, independent theorem that can be used directly by IV.14 and later diameter arguments.

Chapter 14

Proposition IV.14

14.1 Introduction

Forder IV.14 is the first proposition in the present circle sequence whose printed proof is genuinely short while the formal proof architecture is substantial.

Let A, B, C, D lie on one circle with centre K . Assume that AC is a diameter and that B and D lie on opposite sides of the diameter line. Forder's conclusion is

$$\angle BAD + \angle BCD = \pi.$$

In the production formalization this is expressed synthetically: the side BC is extended through C to a point X , and one proves

$$\angle BAD \cong \angle XCD.$$

Thus $\angle BAD$ is congruent to the angle supplementary to $\angle BCD$.

The public theorem is

```
hilbert_IV14_diameter_supplement
```

in

```
CGJteamLab/Proposition4_14.lean
```

The proof depends directly on IV.13, but it also requires substantial reusable circle/order infrastructure from `HilbertInterfaceIV`. In particular, the formal development must prove an order fact that the source diagram leaves implicit: the chord BD meets the diameter AC at a point lying strictly inside the diameter.

14.2 Classical Configuration

Let AC be a diameter of the circle with centre K . The points B and D lie on opposite sides of the diameter line.

The source proof decomposes both target angles by the diameter:

$$\angle BAD = \angle BAC + \angle CAD,$$

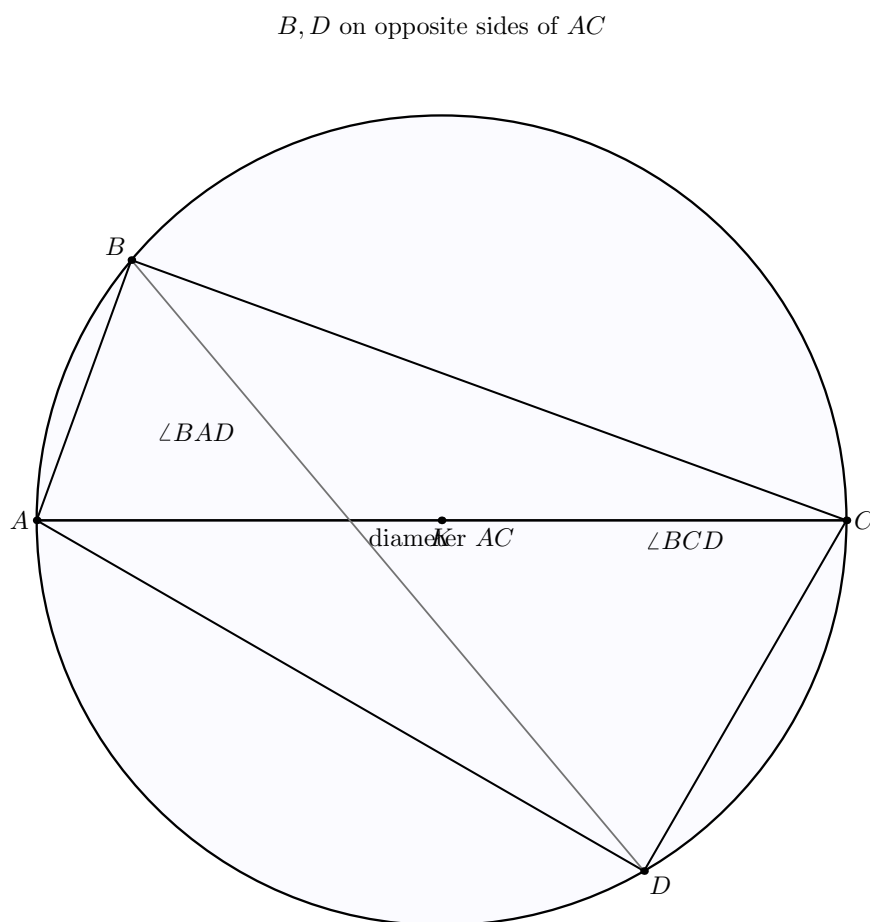


Figure 14.1: Forder IV.14. The diameter is AC ; the circle points B, D lie on opposite sides of its carrier. The theorem asserts that $\angle BAD$ and $\angle BCD$ are supplementary.

and

$$\angle BCD = \angle BCA + \angle ACD.$$

Because AC is a diameter, IV.13 gives the two right angles

$$\angle ABC, \quad \angle ADC.$$

Consequently the two acute-angle pairs satisfy

$$\angle BAC + \angle BCA = \frac{\pi}{2},$$

and

$$\angle CAD + \angle ACD = \frac{\pi}{2}.$$

Adding the decompositions yields

$$\angle BAD + \angle BCD = \pi.$$

This is almost the whole printed proof. The formal reconstruction must replace all three uses of angular arithmetic:

1. the two “sum to a right angle” statements;
2. the decomposition of $\angle BAD$ through the ray AC ;
3. the decomposition of the supplementary angle to $\angle BCD$.

All three are reconstructed from explicit rays, side-of-line information, right-angle congruence, and synthetic angle addition/subtraction.

14.3 Forder’s Statement and Proof

Forder’s statement is:

If A, B, C, D lie on a circle with centre O , A, O, C are collinear, and B, D lie on opposite sides of AC , then the angles BAD and BCD are supplementary.

His proof invokes IV.13 twice. The angles subtending the diameter AC at B and D are right. Therefore the two acute angles of triangle ABC fill one right angle and the two acute angles of triangle ADC fill another.

The fact that B, D lie on opposite sides of AC implies that the segments BD and AC meet. The diagram then shows that AC is an interior divider of $\angle BAD$, while CA is an interior divider of $\angle BCD$. Forder therefore combines the two pairs of acute angles and obtains one straight angle [1].

The formalization follows that architecture but cannot use the phrases “the diagram shows” or “add the two equalities”. Those are exactly the places where the proof expands.

14.4 Formal Statement

IV.14 is slightly different from its neighbours in the production API. There is no separate proposition abbreviation `HilbertForderIV14...`. The public theorem is exported directly:

```
theorem hilbert_IV14_diameter_supplement
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H]
  (K R A B C D : Geo.Point)
  (diameter : Geo.Line)
  (hAC : Ne A C)
  (hAdiam : H.OnLine A diameter)
  (hCdiam : H.OnLine C diameter)
  (hKdiam : H.OnLine K diameter)
  (hAcircle : HilbertCircle Geo K R A)
  (hBcircle : HilbertCircle Geo K R B)
  (hCcircle : HilbertCircle Geo K R C)
  (hDcircle : HilbertCircle Geo K R D)
  (hOppBD : HilbertOppositeSide Geo B D diameter) :
  exists X : Geo.Point,
    BookZeroSupplement Geo B C D D X /\
    Geo.AngleCongruent B A D X C D
```

The conclusion contains both pieces of the synthetic supplement statement.

First,

```
BookZeroSupplement Geo B C D D X
```

records the straight-line extension $B - C - X$ relative to the angle BCD .

Second,

```
Geo.AngleCongruent B A D X C D
```

states that the angle at A is congruent to that adjacent supplementary angle at C .

Thus the theorem does not assert a numerical equation of angle magnitudes. It constructs the supplement geometry.

14.5 Production Architecture

The current module imports only

```
CGJteamLab.HilbertInterfaceIV
CGJteamLab.Proposition4_13
```

The proposition-specific proof is still organized by private implementation lemmas. Their stage suffixes are internal historical names, not public API:

```

hilbert_forder_diameter_right_decomposition_stage36
hilbert_forder_IV14_diameter_rays_inside_stage38
hilbert_forder_IV14_prescribed_perpendicular_stage39
hilbert_forder_IV14_perpendicular_inside_supplement_stage40
hilbert_forder_IV14_first_component_stage41
hilbert_forder_IV14_second_component_stage42
hilbert_forder_IV14_diameter_supplement_stage43

```

Two facts discovered during the IV.14 development were promoted to the reusable interface because their statements are proposition-independent:

```

hilbert_right_triangle_third_angle_congruent
hilbert_circle_diameter_chord_intersection_inside

```

The second of these is used directly by the production IV.14 proof. The first is a reusable by-product of the development, but the final `Proposition4_14.lean` proof does not require it as a local step.

14.6 Right-Angle Decomposition from IV.13

A recurring local block in IV.14 begins with a diameter AC and a circle point B off the diameter. IV.13 yields

$$\angle ABC \text{ right.}$$

The proof extends AB through B to a point E :

$$A - B - E.$$

The adjacent angle CBE is therefore right as well.

Euclid I.32 is then applied to triangle CAB with AB produced through B . It constructs a point P such that

$$C - P - E$$

and supplies the component congruences

$$\angle ACB \cong \angle CBP, \quad \angle CAB \cong \angle PBE.$$

Hence the right angle CBE is decomposed synthetically into the two acute angles of triangle ABC .

This local theorem replaces the numerical identity

$$\angle ACB + \angle CAB = \frac{\pi}{2}.$$

The important point is methodological: even a familiar “the acute angles of a right triangle are complementary” statement is realized by an explicit exterior construction and two angle congruences.

14.7 The Hidden Order Lemma: the Chord Crosses the Diameter Internally

From the opposite-side hypothesis, the line segment BD meets the diameter carrier at some point Y :

$$B - Y - D.$$

For the later angle decomposition it is not enough to know that Y lies somewhere on the line AC . One must prove

$$A - Y - C.$$

internal chord–diameter crossing

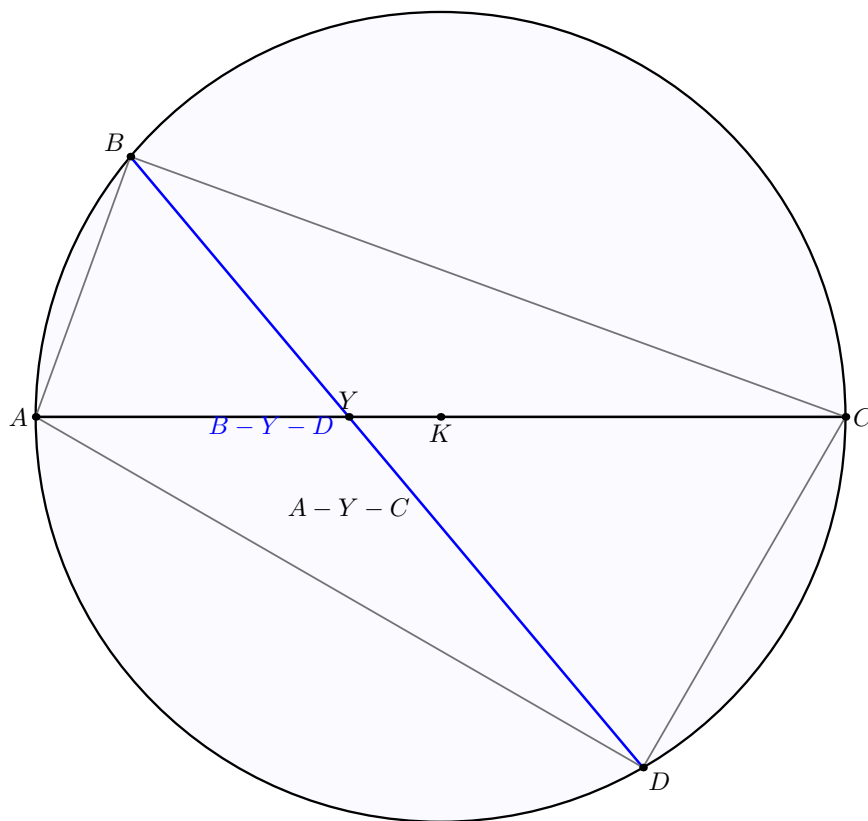


Figure 14.2: The hidden order fact in IV.14. The opposite-side hypothesis gives a point Y with $B - Y - D$ on the diameter carrier. The formal proof must show the stronger statement $A - Y - C$, so the diameter rays are genuine interior dividers of the two target angles.

The reusable theorem is now

```
hilbert_circle_diameter_chord_intersection_inside
```

Its proof has two branches.

If the centre K lies on chord BD , uniqueness of the intersection of the chord carrier and the diameter carrier forces

$$Y = K.$$

Since K is the midpoint of diameter AC , the conclusion follows.

Otherwise Y is a proper interior point of chord BD . A reusable circle theorem gives

$$KY < KB.$$

All radii are congruent, so

$$KY < KA.$$

If Y lay beyond either endpoint of the diameter, the order geometry would force the reverse strict inequality. Segment-order trichotomy therefore leaves only

$$A - Y - C.$$

This is a particularly good example of formalization exposing a hidden diagrammatic premise. No continuity or coordinate argument is needed; the proof is purely synthetic.

14.8 From the Crossing Point to Interior Rays

Once

$$B - Y - D, \quad A - Y - C$$

are known, the proposition-specific helper

```
hilbert_forder_IV14_diameter_rays_inside_stage38
```

converts the order configuration into the two statements

$$\overrightarrow{AC} \text{ is interior to } \angle BAD$$

and

$$\overrightarrow{CA} \text{ is interior to } \angle BCD.$$

In Lean these are

```
HilbertRayMeetsSegment Geo A C B D
HilbertRayMeetsSegment Geo C A B D
```

This is exactly the information needed later by the angle-addition API. The theorem contains no angle calculation at all; it is a pure incidence/order bridge between the circle configuration and the angle decomposition.

14.9 Constructing a Perpendicular on the Prescribed Side

The formal proof next creates a ray CP perpendicular to the diameter at C , specifically on the same side of the diameter as D .

The construction uses IV.13 to obtain the right angle

$$\angle ABC.$$

Hilbert's angle-construction theorem copies this angle to the vertex C with first arm CA , choosing the side that contains D . The resulting point P satisfies

$$P, D \text{ are on the same side of } AC$$

and

$$\angle ACP \text{ is right.}$$

The private production helper is

```
hilbert_forder_IV14_prescribed_perpendicular_stage39
```

This step is important because the eventual supplementary angle to $\angle BCD$ will be decomposed through CP . The point P is not an arbitrary perpendicular witness: its half-plane is prescribed by the location of D .

14.10 Constructing the Supplementary Ray

Extend BC through C to a point X :

$$B - C - X.$$

This is the point that will appear in the final `BookZeroSupplement` witness.

The proof now needs a precise nesting of three rays at C :

$$CP \text{ inside } \angle ACX,$$

$$CD \text{ inside } \angle ACP,$$

and

$$CP \text{ inside } \angle XCD.$$

These statements are the core output of

```
hilbert_forder_IV14_perpendicular_inside_supplement_stage40
```

The proof is not based on a visual comparison of slopes. It uses strict angle comparison.

For example, Euclid I.16 gives an inequality showing that the right angle copied to ACP is smaller than the larger exterior angle ACX . This places CP inside ACX .

Similarly, in the right triangle ADC , the acute angle at C is strictly smaller than the right angle ACP . This places CD inside ACP .

Nested-interior-ray lemmas then combine these order relations to obtain the final position of CP inside XCD .

Thus a single source picture expands into a genuine chain of synthetic angle inequalities and ray-order statements.

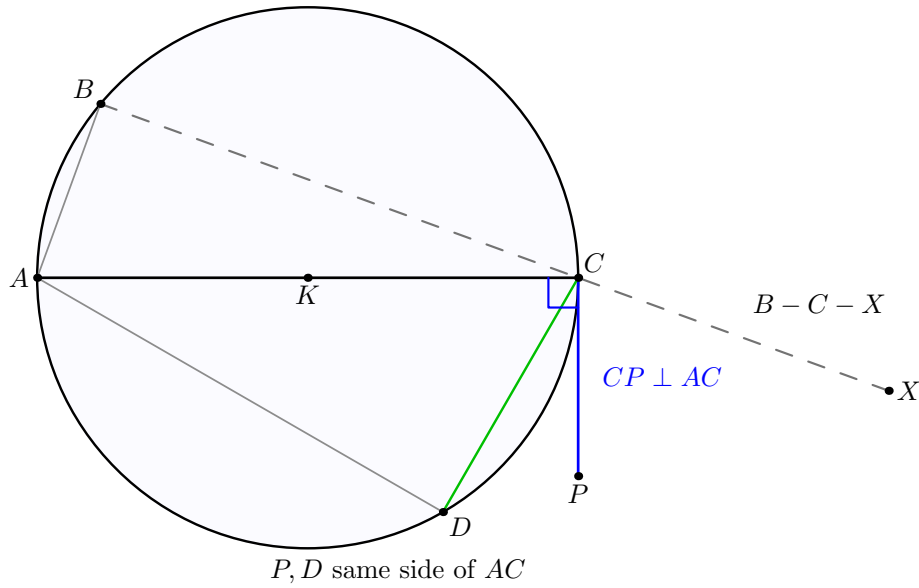


Figure 14.3: The auxiliary configuration at C . The ray CX extends CB ; the ray CP is perpendicular to the diameter and chosen on the side containing D . The proof establishes the nested order CD inside ACP and CP inside XCD .

14.11 The First Component Congruence

The first target component is

$$\angle BAC \cong \angle XCP.$$

The proof again applies Euclid I.32, now to triangle ABC with BC produced through C to X . It obtains an interior splitter CQ of the exterior angle ACX and congruences

$$\angle BAC \cong \angle ACQ,$$

$$\angle ABC \cong \angle QCX.$$

The angles ABC and ACP are both right, hence congruent. Therefore the right-hand component XCQ of the exterior decomposition is congruent to ACP .

At this point the proof compares two decompositions of congruent whole angles:

$$\angle ACX \quad \text{and} \quad \angle XCA.$$

The corresponding right components are congruent, so synthetic angle subtraction yields

$$\angle ACQ \cong \angle XCP.$$

Combining with I.32 gives

$$\boxed{\angle BAC \cong \angle XCP}.$$

The private helper closing this block is

hilbert_forder_IV14_first_component_stage41

14.12 The Second Component Congruence

The second target component is

$$\angle CAD \cong \angle PCD.$$

This time the proof works with triangle ACD .

IV.13 gives

$$\angle ADC \text{ right.}$$

Extend CD through D to E :

$$C - D - E.$$

The adjacent angle

$$\angle ADE$$

is right.

Apply I.32 to triangle ACD with CD produced to E . The construction gives a splitter DQ and

$$\angle CAD \cong \angle ADQ,$$

$$\angle ACD \cong \angle QDE.$$

The whole angle ADE is right, and the auxiliary angle PCA is also right because $CP \perp AC$. Hence the wholes are congruent.

The component EDQ is congruent to ACD . Synthetic subtraction of the corresponding components therefore gives

$$\angle ADQ \cong \angle PCD.$$

Combining with I.32 yields

$$\boxed{\angle CAD \cong \angle PCD}.$$

This is the output of

```
hilbert_forder_IV14_second_component_stage42
```

The two component lemmas mirror one another, but they are not literally the same proof: the first uses the extension $B - C - X$, while the second uses a new extension $C - D - E$.

14.13 Final Assembly

At A , the ray AC lies inside $\angle BAD$. Hence

$$\angle BAD$$

is decomposed into

$$\angle BAC \quad \text{and} \quad \angle CAD.$$

At C , the ray CP lies inside $\angle XCD$. Hence

$$\angle XCD$$

is decomposed into

$$\angle XCP \text{ and } \angle PCD.$$

The two component congruences proved above are

$$\angle BAC \cong \angle XCP$$

and

$$\angle CAD \cong \angle PCD.$$

Synthetic angle addition therefore gives

$$\angle BAD \cong \angle XCD.$$

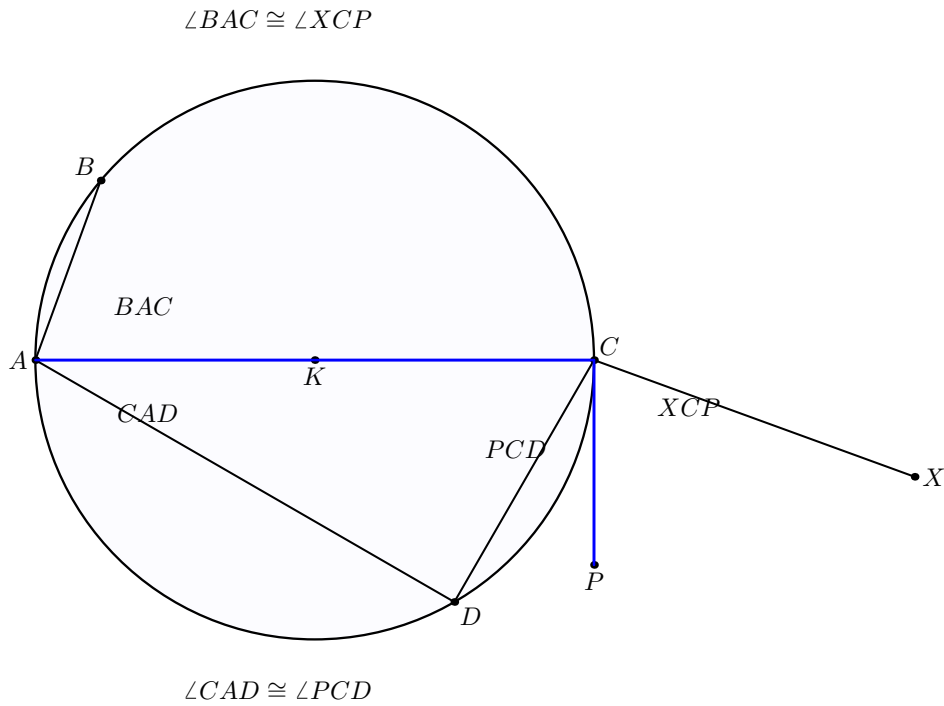


Figure 14.4: Final component matching in IV.14. The diameter ray AC decomposes $\angle BAD$, while the auxiliary perpendicular CP decomposes the supplementary angle $\angle XCD$. The two corresponding component pairs are proved congruent separately and then reassembled.

Finally, $B - C - X$ is exactly the straight-line extension required by the Book Zero supplement predicate. The theorem returns

$$X$$

together with

$$\text{Supplement}(BCD, DCX)$$

and

$$\angle BAD \cong \angle XCD.$$

This is the synthetic content of Forder's equation

$$\angle BAD + \angle BCD = \pi.$$

14.14 Lean Formalization

The public theorem is deliberately thin:

```

theorem hilbert_IV14_diameter_supplement
  ...
  : exists X : Geo.Point,
    BookZeroSupplement Geo B C D D X /\
    Geo.AngleCongruent B A D X C D :=
  hilbert_forder_IV14_diameter_supplement_stage43
  ...

```

All proposition-specific detail is private to `Proposition4_14.lean`.

The proof architecture is:

```

IV.13 right angles
  |
  v
diameter/right-angle decomposition
  |
  +-----+
  |                                     |
  v                                     v
internal chord/diameter      prescribed perpendicular
intersection                  CP
  |                             |
  v                             v
diameter rays interior      extend BC through C to X
  |                             |
  +-----+-----+
  |                                     |
  v                                     v
nested ray configuration
  |
  +-----+-----+
  |                                     |
  v                                     v
first component      second component
BAC ~= XCP           CAD ~= PCD
  |                   |
  +-----+-----+
  |
  v
angle addition
  |
  v
BAD ~= XCD
  |
  v
BookZeroSupplement

```

This is considerably richer than the three-line angle calculation in the source, but every extra step has a specific synthetic role.

14.15 Reusable Interface Ownership

Two general results associated with this development belong to `HilbertInterfaceIV`.

14.15.1 Diameter–chord intersection

The theorem

```
hilbert_circle_diameter_chord_intersection_inside
```

states a general fact about a chord connecting opposite sides of a diameter. Its statement does not mention IV.14 and it is reusable independently.

14.15.2 Third angle in right triangles

The theorem

```
hilbert_right_triangle_third_angle_congruent
```

says that two right triangles with one pair of corresponding acute angles congruent have their remaining acute angles congruent.

It was isolated during the IV.14 development and belongs semantically in the interface. The final production proof of IV.14 uses the more direct decomposition/subtraction route described above, so this theorem is not a hidden proposition-level dependency.

This distinction is important for documentation: provenance during proof development is not the same thing as ownership in the final library.

14.16 Relation to IV.13

IV.14 imports `Proposition4_13.lean` directly.

IV.13 is used repeatedly:

- to prove the diameter angle at B right;
- to prove the diameter angle at D right;
- to provide a right angle that can be copied to construct the prescribed perpendicular at C .

Thus IV.14 is the first proposition in this block whose production module has an explicit numbered predecessor dependency:

```
HilbertInterfaceIV
Proposition4_13
-> Proposition4_14
```

IV.12 is not required by this proof.

14.17 Hilbert, Book Zero, and Earlier Euclid Components

14.17.1 Incidence and plane separation

The opposite-side hypothesis is structural. It supplies the crossing point of chord BD with the diameter line and controls the side on which the copied perpendicular CP is constructed.

14.17.2 Order

Betweenness appears at every major transition:

$$A - Y - C, \quad B - Y - D, \quad B - C - X, \quad C - D - E.$$

These order relations are converted into interior-ray witnesses used by the angle-decomposition API.

14.17.3 Congruence

Circle membership supplies congruent radii and IV.13 supplies right angles. The theorem that all right angles are congruent is then used repeatedly to match components across different vertices.

14.17.4 Hilbert angle construction

The prescribed perpendicular at C is obtained by copying an existing right angle to a specified side of the diameter. This is the synthetic substitute for choosing a coordinate direction perpendicular to AC .

14.17.5 Euclid I.16 and I.32

I.16 supplies strict angle inequalities needed to prove ray interiority in the nested configuration.

I.32 is used to create explicit decompositions of exterior angles into remote interior components. It is the main bridge from the classical right-triangle picture to the synthetic angle-decomposition machinery.

14.17.6 Book Zero supplement

The final theorem does not use a numerical straight angle. The supplement is represented by a point X on the continuation of BC , precisely as encoded by `BookZeroSupplement`.

14.18 Foundational Boundary

IV.14 introduces no new axiom and uses no coordinate geometry.

The difficult-looking step that a chord crossing a diameter does so internally is proved from:

- plane separation,
- segment order,
- circle-radius congruence,
- the theorem that an interior point of a chord lies strictly inside the circle.

The remainder of the proof uses the already established Euclidean-plane angle infrastructure.

The formal statement therefore sits cleanly within the same `HilbertEuclideanPlane` framework as IV.13.

14.19 Classical and Formal Dependency Structures

Forder’s visible proof is essentially

$$\begin{array}{c}
 AC \text{ diameter} \\
 \downarrow \\
 \angle ABC, \angle ADC \text{ right (IV.13)} \\
 \downarrow \\
 \begin{array}{l}
 \angle BAC + \angle BCA = \frac{\pi}{2} \\
 \angle CAD + \angle ACD = \frac{\pi}{2}
 \end{array} \\
 \downarrow \\
 AC \text{ and } BD \text{ cross} \\
 \downarrow \\
 \angle BAD + \angle BCD = \pi.
 \end{array}$$

The formal reconstruction expands this to

$$\begin{array}{c}
 \text{IV.13} \\
 \downarrow \\
 \text{right-angle decompositions} \\
 \downarrow \\
 \text{hilbert_circle_diameter_chord_intersection_inside} \\
 \downarrow \\
 AC, CA \text{ certified as interior dividing rays} \\
 \downarrow \\
 \text{prescribed perpendicular } CP \\
 \downarrow \\
 B - C - X \text{ and nested interior rays} \\
 \downarrow \\
 \begin{array}{l}
 \angle BAC \cong \angle XCP \\
 \angle CAD \cong \angle PCD
 \end{array} \\
 \downarrow \\
 \text{synthetic angle addition} \\
 \downarrow \\
 \angle BAD \cong \angle XCD \\
 \downarrow \\
 \text{BookZeroSupplement.}
 \end{array}$$

At module level:

```

HilbertInterfaceIV
  |
Proposition4_13
  |
  v
Proposition4_14

```

14.20 Diagrammatic Audit

Four figures are justified for IV.14.

1. The source configuration shows the theorem itself.
2. The chord–diameter intersection figure isolates the hidden order lemma.
3. The perpendicular/supplement figure explains the auxiliary ray CP and the extension $B - C - X$.
4. The component figure displays the final two-by-two angle matching used by synthetic addition.

These figures correspond to genuinely different logical layers. Combining them into one dense diagram would conceal the formal proof structure.

14.21 Conclusion

Forder IV.14 is a compact classical argument whose formal reconstruction reveals three distinct kinds of hidden structure.

First, the crossing of BD and AC must be strengthened to an internal crossing of the diameter. Second, the supplementary relation must be realized by an actual extension point X . Third, the angle arithmetic must be replaced by explicit decompositions and componentwise congruence.

The resulting proof remains fully synthetic. It uses IV.13, plane separation, betweenness, angle construction, Euclid I.16 and I.32, and the project’s angle addition/subtraction machinery. The public theorem exposes only the natural geometric conclusion, while the proposition-specific construction remains private to `Proposition4_14.lean`.

This proposition is therefore a good model of the purpose of the present documentation: not merely to record that Lean accepted the theorem, but to show precisely which pieces of geometry are compressed into the classical diagram and how they are recovered in the formal proof.

Chapter 15

Proposition IV.15

15.1 Introduction

Forder IV.15 is the general cyclic-quadrilateral supplement theorem: if A, B, C, D form a cyclic quadrilateral, then the opposite angles

$$\angle BAD \quad \text{and} \quad \angle BCD$$

are supplementary [1].

In the source this proposition is proved from IV.12.1, IV.13, and IV.14 by introducing the centre of the circumcircle and, in the generic case, an antipodal point. The formal reconstruction follows exactly that geometric strategy, but it makes explicit several order facts that Forder leaves in the diagram.

The production file is

```
CGJteamLab/Proposition4_15.lean
```

and the public theorem is

```
hilbert_IV15_cyclic_supplement
```

IV.15 is also the first proposition in this Book IV block whose formal proof uses all three earlier numbered results:

$$\text{IV.12.1,} \quad \text{IV.13,} \quad \text{IV.14.}$$

15.2 Classical Configuration

Let A, B, C, D be four points on one circle. We use the diagonal BD as the reference carrier. In the convex cyclic configuration the other two vertices A, C lie on opposite sides of BD .

The production interface does not leave this convexity information implicit. It asks for an explicit line `diagonal` through B, D and the hypothesis

```
HilbertOppositeSide Geo A C diagonal
```

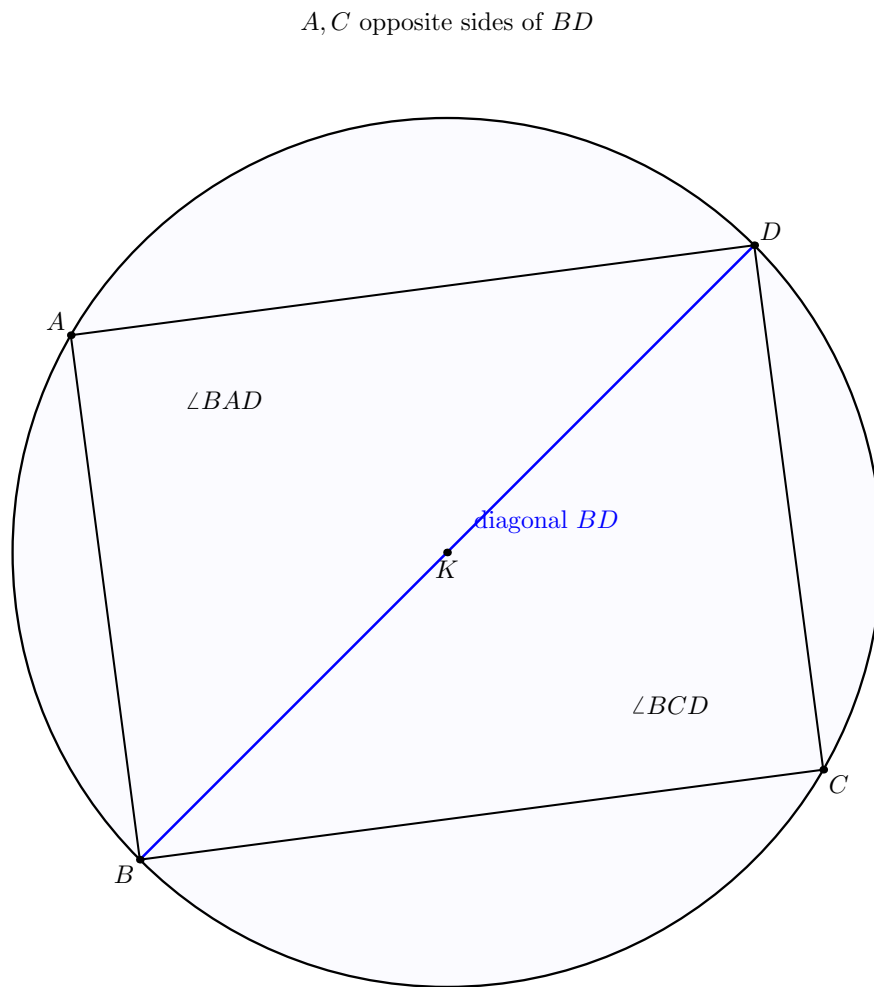


Figure 15.1: Forder IV.15. The cyclic quadrilateral is organized around the diagonal BD . The vertices A, C lie on opposite sides of that diagonal, and the target is that $\angle BAD$ and $\angle BCD$ are supplementary.

This is the exact order information needed by the proof.

15.3 Forder's Statement and Proof

Forder states:

If $ABCD$ is a cyclic quadrilateral, then $\angle BAD + \angle BCD = \pi$.

He first notes that A and C lie on opposite sides of the diagonal BD . Let O be the centre of the circle.

If O lies on BD , then BD is a diameter. Both $\angle BAD$ and $\angle BCD$ are right by IV.13, so they are supplementary.

Suppose now that O is off BD . Since A, C lie on opposite sides, the centre lies on the same side of BD as exactly one of them. Forder chooses the case in which O is on the same side as C . Let E be the point of the circle antipodal to A , so that

$$A - O - E.$$

The diameter AE crosses the diagonal BD , and the relevant order relations imply that B, D lie on opposite sides of AE . IV.14 applied to the cyclic quadrilateral A, B, E, D gives

$$\angle BAD + \angle BED = \pi.$$

On the other hand C, O, E lie on the same side of BD , so IV.12.1 gives

$$\angle BED \cong \angle BCD.$$

Substitution yields

$$\angle BAD + \angle BCD = \pi.$$

The printed proof handles the alternative placement of the centre by symmetry. The formal proof makes that symmetry explicit by relabelling the cyclic quadrilateral.

15.4 Formal Statement

The public proposition contract is

```
def HilbertForderIV15CyclicSupplement
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall A B C D : Geo.Point,
    forall diagonal : Geo.Line,
      Ne B D ->
        H.OnLine B diagonal ->
          H.OnLine D diagonal ->
            HilbertConcyclic4 Geo A B C D ->
              HilbertOppositeSide Geo A C diagonal ->
                Not (PrimCollinear Geo B A D) ->
```

```

Not (PrimCollinear Geo B C D) ->
exists X : Geo.Point,
  BookZeroSupplement Geo
    B C D
    D X /\
  Geo.AngleCongruent
    B A D
    X C D

```

The theorem therefore states the supplement synthetically. The witness X satisfies the straight-line extension condition associated with $\angle BCD$, and the angle

$$\angle XCD$$

is congruent to the opposite cyclic angle

$$\angle BAD.$$

The closed production theorem is

```

theorem hilbert_IV15_cyclic_supplement
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV15CyclicSupplement (Geo := Geo)

```

15.5 Recovering an Explicit Circumcircle

The hypothesis

```
HilbertConcyclic4 Geo A B C D
```

does not expose the centre directly. The first step of the production proof is therefore

```

rcases
  hilbert_concyclic4_circle
    Geo A B C D hCyclic
with
  <K, hAcircle, hBcircle, hCcircle, hDcircle>

```

This produces an explicit centre K and circle membership witnesses for all four vertices.

The rest of the proof is a classification of the position of K relative to the diagonal BD .

15.6 Case I: the Centre Lies on the Diagonal

Assume

$$K \in BD.$$

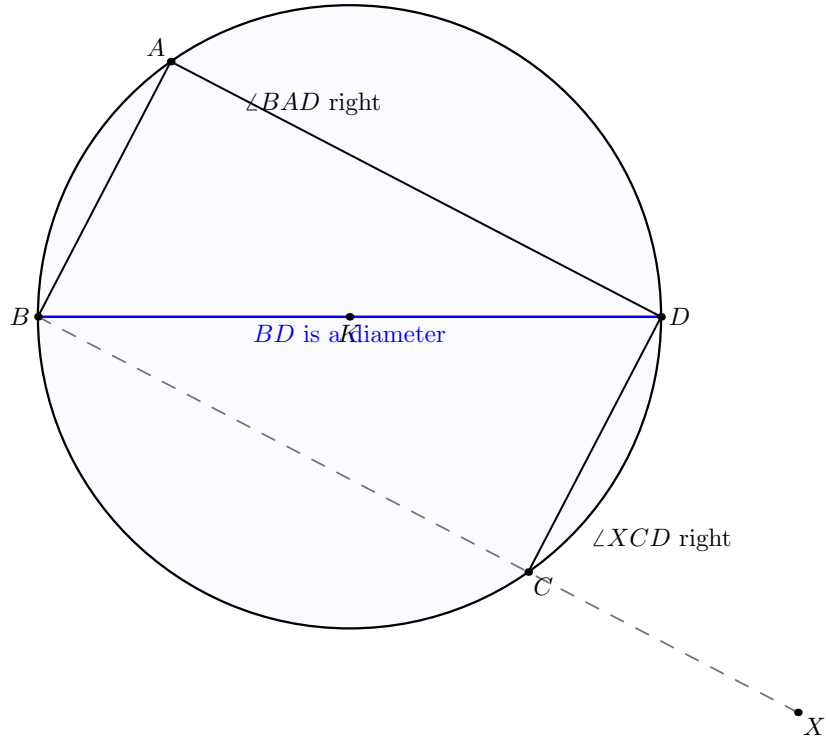


Figure 15.2: The exceptional branch $K \in BD$. The diagonal BD is a diameter, so both opposite angles BAD and BCD are right by IV.13.

Then BD is a diameter of the circumcircle.

IV.13 gives

$$\angle BAD \text{ right}, \quad \angle BCD \text{ right}.$$

The formal proof extends BC through C to X :

$$B - C - X.$$

The right angle at C is transported across the straight carrier so that

$$\angle XCD$$

is also right. Since all right angles are congruent,

$$\angle BAD \cong \angle XCD.$$

The point X therefore witnesses

$$\text{Supplement}(BCD, DCX),$$

and the IV.15 conclusion follows immediately.

This branch is implemented by

```
hilbert_forder_IV15_center_on_diagonal_stage44
```

and depends essentially only on IV.13 and the Book Zero notion of a supplement.

15.7 The Generic Case: the Centre Is Off the Diagonal

Assume now

$$K \notin BD.$$

Since A and C lie on opposite sides of BD , exactly one of the two vertices is on the same side of the diagonal as K .

The proof first treats the branch

$$C \sim K \text{ relative to } BD.$$

This is the branch that corresponds directly to Forder's printed diagram.

15.8 Constructing the Antipode of A

Let E be the antipode of A :

$$A - K - E,$$

with E on the same circle.

The reusable theorem is

```
hilbert_circle_antipode
```

The line AE is therefore a diameter.

The formal proof next establishes that A and E lie on opposite sides of the diagonal BD . The argument is pure order transport.

Because A, C are opposite sides and C, K are same side, A, K are opposite sides. Let Y be their crossing point on BD :

$$A - Y - K.$$

Since also

$$A - K - E,$$

betweenness transitivity gives

$$A - Y - E.$$

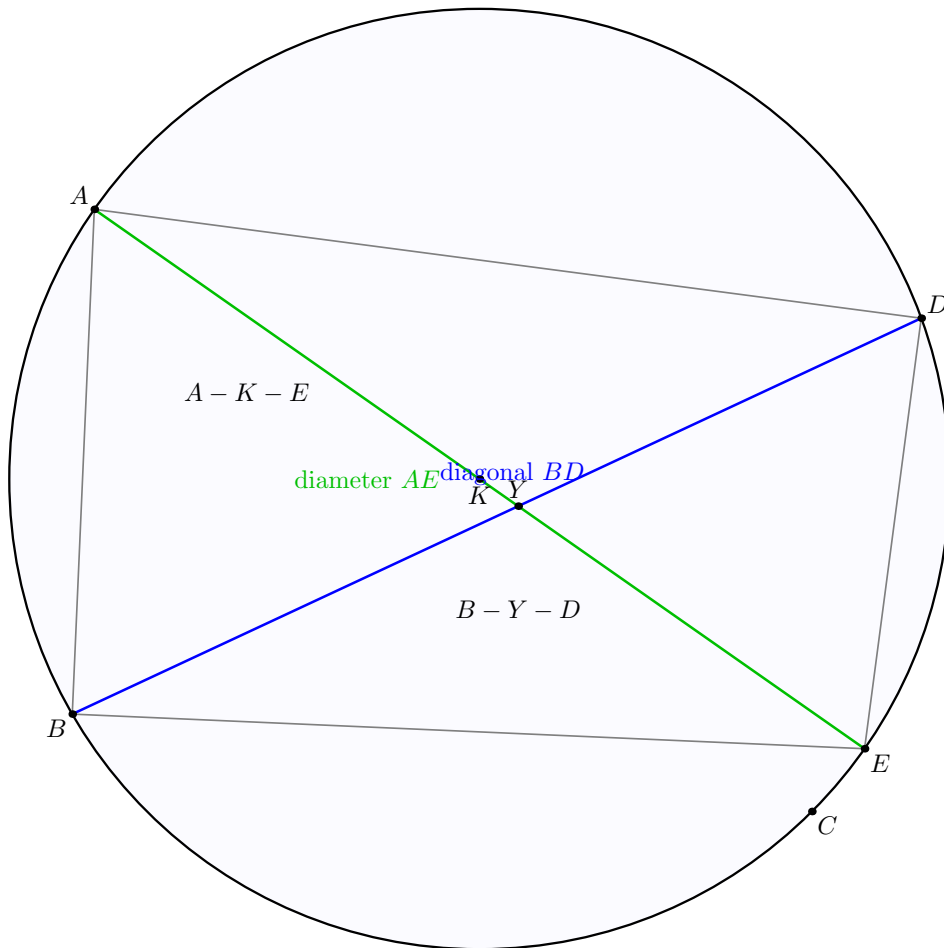
Thus the same point Y witnesses that A, E lie on opposite sides of BD .

The proposition-specific helper is

```
hilbert_IV15_antipode_opposite_diagonal_stage45
```

15.9 Where Is the Intersection Point on the Diameter?

The crossing point Y lies on both the diagonal BD and the diameter AE . To control the position of B, D relative to AE , the proof first needs to know that Y lies inside the circle unless it is the centre.



C, K, E same side of BD

Figure 15.3: Generic IV.15 branch. The centre K lies on the same side of the diagonal BD as C . The point E is the antipode of A , so $A - K - E$. The diameter AE crosses BD at Y .

From

$$A - K - E, \quad A - Y - E, \quad KA \cong KE,$$

the local lemma

```
hilbert_IV15_diameter_inner_point_stage46
```

proves

$$Y = K \quad \text{or} \quad KY < KA.$$

The proof is a trichotomy on the collinear points A, K, Y .

If $A - K - Y$, then $K - Y - E$, so

$$KY < KE \cong KA.$$

The order $K - A - Y$ is incompatible with $A - K - E$ and $A - Y - E$.

If $A - Y - K$, then $K - Y - A$ directly gives

$$KY < KA.$$

Thus the intersection point is either the centre or strictly inside the circle.

15.10 Why B, D Must Be on Opposite Sides of the New Diameter

The key order theorem of the generic branch is that the chord endpoints B, D lie on opposite sides of the diameter AE .

The difficulty is subtle. We know only that Y lies on the chord carrier BD and on the diameter AE . To prove opposite-side status we must show

$$B - Y - D.$$

The points B, Y, D are collinear, so order trichotomy gives three possibilities:

$$B - Y - D, \quad Y - B - D, \quad B - D - Y.$$

The first is the desired order.

The other two would put Y outside the chord. Since

$$KB \cong KD,$$

the two exterior-order lemmas

```
hilbert_IV15_chord_exterior_farther_left_stage47
hilbert_IV15_chord_exterior_farther_right_stage47
```

show respectively that an exterior Y would satisfy

$$KD < KY \quad \text{or} \quad KB < KY.$$

But the previous diameter lemma gives

$$Y = K \quad \text{or} \quad KY < KA,$$

while all circle radii satisfy

$$KA \cong KB \cong KD.$$

Both exterior orders therefore contradict the radius comparison.

Hence only

$$B - Y - D$$

is possible, and the diameter AE genuinely separates B from D .

The proposition-specific theorem is

hilbert_IV15_chord_opposite_diameter_stage48

This is one of the places where the formal proof contains much more order geometry than the source text: Forder's figure treats this separation as visually evident.

15.11 Applying IV.14 to the Antipodal Quadrilateral

Now A, E are endpoints of a diameter, while B, D lie on opposite sides of that diameter. IV.14 applies to the four circle points

$$A, B, E, D.$$

It constructs a point X such that

$$\angle BAD \cong \angle XED$$

and $\angle XED$ is a supplement of

$$\angle BED.$$

In source notation this is exactly

$$\angle BAD + \angle BED = \pi.$$

This is the first half of Forder's generic argument.

15.12 Transporting the Same-Segment Angle

We next compare $\angle BED$ with the original angle $\angle BCD$.

From A, K being opposite sides of BD and the order

$$A - K - E,$$

the point E lies on the same side of BD as K .

Since the current branch assumes that C, K are also on the same side of BD , both C and E are in the same segment determined by chord BD .

IV.12.1 therefore gives

$$\angle BED \cong \angle BCD.$$

The production proof uses

```
hilbert_IV12_same_segment
```

directly.

The same-side transport from $A - K - E$ is now supplied by the reusable interface theorem

```
hilbert_oppositeSide_tail_sameSide
```

rather than by a proposition-specific stage helper.

15.13 Transporting Supplements

The proof now has:

$$\angle BAD \cong \text{a supplement of } \angle BED,$$

and

$$\angle BED \cong \angle BCD.$$

The target, however, is not merely to say that congruent angles have equal numerical supplements. We must construct the actual supplement at C .

Extend BC through C to X_c :

$$B - C - X_c.$$

This supplies

```
BookZeroSupplement Geo B C D D Xc
```

Book Zero theorem 43 on supplements then transports the congruence

$$\angle BED \cong \angle BCD$$

to a congruence of their supplements. After reorienting the angle arms, the formal chain is

$$\angle BAD \cong \angle XED \cong \angle X_cCD.$$

Therefore

$$\boxed{\angle BAD \cong \angle X_cCD}.$$

This completes the branch $C \sim K$.

The proposition-specific theorem is

```
hilbert_forder_IV15_sameSide_CK_stage50
```

15.14 The Other Side of the Diagonal: Explicit Symmetry

The source proof simply chooses the case in which the centre is on the same side of BD as C . Formally, when this is false we must close the alternative branch.

Assume

$$C \not\sim K.$$

Because both are off the diagonal, they are opposite sides:

$$C \mid K.$$

Since $A \mid C$, the two points A, K are on the same side of BD .

The reusable theorem

```
hilbert_two_opposites_sameSide_with_carrier
```

performs this plane-separation inference.

Now apply the already proved same-side- K branch after the cyclic relabelling

$$(A, B, C, D) \mapsto (C, D, A, B).$$

The relabelled theorem returns a supplement relation based at A and an angle congruence

$$\angle DCB \cong \angle XAB.$$

After reversing the first angle this is

$$\angle BCD \cong \angle XAB.$$

The same witness also shows that $\angle BAD$ is the supplement of $\angle XAB$. The proof then constructs the target supplement of $\angle BCD$ at C and invokes Book Zero supplement transport once more.

The result is

$$\angle BAD \cong \angle X_c CD.$$

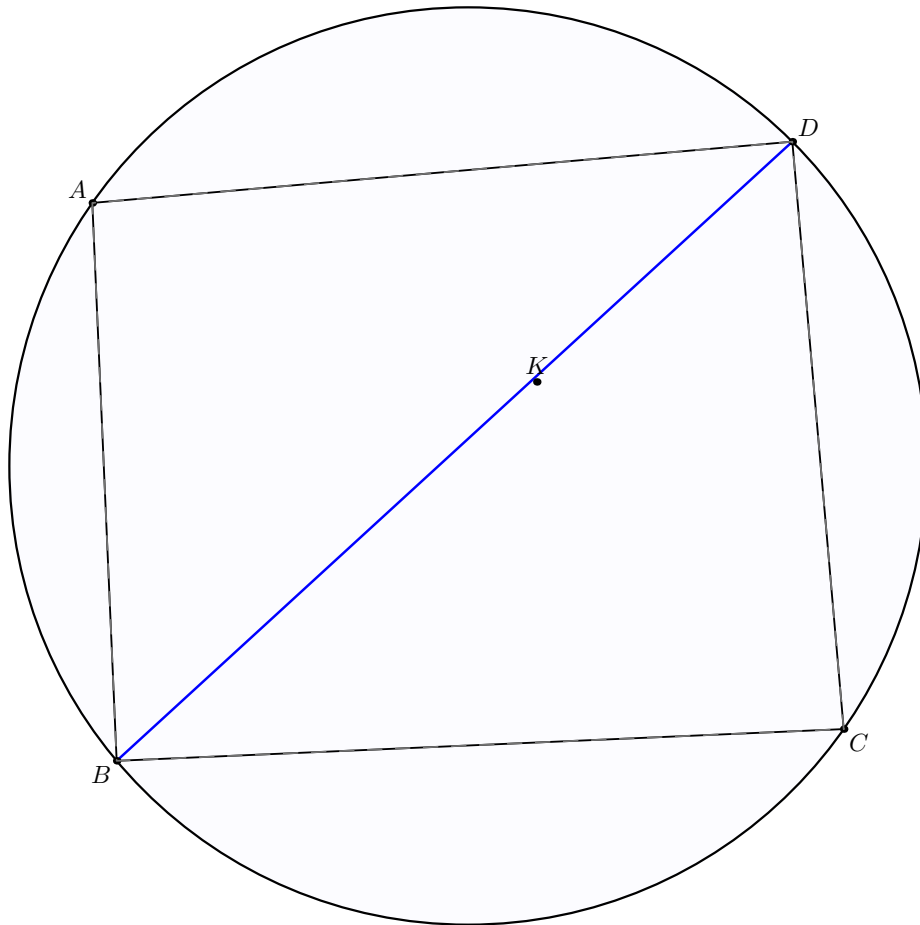
This explicit relabelling is the formal replacement for the phrase “the other case is symmetric”.

15.15 Completion of IV.15

The public proof has three branches.

1. $K \in BD$: the diagonal is a diameter, so both target angles are right by IV.13.
2. $K \notin BD$ and C, K are on the same side: construct the antipode of A , apply IV.14, then IV.12.1, and transport supplements.
3. $K \notin BD$ and C, K are not on the same side: prove A, K same side, cyclically relabel the quadrilateral, apply the completed generic branch, and transport the result back.

K on the A -side of BD



apply completed branch to

$$(A, B, C, D) \mapsto (C, D, A, B)$$

Figure 15.4: The branch not drawn by Forder. If K lies on the same side of BD as A rather than C , the proof applies the completed generic branch to the cyclic relabelling $(A, B, C, D) \mapsto (C, D, A, B)$, then transports the resulting supplement back to the original orientation.

This case split is implemented by

```
hilbert_forder_IV15_complete_stage52
```

and the public theorem is the thin production wrapper

```
theorem hilbert_IV15_cyclic_supplement
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV15CyclicSupplement (Geo := Geo) :=
  hilbert_forder_IV15_complete_stage52 Geo
```

15.16 Lean Formalization

The proposition-specific implementation layer consists of:

```
hilbert_forder_IV15_center_on_diagonal_stage44
hilbert_IV15_antipode_opposite_diagonal_stage45
hilbert_IV15_diameter_inner_point_stage46
hilbert_IV15_chord_exterior_farther_left_stage47
hilbert_IV15_chord_exterior_farther_right_stage47
hilbert_IV15_chord_opposite_diameter_stage48
hilbert_forder_IV15_sameSide_CK_stage50
hilbert_forder_IV15_complete_stage52
```

These names remain private implementation details of `Proposition4_15.lean`.

The generic infrastructure used by those helpers includes:

```
hilbert_concyclic4_circle
hilbert_circle_antipode
hilbert_circle_center_midpoint_of_chord
hilbert_sameRay_beyond_common_middle
hilbert_oppositeSide_tail_sameSide
hilbert_two_opposites_sameSide_with_carrier
bookZero_43_supplements
```

The numbered dependencies are explicit:

```
hilbert_IV12_same_segment
hilbert_IV13_diameter_right_angle
hilbert_IV14_diameter_supplement
```

This is the intended production architecture: proposition-specific order lemmas remain local, while general side-of-line and circle facts are owned by the interface.

15.17 Relation to IV.12.1, IV.13, and IV.14

IV.15 is where the preceding circle results first assemble into a single theorem.

15.17.1 Use of IV.13

If the centre lies on diagonal BD , IV.13 closes the theorem immediately: both opposite angles stand on the diameter BD .

15.17.2 Use of IV.14

In the generic branch the antipodal diameter AE is introduced precisely so that IV.14 can be applied to A, B, E, D .

15.17.3 Use of IV.12.1

Once E is shown to lie in the same segment of chord BD as C , IV.12.1 replaces the auxiliary angle

$$\angle BED$$

by the original target angle

$$\angle BCD.$$

Thus the classical proof architecture is exactly:

$$\text{diameter case (IV.13)} \quad \text{or} \quad (\text{IV.14} + \text{IV.12.1}).$$

15.18 Hilbert and Book Zero Components

15.18.1 Plane separation

The entire generic proof is organized by same-side and opposite-side relations with respect to the diagonal BD and the antipodal diameter AE .

The formal proof must transport these relations through betweenness and through the antipodal extension $A - K - E$.

15.18.2 Betweenness and trichotomy

The key order configurations are

$$A - K - E, \quad A - Y - E, \quad B - Y - D.$$

The last one is not assumed; it is derived by trichotomy and metric contradiction.

15.18.3 Segment comparison

The proof that exterior orders on B, Y, D are impossible uses strict segment comparison: an exterior point on the chord carrier would be farther from the centre than a circle radius, contradicting the fact that Y is inside the antipodal diameter.

15.18.4 Book Zero supplements

The final conclusion is not an arithmetic equality

$$\alpha + \beta = \pi.$$

Instead two explicit supplement configurations are compared by

```
bookZero_43_supplements
```

This is the synthetic mechanism behind the substitution of congruent inscribed angles into supplementary relations.

15.19 Foundational Boundary

IV.15 introduces no new axiom.

The proof uses:

- circle existence already packaged in `HilbertConcyclic4`;
- incidence and line uniqueness;
- plane separation;
- order trichotomy and betweenness transitivity;
- segment congruence and strict comparison;
- the established numbered theorems IV.12.1, IV.13, and IV.14;
- Book Zero supplement transport.

No numerical angle measure, coordinate geometry, or continuity argument is introduced.

15.20 Classical and Formal Dependency Structures

The source proof is compact:

$$\begin{array}{c}
 ABCD \text{ cyclic, } A, C \text{ opposite sides of } BD \\
 \downarrow \\
 \text{centre } K \\
 \downarrow \\
 \left\{ \begin{array}{l} K \in BD \implies \text{IV.13,} \\ K \notin BD \implies \text{choose antipode } E \end{array} \right. \\
 \downarrow \\
 \text{IV.14 on } A, B, E, D \\
 \downarrow \\
 \angle BAD + \angle BED = \pi \\
 \downarrow \\
 \text{IV.12.1: } \angle BED \cong \angle BCD \\
 \downarrow \\
 \angle BAD + \angle BCD = \pi.
 \end{array}$$

The formal generic branch expands the middle:

$$\begin{array}{c}
 C \sim K \text{ across } BD \\
 \downarrow \\
 A - K - E \\
 \downarrow \\
 A \mid E \text{ across } BD \\
 \downarrow \\
 Y = AE \cap BD \\
 \downarrow \\
 Y = K \text{ or } KY < KA \\
 \downarrow \\
 \text{exclude } Y - B - D \text{ and } B - D - Y \\
 \downarrow \\
 B - Y - D \\
 \downarrow \\
 B \mid D \text{ across diameter } AE \\
 \downarrow \\
 \text{IV.14} \\
 \downarrow \\
 E \sim C \text{ across } BD \\
 \downarrow \\
 \text{IV.12.1} \\
 \downarrow \\
 \text{Book Zero supplement transport.}
 \end{array}$$

The module dependency graph is

```

HilbertInterfaceIV
|
+--> Proposition4_12
|

```

```

+--> Proposition4_13
|
+--> Proposition4_14
    |
    v
    Proposition4_15

```

More precisely, `Proposition4_15.lean` imports all three numbered predecessors because each has a distinct mathematical role.

15.21 Diagrammatic Audit

Four figures are useful for IV.15.

1. The source cyclic quadrilateral and diagonal BD .
2. The special case $K \in BD$, where the diagonal is a diameter.
3. The generic antipodal construction $A - K - E$ with the crossing point Y .
4. The relabelled symmetry branch that Lean handles explicitly rather than by a verbal “without loss of generality”.

The difficult order proof excluding exterior positions of Y is better explained in text and inequalities than by adding yet another crowded figure.

15.22 Conclusion

IV.15 is the first proposition in the Forder circle block where the formal proof reveals the full cost of a classical symmetry argument.

The centre-on-diagonal case is short and uses IV.13. The generic case introduces an antipodal diameter, proves that the original diagonal crosses that diameter internally, applies IV.14, and then uses IV.12.1 to replace the auxiliary inscribed angle by the target angle. The final substitution is performed through explicit Book Zero supplement witnesses.

Most importantly, the alternative placement of the centre is not dismissed as “symmetric”. The Lean proof performs the cyclic relabelling and transports the supplement back to the original orientation.

The result is a production theorem whose public statement is simple but whose documentation exposes the actual incidence, order, congruence, and symmetry machinery hidden in the printed diagram.

Chapter 16

Proposition IV.16

16.1 Introduction

Forder IV.16 collects the two fundamental “same chord” configurations for four points on one circle.

Let A, B, C, D lie on a circle. The chord AB separates the plane into two half-planes. There are two geometrically different possibilities.

If C, D lie on opposite sides of AB , then the two angles

$$\angle ACB, \quad \angle ADB$$

are supplementary.

If C, D lie on the same side of AB , then

$$\angle ACB \cong \angle ADB.$$

Forder states both conclusions in one proposition [1]. The production library deliberately exposes them as two separate proposition interfaces and two separate public theorems:

```
hilbert_IV16_opposite_side  
hilbert_IV16_same_side
```

This separation is mathematically useful. Later proofs—especially IV.17, IV.19, and IV.20—need to invoke one side configuration at a time. The production API therefore preserves the actual branching structure rather than hiding it inside a disjunction.

The implementation lives in

```
CGJteamLab/Proposition4_16.lean
```

16.2 Forder’s Statement

Forder IV.16 says:

If A, B, C, D lie on a circle and C, D are on opposite sides of AB , then the angles ACB, ADB are supplementary; if C, D are on the same side of AB , then these angles are congruent.

His proof treats the two halves differently.

The opposite-side half follows immediately from IV.15 by viewing C, A, D, B as a cyclic quadrilateral. The opposite angles of that quadrilateral are precisely

$$\angle ACB \quad \text{and} \quad \angle ADB.$$

For the same-side half, Forder distinguishes the position of the centre relative to chord AB .

If the centre lies on AB , then AB is a diameter and both inscribed angles are right by IV.13.

If C, D and the centre lie in the same half-plane, then the result is exactly IV.12.1.

If C, D lie on one side of AB and the centre lies on the other, an antipodal point X is introduced. Both

$$\angle ACB \quad \text{and} \quad \angle ADB$$

become supplements of the same angle

$$\angle AXB.$$

Hence they are congruent.

That three-case source argument is preserved explicitly in the formal proof.

16.3 Formal Interfaces

The two proposition contracts are defined in `HilbertInterfaceIV.lean`.

16.3.1 Same-side interface

```
def HilbertForderIV16SameSide
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H]: Prop :=
  forall K R A B C E : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
        H.OnLine A chord ->
        H.OnLine B chord ->
        HilbertCircle Geo K R A ->
        HilbertCircle Geo K R B ->
        HilbertCircle Geo K R C ->
        HilbertCircle Geo K R E ->
        HilbertSameSide Geo C E chord ->
        Geo.AngleCongruent
          A C B
          A E B
```

The conclusion is direct angle congruence.

16.3.2 Opposite-side interface

```

def HilbertForderIV16OppositeSide
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C E : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
      H.OnLine A chord ->
      H.OnLine B chord ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R B ->
      HilbertCircle Geo K R C ->
      HilbertCircle Geo K R E ->
      HilbertOppositeSide Geo C E chord ->
      exists X : Geo.Point,
        BookZeroSupplement Geo
          A E B
          B X /\
        Geo.AngleCongruent
          A C B
          X E B

```

The supplementary relation is therefore constructive. The witness X extends AE through E , and the theorem says that the adjacent angle

$$\angle XEB$$

is congruent to

$$\angle ACB.$$

The production endpoints are:

```

theorem hilbert_IV16_opposite_side
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV16OppositeSide (Geo := Geo)

theorem hilbert_IV16_same_side
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV16SameSide (Geo := Geo)

```

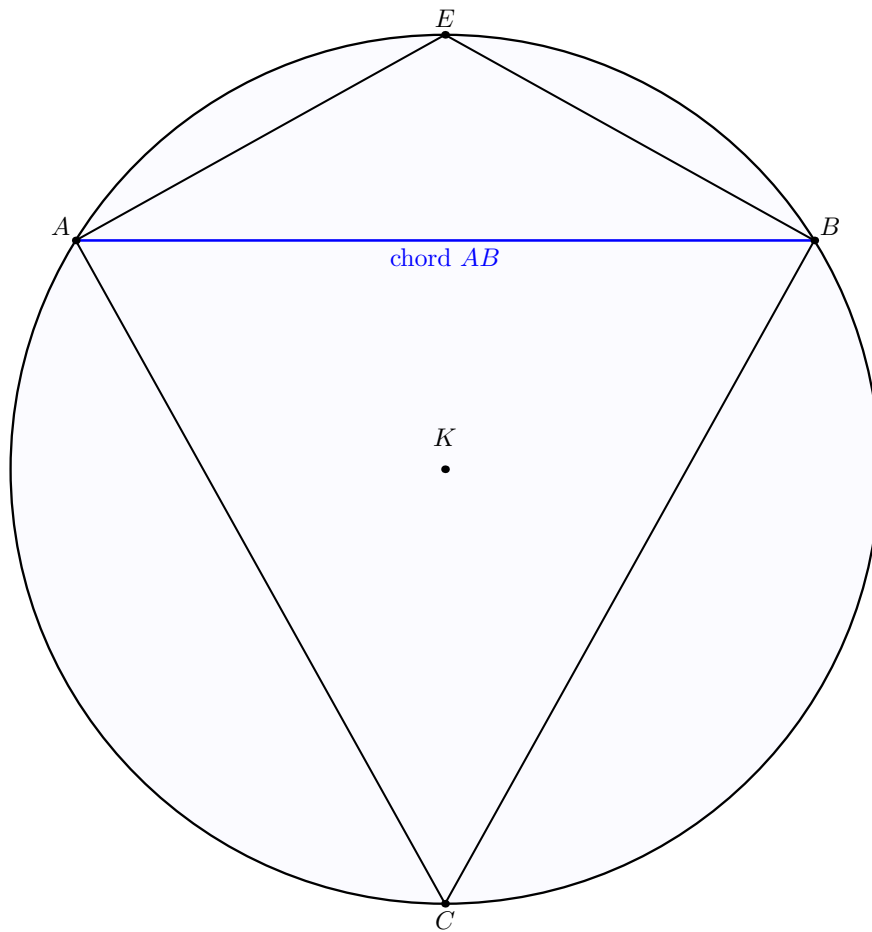
16.4 Opposite-Side Half

The opposite-side half is not an independent new circle theorem. It is an immediate reformulation of IV.15.

Suppose C, E lie on opposite sides of chord AB . Consider the cyclic quadrilateral

$$C - A - E - B.$$

apply IV.15 to $C - A - E - B$



C, E opposite sides of AB

Figure 16.1: Opposite-side half of IV.16. The four circle points are viewed in the cyclic order $C - A - E - B$. IV.15 applied to this quadrilateral says that the opposite angles $\angle ACB$ and $\angle AEB$ are supplementary.

Its diagonal is still the original chord AB . The two opposite vertices are C, E , and by assumption they lie on opposite sides of that diagonal.

The two opposite angles in IV.15 become:

$$\angle ACB$$

at C , and

$$\angle AEB$$

at E .

IV.15 therefore constructs X with

$$A - E - X$$

and

$$\angle ACB \cong \angle XEB.$$

This is exactly the definition of `HilbertForderIV16OppositeSide`.

The reduction is encapsulated by the private production helper

```
hilbert_forder_IV16_opposite_of_IV15_stage13
```

and the closed public theorem simply supplies IV.15:

```
theorem hilbert_IV16_opposite_side
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV16OppositeSide (Geo := Geo)
```

This branch is therefore conceptually one line:

$$\boxed{\text{IV.16 opposite-side} = \text{IV.15 on } C, A, E, B.}$$

16.5 Same-Side Half: Why Three Cases Are Needed

Now assume that C, D lie on the same side of chord AB .

The conclusion is

$$\angle ACB \cong \angle ADB.$$

The source proof divides according to the position of the centre K relative to the chord AB . The formal proof follows exactly that classification.

The three cases are:

1. K lies on AB ;
2. K is off AB and lies on the same side as C, D ;
3. K is off AB and lies on the opposite side from C, D .

The first is the diameter case, the second is IV.12.1 directly, and the third is the only genuinely new reduction.

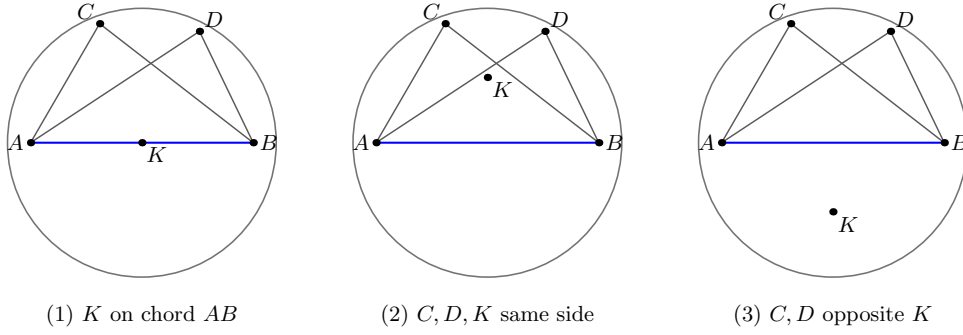


Figure 16.2: The three geometric cases in the same-side half of IV.16: (1) the centre K lies on AB ; (2) C, D, K occupy one half-plane; (3) C, D occupy one half-plane while K lies in the other.

16.6 Case I: the Centre Lies on the Chord

Assume

$$K \in AB.$$

Then AB is a diameter.

IV.13 applied at C gives

$$\angle ACB \text{ right.}$$

IV.13 applied at D gives

$$\angle ADB \text{ right.}$$

The theorem

```
hilbert_all_right_angles_congruent
```

then yields

$$\angle ACB \cong \angle ADB.$$

No same-segment theorem is required in this branch.

This case is a useful reminder that the same-side theorem must include the possibility that the centre lies on the chord carrier. The relation

$$\text{“}C \text{ and } K \text{ are on the same side of } AB\text{”}$$

is not even defined when K lies on the line, so the diameter case cannot be subsumed under IV.12.1.

16.7 Case II: the Centre Lies in the Same Half-Plane

Assume now that K is off chord AB and

$$C \sim K$$

with respect to the chord carrier.

Since C, D are on the same side of AB , symmetry and transitivity of same-side relations give

$$D \sim K.$$

All three points C, D, K therefore lie in the same half-plane determined by AB .

This is precisely the hypothesis of IV.12.1. Hence

$$\boxed{\angle ACB \cong \angle ADB}.$$

The formal proof in this branch is correspondingly short:

```

have hSameDC :
  HilbertSameSide Geo D C chord := ...

have hSameDK :
  HilbertSameSide Geo D K chord := ...

exact
  hilbert_IV12_same_segment
    Geo K R A B C D chord ...

```

Thus the ordinary “same segment” theorem is recovered as the middle branch of the more general IV.16 same-side statement.

16.8 Case III: the Centre Lies on the Opposite Side

The remaining case is the important one.

Assume:

$$C \sim D$$

relative to chord AB , but

$$C \not\sim K.$$

Since both C and K are off the chord, they are on opposite sides:

$$C \mid K.$$

Because D is on the same side as C , it follows that

$$D \mid K.$$

Now construct the antipode X of C :

$$C - K - X.$$

The next task is to prove

$$C \mid X$$

relative to AB .

The opposite-side witness for C, K supplies a point Y with

$$C - Y - K$$

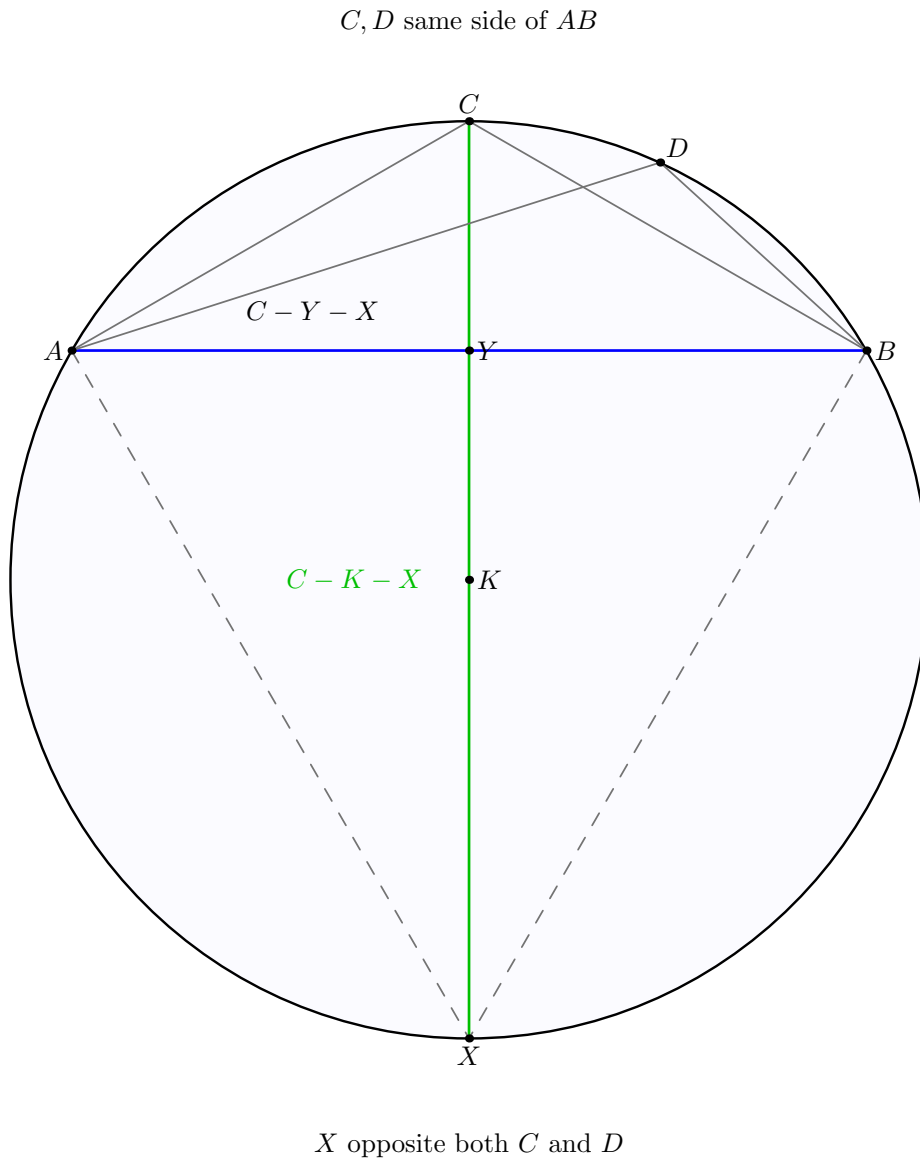


Figure 16.3: Third same-side case. The centre K lies on the opposite side of chord AB from C, D . The antipode X of C satisfies $C - K - X$ and lies on the K -side. Therefore both pairs C, X and D, X are opposite-side configurations for chord AB .

on chord AB . Since

$$C - K - X,$$

inner betweenness transitivity gives

$$C - Y - X.$$

Thus the same point Y witnesses that C and X are on opposite sides of the chord.

Because $D \sim C$, transport of opposite-side relations gives

$$D \mid X.$$

At this point the already proved opposite-side half of IV.16 can be applied twice.

16.9 Two Supplements of the Same Angle

Apply the opposite-side theorem to C, X . It produces a supplement of

$$\angle AXB$$

congruent to

$$\angle ACB.$$

Concretely there exists U such that

$$\text{Supplement}(\angle AXB, \angle BXU)$$

and

$$\angle ACB \cong \angle UXB.$$

Apply the opposite-side theorem again to D, X . It produces V such that

$$\text{Supplement}(\angle AXB, \angle BVX)$$

and

$$\angle ADB \cong \angle VXB.$$

Therefore $\angle UXB$ and $\angle VXB$ are two supplements of the same proper angle $\angle AXB$.

Book Zero theorem 43 on supplements gives

$$\angle UXB \cong \angle VXB.$$

Transitivity now yields

$$\angle ACB \cong \angle UXB \cong \angle VXB \cong \angle ADB.$$

Hence

$$\boxed{\angle ACB \cong \angle ADB}.$$

This is Forder's source idea expressed without numerical angle arithmetic: two angles supplementary to the same angle are congruent.

16.10 Why the Antipode Is the Correct Auxiliary Point

The point X is not chosen merely because it lies conveniently on the circle.

The antipodal order

$$C - K - X$$

has two essential consequences.

First, it places X on the opposite side of chord AB from C when the centre is already opposite C .

Second, because D lies on the same side as C , the same X is also opposite D .

Thus a single auxiliary circle point creates two opposite-side IV.16 configurations:

$$(C, X) \quad \text{and} \quad (D, X).$$

The resulting supplements share the same reference angle

$$\angle AXB.$$

That common reference is exactly what allows Book Zero supplement uniqueness to close the proof.

16.11 Lean Formalization

The production file contains a small public surface:

```
hilbert_IV16_opposite_side
hilbert_IV16_same_side
```

The proposition-specific reduction layer is represented by private helpers whose historical stage suffixes are not part of the public API:

```
hilbert_forder_IV16_opposite_of_IV15_stage13
hilbert_forder_IV16_same_of_IV12_1_IV13_IV15_stage15
hilbert_forder_IV16_opposite_complete_stage53
hilbert_forder_IV16_same_complete_stage54
```

Conceptually the closed wrappers amount to

```
theorem hilbert_IV16_opposite_side ... :=
  hilbert_forder_IV16_opposite_of_IV15_stage13
  Geo
  (hilbert_IV15_cyclic_supplement Geo)

theorem hilbert_IV16_same_side ... :=
  hilbert_forder_IV16_same_of_IV12_1_IV13_IV15_stage15
  Geo
  (hilbert_IV12_same_segment Geo)
  (hilbert_IV13_diameter_right_angle Geo)
```

```
(hilbert_IV15_cyclic_supplement Geo)
```

The exact internal production implementation may inline or wrap these private declarations, but the mathematical dependency is the one displayed above.

16.12 Production Dependency Structure

The two halves have different direct logical dependencies.

The opposite-side half is:

$$\text{IV.15} \implies \text{IV.16 opposite-side.}$$

The same-side half is:

$$\text{IV.12.1} + \text{IV.13} + \text{IV.15} \implies \text{IV.16 same-side.}$$

At module level the production file imports:

```
HilbertInterfaceIV
Proposition4_12
Proposition4_13
Proposition4_15
  -> Proposition4_16
```

There is no direct IV.14 import because IV.15 already encapsulates the supplement theorem needed downstream.

16.13 Relation to the Previous Propositions

16.13.1 IV.12.1

IV.12.1 solves exactly the same-side case in which the centre lies in the same segment as C, D .

16.13.2 IV.13

IV.13 solves the exceptional case in which chord AB passes through the centre and is therefore a diameter.

16.13.3 IV.15

IV.15 has two roles.

First, it immediately gives the opposite-side half of IV.16.

Second, through that opposite-side half, it is used twice in the antipodal case of the same-side proof.

Thus IV.16 is not merely “a restatement of IV.15”. One half is a direct corollary, while the other combines the earlier same-segment theorem, diameter theorem, and supplement theorem according to the position of the centre.

16.14 Hilbert and Book Zero Components

16.14.1 Plane separation

Same-side and opposite-side relations are the organizing language of the entire proposition. The case split is fundamentally a classification of the centre relative to the chord.

The formal proof uses:

- symmetry and transitivity of same-side;
- conversion of “not same side” into opposite-side when both points are off the carrier;
- transport of opposite-side relations across a same-side pair.

16.14.2 Betweenness

The antipodal relation

$$C - K - X$$

is combined with the crossing witness

$$C - Y - K$$

to obtain

$$C - Y - X.$$

This is the exact order step that turns the antipodal construction into an opposite-side statement.

16.14.3 Congruence of right angles

In the diameter branch both target angles are right by IV.13. The proof uses the general theorem that all right angles are congruent.

16.14.4 Book Zero supplements

The difficult same-side branch closes because two constructed angles are supplements of the same proper angle AXB . Book Zero theorem 43 turns that shared supplement relation into angle congruence.

No numerical representation of a supplement is used.

16.15 Foundational Boundary

IV.16 introduces no new axiom and no new circle-existence principle.

Its content is assembled from:

- the existing circle predicate;
- plane separation and betweenness;
- the antipode construction;
- IV.12.1;
- IV.13;
- IV.15;
- Book Zero supplement transport.

The theorem therefore shows especially clearly how a later classical result can be built by recombining earlier synthetic layers without adding new foundational commitments.

16.16 Classical and Formal Dependency Diagrams

The opposite-side half is minimal:

$$\begin{array}{c}
 C \mid D \text{ across chord } AB \\
 \downarrow \\
 C - A - D - B \text{ cyclic quadrilateral} \\
 \downarrow \\
 \text{IV.15} \\
 \downarrow \\
 \angle ACB \text{ supplementary to } \angle ADB.
 \end{array}$$

The same-side half is:

$$\begin{array}{c}
 C \sim D \text{ across } AB \\
 \downarrow \\
 \text{position of } K \\
 \downarrow \\
 \left\{ \begin{array}{ll}
 K \in AB & \xRightarrow{\text{IV.13}} \text{both angles right,} \\
 C \sim K & \xRightarrow{\text{IV.12.1}} \angle ACB \cong \angle ADB, \\
 C \mid K & \implies C - K - X \implies \begin{cases} C \mid X, \\ D \mid X \end{cases} \\
 & \implies \text{two IV.16 opposite-side supplements of } \angle AXB \\
 & \implies \angle ACB \cong \angle ADB.
 \end{array} \right.
 \end{array}$$

The module dependency graph is:

$$\begin{array}{ccccc}
 \text{IV.12} & \text{IV.13} & \text{IV.15} & & \\
 & \searrow & \downarrow & \swarrow & \\
 & & \text{IV.16} & &
 \end{array}$$

16.17 Diagrammatic Audit

Three figures are sufficient.

1. The opposite-side figure shows the direct IV.15 relabelling.
2. The three-case figure records the complete classification of the centre for the same-side half.
3. The antipode figure explains the only nontrivial same-side branch and makes visible why the same point X creates two opposite-side configurations.

No separate figure is needed for the diameter case; it is already a standard IV.13 configuration.

16.18 Conclusion

Order IV.16 contains two theorems with quite different proof structures.

The opposite-side half is a direct corollary of IV.15 after relabelling the cyclic quadrilateral.

The same-side half requires a complete classification of the centre. If the centre lies on the chord, IV.13 gives two right angles. If it lies in the same half-plane, IV.12.1 applies directly. If it lies in the opposite half-plane, an antipodal point converts the problem into two applications of the opposite-side theorem, and Book Zero supplement uniqueness identifies the two target angles.

The production API reflects this structure by exporting separate same-side and opposite-side theorems. That choice is not merely organizational: it provides the exact interfaces required by the tangent and converse theorems that follow.

Chapter 17

Proposition IV.17

17.1 Introduction

Former IV.17 is the converse tangent–chord theorem.

Let X, A, Y be points of one circle, let AT be a candidate line through the circle point A , and assume that T and Y lie on opposite sides of the chord AX . If

$$\angle XAT \cong \angle XYA,$$

then the line AT is tangent to the circle at A [1].

The theorem is a converse. The angle relation itself does not assume tangency; it is used to exclude every possible secant configuration.

The formal proof is therefore naturally organized by the circle-line dichotomy:

second circle intersection or tangent.

If the tangent alternative occurs, the theorem is finished. If there is a second circle point S on the line AT , the proof must show that every possible order of S relative to the ray AT is impossible.

That produces the two characteristic branches of IV.17:

1. S lies on the same ray from A as T ;
2. S lies on the opposite ray, equivalently $S - A - T$.

The first branch is contradicted through the *opposite-side* half of IV.16 and a supplement argument. The second is contradicted through the *same-side* half of IV.16 and the exterior-angle theorem.

The production implementation is

```
CGJteamLab/Proposition4_17.lean
```

with public endpoint

```
hilbert_IV17_tangent_converse
```

17.2 Classical Configuration

The basic configuration consists of chord AX , the circle point Y , and a candidate line AT through A . The side condition is

$$T \mid Y$$

relative to the chord AX .

$$\angle XAT \cong \angle XYA$$

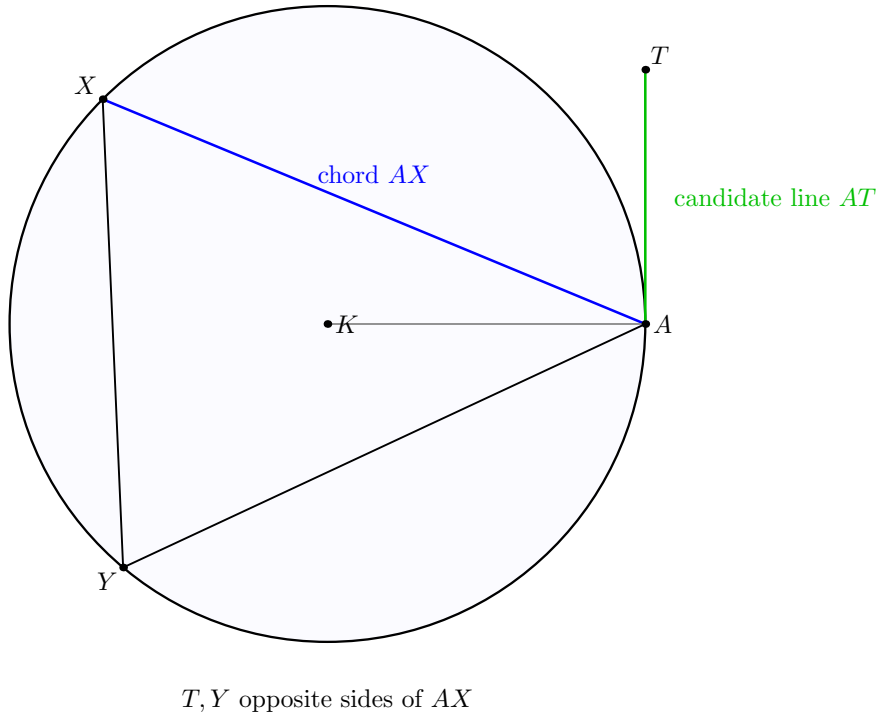


Figure 17.1: Forder IV.17. The line AT is a candidate tangent at A . The point T lies on the opposite side of chord AX from the circle point Y , and the hypothesis is $\angle XAT \cong \angle XYA$.

The theorem says that this angle condition forces AT to be the tangent carrier.

The proof does not construct the tangent independently and then compare lines. Instead, it proves that AT cannot be a secant.

17.3 Forder's Statement and Proof

Forder states, in substance:

If $\angle XAT = \angle XYA$ and Y, T are on opposite sides of AX , then the circle through X, A, Y touches AT at A .

His proof is by contradiction. If AT is not tangent, it meets the circle again at a point $S \neq A$.

Then IV.16 determines the relation between

$$\angle XSA$$

and

$$\angle XYA$$

according to whether S and Y lie on the same or opposite sides of the chord AX .

The order of S relative to T changes which side relation holds.

In one case $\angle XSA$ becomes supplementary to $\angle XYA$, while the hypothesis and the collinearity of A, S, T identify $\angle XYA$ with the angle at A . This forces an interior triangle angle to be congruent to its own supplement, contradicting the triangle angle inequality.

In the other case $\angle XSA \cong \angle XYA$, and the hypothesis gives

$$\angle XSA \cong \angle XAT.$$

But $\angle XAT$ is an exterior angle of triangle XSA , again contradicting the exterior-angle theorem.

Thus the secant alternative is impossible, so AT is tangent.

17.4 Formal Interface

The production predicate is:

```
def HilbertForderIV17TangentConverse
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H] : Prop :=
  forall K R X A Y T : Geo.Point,
    forall chord tangent : Geo.Line,
      Ne X A ->
        H.OnLine X chord ->
        H.OnLine A chord ->
        H.OnLine A tangent ->
        H.OnLine T tangent ->
        Ne T A ->
        HilbertCircle Geo K R X ->
        HilbertCircle Geo K R A ->
        HilbertCircle Geo K R Y ->
        HilbertOppositeSide Geo T Y chord ->
        Geo.AngleCongruent X A T X Y A ->
        HilbertCircleTangentAt
          (Geo := Geo) K A tangent
```

Several design choices are visible in the interface.

First, the chord AX and the candidate tangent carrier are explicit `Geo.Line` objects.

Second, the side condition is not encoded by an orientation convention but by

```
HilbertOppositeSide Geo T Y chord
```

Third, the conclusion uses the project's tangent predicate

```
HilbertCircleTangentAt (Geo := Geo) K A tangent
```

rather than a numerical slope or perpendicular equation.

The public theorem is:

```
theorem hilbert_IV17_tangent_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV17TangentConverse (Geo := Geo)
```

17.5 The Tangency Predicate

The relevant interface definition is

```
def HilbertCircleTangentAt
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H]
  (K A : Geo.Point)
  (line : Geo.Line) : Prop :=
  exists T : Geo.Point,
    H.OnLine T line /\
    Ne T A /\
    Not (PrimCollinear Geo T A K) /\
    HilbertRightAngle Geo T A K
```

Thus a tangent at A is represented synthetically by a carrier through A containing a point whose ray makes a right angle with the radius AK .

IV.17 does not prove tangency by directly constructing such a perpendicular. Instead, the general circle-line dichotomy eventually returns this tangent predicate after all secant alternatives have been eliminated.

17.6 The Circle-Line Dichotomy

The reusable theorem

```
hilbert_circle_line_second_or_tangent
```

says that a line through a noncentral circle point A is either:

1. a secant, containing a second circle point $S \neq A$, or
2. tangent at A .

The top-level IV.17 proof begins by showing

$$K \neq A.$$

This follows because A and X are distinct circle points with equal radii from K ; if $K = A$, the segment AX would be congruent to the null segment AA , forcing $A = X$.

The dichotomy can then be applied to the candidate carrier AT .

If it returns tangency, that is exactly the theorem's conclusion.

The rest of IV.17 is therefore a proof that the secant alternative is impossible.

17.7 Normalizing the Secant Order

Suppose the candidate line contains a second circle point $S \neq A$.

There are three superficial possibilities along the carrier: $S = T$, S, T distinct on the same ray from A , or S on the opposite ray.

The equality case $S = T$ is absorbed into the same-ray branch through reflexivity of the same-ray relation.

For distinct S, T , the reusable production theorem

```
hilbert_secant_ray_order_split
```

normalizes the order into exactly two alternatives:

$$\text{SameRay}(A, S, T)$$

or

$$T - A - S.$$

The second relation is immediately reversed to

$$S - A - T$$

for the exterior-angle argument.

This theorem is architecturally important. During development it carried an IV.19 stage name because it was first isolated there. In the production split it was promoted to the semantic, proposition-independent name `hilbert_secant_ray_order_split` in `HilbertInterfaceIV`.

Without that promotion, IV.17 would appear to depend on the later IV.19, creating an artificial cycle. The production architecture correctly records the theorem as general line/ray order infrastructure.

17.8 Same-Ray Secant Branch

Assume

$$\text{SameRay}(A, S, T).$$

Because S, T lie on the same ray from the chord point A , they lie on the same side of chord AX .

The theorem hypothesis says that T, Y lie on opposite sides of the chord. Transporting the side relation therefore gives

$$S \mid Y.$$

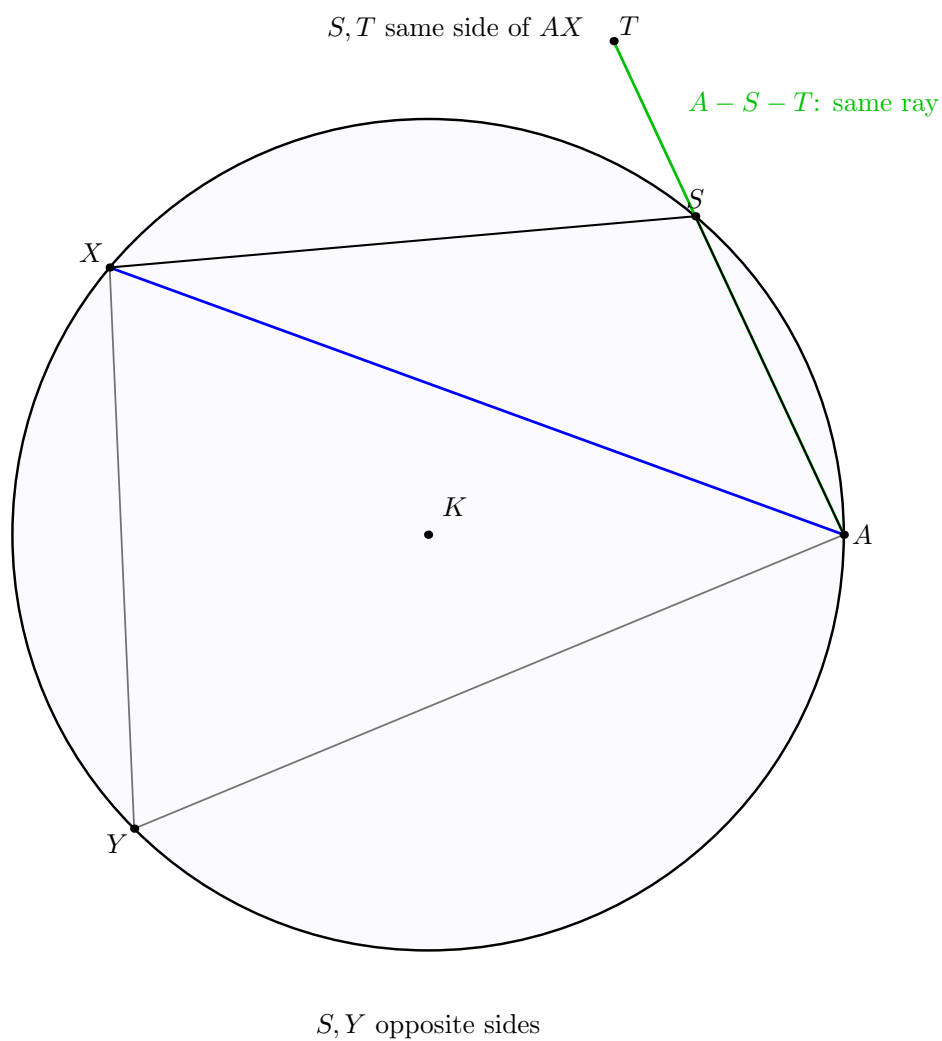


Figure 17.2: The impossible same-ray secant branch. The second circle intersection S and the point T lie on the same ray from A . Consequently S, T are on the same side of chord AX , while T, Y are opposite; hence S, Y are opposite.

Now apply the opposite-side half of IV.16 to the circle points

$$X, A, S, Y.$$

It constructs a point U such that

$$\angle XSA \cong \angle UYA$$

and $\angle UYA$ is a supplement of

$$\angle XYA.$$

Meanwhile, because AS and AT are the same ray,

$$\angle XAS = \angle XAT.$$

The hypothesis of IV.17 gives

$$\angle XAT \cong \angle XYA,$$

hence

$$\angle XAS \cong \angle XYA.$$

After reversing the first angle:

$$\angle SAX \cong \angle XYA.$$

17.9 Transporting the Supplement in the Same-Ray Branch

Extend SA through A to a point V :

$$S - A - V.$$

Then $\angle XAV$ is a supplement of $\angle SAX$.

From IV.16, $\angle UYA$ is a supplement of $\angle XYA$.

Since

$$\angle SAX \cong \angle XYA,$$

Book Zero supplement transport gives

$$\angle XAV \cong \angle UYA.$$

But IV.16 also gave

$$\angle XSA \cong \angle UYA.$$

Therefore

$$\angle XSA \cong \angle XAV.$$

Now $S - A - V$, so $\angle XAV$ is the exterior angle at A of the proper triangle XSA .

The formal contradiction is supplied by the reusable theorem now owned by the interface:

```
hilbert_triangle_angle_not_supplement
```

It is the synthetic Euclid I.17 fact that an interior angle of a triangle cannot be congruent to the corresponding exterior supplement.

Hence the same-ray secant branch is impossible.

The proposition-specific private helper is

hilbert_IV17_sameRay_secant_impossible_stage57

17.10 Opposite-Ray Secant Branch

Assume now

$$S - A - T.$$

S, T opposite sides of AX

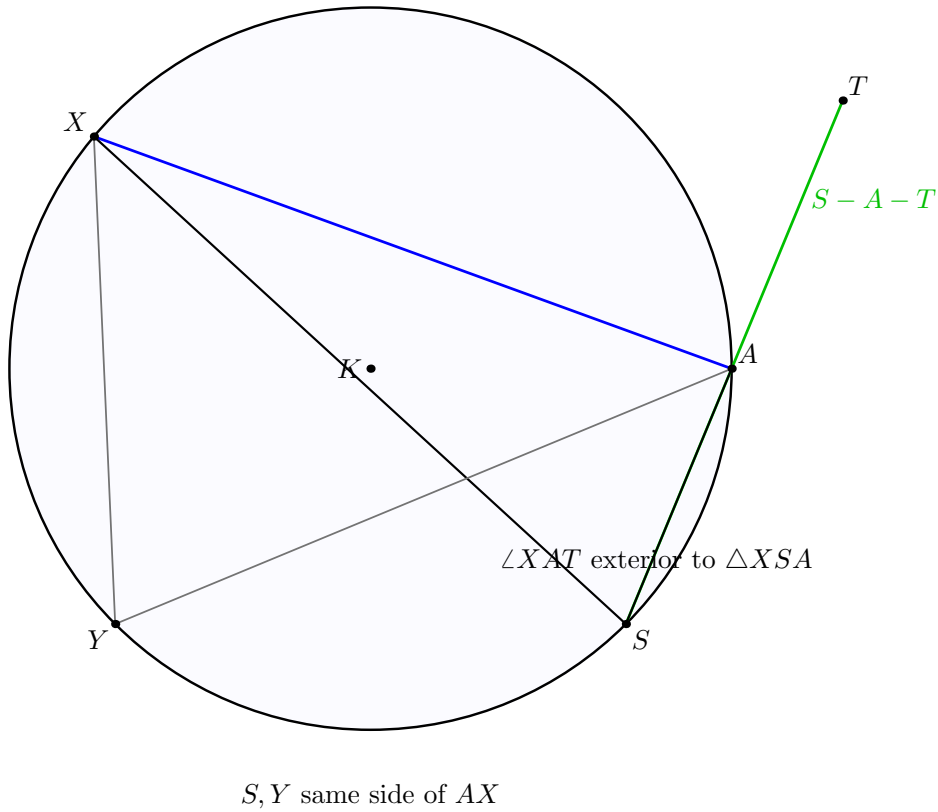


Figure 17.3: The impossible opposite-ray secant branch. Here $S - A - T$. Thus S, T are on opposite sides of the chord AX . Since T, Y are also opposite, S, Y lie on the same side of the chord.

Because A lies on the chord AX , the betweenness

$$S - A - T$$

shows directly that S, T lie on opposite sides of the chord.

The theorem hypothesis already gives

$$T \mid Y.$$

Thus T is opposite both S and Y , and the plane-separation theorem “two opposites are same side” gives

$$S \sim Y.$$

Apply the same-side half of IV.16 to X, A, S, Y :

$$\angle XSA \cong \angle XYA.$$

Using the hypothesis

$$\angle XAT \cong \angle XYA$$

and symmetry/transitivity gives

$$\angle XSA \cong \angle XAT.$$

But when

$$S - A - T,$$

the angle $\angle XAT$ is an exterior angle of triangle XSA .

The general exterior-angle theorem says that it cannot be congruent to the remote interior angle $\angle XSA$.

This contradiction eliminates the opposite-ray secant branch.

The proposition-specific helper is

```
hilbert_IV17_oppositeRay_secant_impossible_stage58
```

17.11 The Two Contradictions Compared

The two branches use different halves of IV.16 and different final contradictions.

Secant order	IV.16 input	Final contradiction
S, T same ray from A	S, Y opposite sides: IV.16 opposite-side	interior angle becomes congruent to its supplementary exterior angle
$S - A - T$	S, Y same side: IV.16 same-side	remote interior angle becomes congruent to the exterior angle

This is not an implementation accident. It is exactly the geometry of Forder's proof.

17.12 Completion of the Tangent Converse

The top-level proof is now short.

Apply `hilbert_circle_line_second_or_tangent` to the line AT .

If the tangent alternative is returned, finish.

If a second circle point S is returned:

- if $S = T$, treat it as the same-ray case;
- otherwise apply `hilbert_secant_ray_order_split`;

- the same-ray result contradicts the first branch theorem;
- the opposite-ray result yields $S - A - T$ and contradicts the second.

Therefore the secant branch is impossible, and

AT is tangent at A .

The private complete theorem is

```
hilbert_forder_IV17_tangent_converse_stage59
```

and the production wrapper exports it as

```
hilbert_IV17_tangent_converse
```

17.13 Lean Formalization

The public surface is:

```
HilbertForderIV17TangentConverse
hilbert_IV17_tangent_converse
```

The proposition-specific private proof layer contains:

```
hilbert_IV17_sameRay_secant_impossible_stage57
hilbert_IV17_oppositeRay_secant_impossible_stage58
hilbert_forder_IV17_tangent_converse_stage59
```

The former stage-56 triangle lemma is no longer conceptually owned by IV.17. Its production form is reusable interface infrastructure:

```
hilbert_triangle_angle_not_supplement
```

Likewise the secant-order split used by the top-level proof is the interface theorem

```
hilbert_secant_ray_order_split
```

The general circle-line dichotomy is

```
hilbert_circle_line_second_or_tangent
```

Thus the production module contains only the contradictions that are specific to IV.17 and the final assembly.

17.14 Relation to IV.16

IV.17 depends directly on both halves of IV.16.

The same-ray secant branch requires:

$$S \mid Y,$$

so it invokes

IV.16 opposite-side.

The opposite-ray secant branch requires:

$$S \sim Y,$$

so it invokes

IV.16 same-side.

This makes IV.17 the first theorem in the tangent block where the split public API of IV.16 pays off directly.

Had IV.16 been packaged only as a single theorem containing a disjunction, both contradiction lemmas would have had to reopen the chord-side classification. The present API avoids that duplication.

17.15 Hilbert and Book Zero Components

17.15.1 Circle-line incidence

The secant/tangent dichotomy is the global structural theorem behind IV.17. It says that no third geometric possibility exists for a line through the circle point A .

17.15.2 Ray order

The distinction

$$\text{SameRay}(A, S, T) \quad \text{versus} \quad S - A - T$$

is the decisive order classification.

The theorem `hilbert_secant_ray_order_split` is therefore not merely a technical convenience; it is the formal version of the source proof's implicit line-order case split.

17.15.3 Plane separation

Same-ray points on a carrier through the chord point lie on the same side of the chord. Opposite-ray points lie on opposite sides.

These relations are then combined with the hypothesis $T \mid Y$ to determine whether S, Y are same-side or opposite-side.

17.15.4 Book Zero supplements

The same-ray branch depends on transport of supplements of congruent proper angles. This replaces the source-level statement that two supplementary angles have equal supplements.

17.15.5 Euclid I.17 and exterior-angle order

The final contradiction in both branches is an angle-order statement for a proper triangle: an exterior angle is strictly larger than the relevant interior angle.

No angle measure is introduced; the contradiction is purely synthetic.

17.16 Foundational Boundary

IV.17 introduces no new tangent axiom.

Tangency is obtained from the previously proved circle-line dichotomy after the secant alternative has been eliminated.

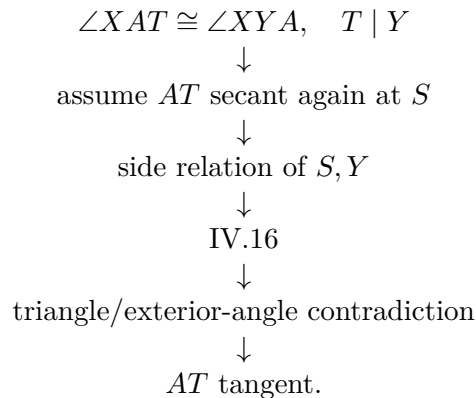
The proof uses:

- circle incidence;
- line and ray order;
- plane separation;
- IV.16 same-side and opposite-side;
- Book Zero supplement transport;
- Euclid I.17 / exterior-angle comparison.

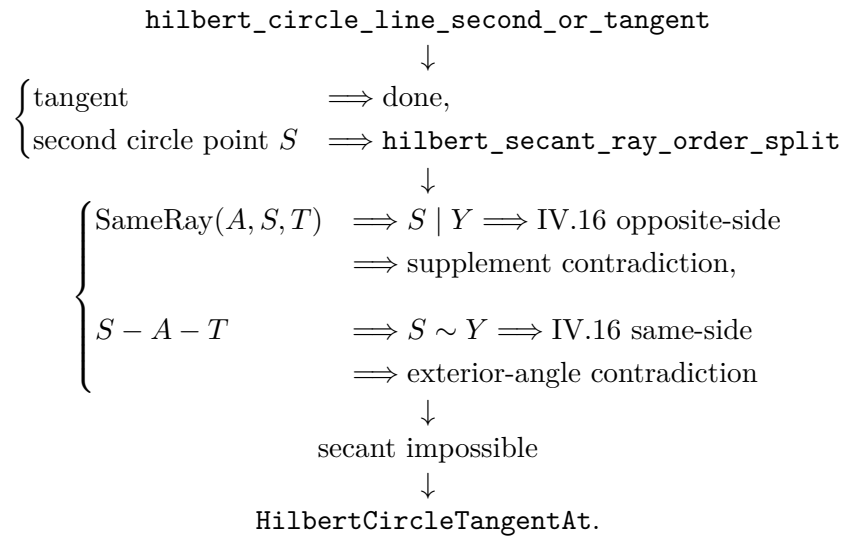
The theorem therefore reconstructs the converse tangent–chord criterion inside the existing Hilbert Euclidean plane theory.

17.17 Classical and Formal Dependency Structures

Forder’s proof can be summarized as



The formal proof makes the line order explicit:



At module level:

```
HilbertInterfaceIV
Proposition4_16
  -> Proposition4_17
```

There is no production dependency on IV.19. The old stage-labelled secant-order helper was promoted to the interface precisely to preserve this acyclic direction.

17.18 Diagrammatic Audit

Three figures are useful.

1. The source tangent-converse configuration.
2. The same-ray secant contradiction.
3. The opposite-ray secant contradiction.

A separate tangent-only figure is unnecessary: the source figure already shows the intended conclusion, while the logical content of the proof lies in excluding the two secant orders.

17.19 Conclusion

Forder IV.17 is a particularly clean example of proof by geometric exhaustion.

The circle-line dichotomy says that the candidate carrier AT is either tangent or secant. In the secant case the second intersection S is classified relative to the ray AT .

If S lies on the same ray, IV.16 opposite-side makes the angle at S a supplement of the prescribed inscribed angle, and Book Zero transport forces an impossible triangle-angle supplement.

If S lies on the opposite ray, IV.16 same-side identifies the angle at S directly with the prescribed inscribed angle, making it congruent to an exterior angle of triangle XSA .

Both secant orders are impossible. Tangency is therefore forced.

The formal proof also illustrates the value of the Book IV refactor: general secant-order and triangle-angle lemmas live in `HilbertInterfaceIV`, while `Proposition4_17.lean` owns only the two proposition-specific contradictions and the final tangent-converse assembly.

Chapter 18

Proposition IV.18

18.1 Introduction

Forder IV.18 is the direct tangent–chord theorem.

Let X, A, Y lie on a circle, let AT be tangent to the circle at A , and assume that T and Y lie on opposite sides of chord AX . Then

$$\angle XAT \cong \angle XYA.$$

In a classical treatment this theorem is often presented as the natural counterpart of IV.17. Forder’s proof is indeed extremely short: construct a ray AT' on the same side of AX as T such that

$$\angle XAT' \cong \angle XYA.$$

By IV.17 the line AT' is tangent at A . Uniqueness of the tangent then forces $AT' = AT$, and hence the required angle equality.

The formal reconstruction follows exactly this route, but it makes two points explicit that the printed proof suppresses:

1. uniqueness of the tangent *carrier line* must be proved synthetically;
2. equality of the two carrier lines is not yet enough to replace T' by T inside an angle: one must show that T, T' lie on the same ray from A .

The production implementation is

```
CGJteamLab/Proposition4_18.lean
```

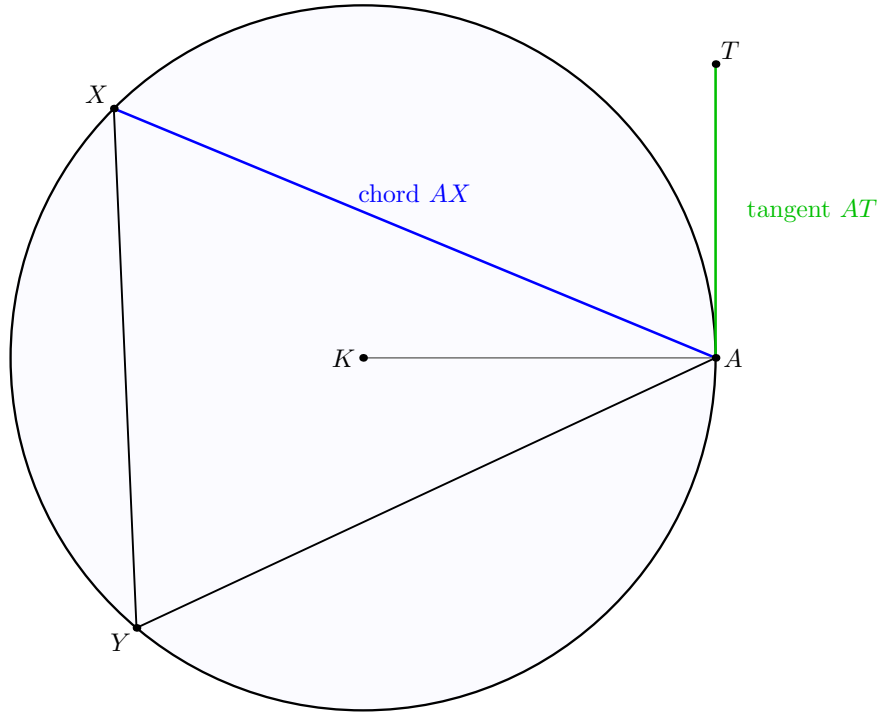
with public endpoint

```
hilbert_IV18_tangent_chord
```

18.2 Classical Configuration

The theorem starts with an actual tangent at A , not with a candidate line.

$$\angle XAT \cong \angle XYA$$



T, Y opposite sides of AX

Figure 18.1: Forder IV.18. The line AT is tangent at A , and T, Y lie on opposite sides of chord AX . The theorem concludes $\angle XAT \cong \angle XYA$.

The source proof introduces a second point T' , lying on the same side of the chord as T , whose ray from A is constructed to satisfy the target angle relation.

IV.17 then turns that angle relation into tangency. Thus the proof derives the direct tangent–chord theorem from its converse plus uniqueness.

18.3 Forder’s Statement and Proof

Forder states:

If AT is tangent to the circle XAY at A , and T, Y are on opposite sides of AX , then $\angle XAT \cong \angle XYA$.

His proof is concise. Choose T' so that

$$\angle XAT' \cong \angle XYA$$

with T, T' on the same side of AX . Then IV.17 says that AT' is tangent at A . By the uniqueness result cited by Forder, the new tangent coincides with the original one, and therefore the corresponding tangent rays determine the same angle [1].

The formal proof preserves that order exactly:

construct $T' \implies \text{IV.17} \implies \text{tangent uniqueness} \implies \text{same ray} \implies \text{angle transport}.$

18.4 Formal Interface

The proposition contract is:

```
def HilbertForderIV18TangentChord
  [H : HilbertIncidence Geo]
  [_HC : @HilbertCongruence Geo H] : Prop :=
  forall K R X A Y T : Geo.Point,
    forall chord tangent : Geo.Line,
      Ne X A ->
      H.OnLine X chord ->
      H.OnLine A chord ->
      H.OnLine A tangent ->
      H.OnLine T tangent ->
      Ne T A ->
      HilbertCircle Geo K R X ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R Y ->
      HilbertCircleTangentAt
        (Geo := Geo) K A tangent ->
      HilbertOppositeSide Geo T Y chord ->
      Geo.AngleCongruent
        X A T
        X Y A
```

Compared with IV.17, the hypotheses are nearly the same. The difference is precisely the direction of implication.

IV.17 assumes the angle congruence and concludes tangency.

IV.18 assumes tangency and concludes the angle congruence.

The production theorem is:

```
theorem hilbert_IV18_tangent_chord
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV18TangentChord (Geo := Geo)
```

18.5 Constructing the Comparison Ray

The first step is to construct a ray through A that already has the desired angle.

Because T, Y lie on opposite sides of chord AX , both are off the chord. Hence the angle

$$\angle XYA$$

is proper.

Hilbert's angle-construction theorem is applied with vertex A , first arm AX , and prescribed side equal to the side containing T .

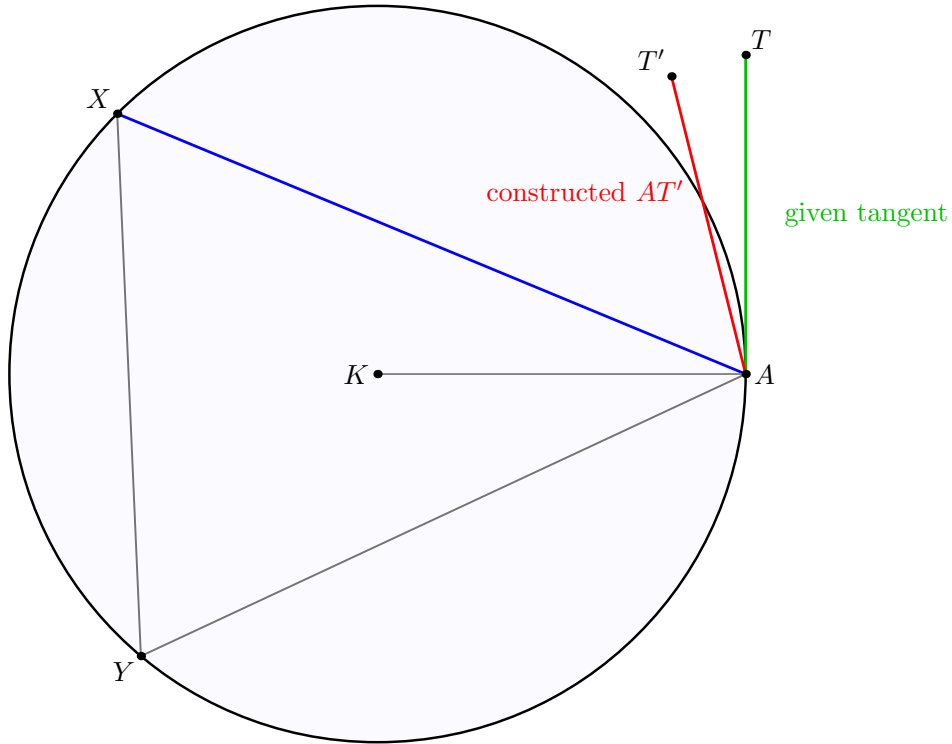
It produces a point T' such that:

$$T' \sim T$$

with respect to chord AX , and

$$\angle XAT' \cong \angle XYA.$$

T, T' same side of AX



IV.17: AT' tangent

uniqueness: carriers coincide

Figure 18.2: The comparison ray AT' . Hilbert angle construction places T' on the same side of chord AX as the given tangent point T and enforces $\angle XAT' \cong \angle XYA$. IV.17 will then show that AT' is itself tangent.

The formal construction is

```
rcases
  HilbertCongruence.angle_construction
    (Geo := Geo)
    X Y A
    X A T
    hXYA
    hXA
    chord
    hXchord
```

```

      hAChord
      hTOff
with
<T', hT'TSame, hXYA_XAT', _hUnique>

```

The point T' is automatically off chord AX , because it lies on the same side of the chord as T . In particular $T' \neq A$.

18.6 The Side Condition for IV.17

To apply IV.17 to AT' , the constructed point T' must lie on the opposite side of chord AX from Y .

The original data give

$$T \mid Y.$$

The construction gives

$$T' \sim T.$$

Transport of opposite-side relations therefore yields

$$T' \mid Y.$$

This is a small step in the printed proof but an essential formal obligation: IV.17 requires the side relation explicitly in its interface.

18.7 Constructing the Carrier of AT'

The point T' determines a unique carrier line through A . Let that line be

$$\ell'.$$

The formal proof obtains it from incidence:

```

rcases
  HilbertPlaneIncidence.line_through
    A T' hAT'
with
<tangent', hAtangent', hT'tangent'>

```

The angle congruence from the construction is reversed into the orientation expected by IV.17:

$$\angle XAT' \cong \angle XYA.$$

All hypotheses of IV.17 are now present.

18.8 Using IV.17

Apply

```
hilbert_IV17_tangent_converse
```

to the new carrier ℓ' .

The conclusion is

ℓ' is tangent to the circle at A .

Thus we now have two tangent carriers at the same circle point:

ℓ and ℓ' ,

where ℓ is the original carrier containing A, T .

The remainder of the proof is a uniqueness argument.

18.9 Uniqueness of the Tangent Carrier

The reusable theorem is

```
hilbert_circle_tangent_carrier_unique
```

It states that if two lines through the same circle point A are tangent to the same circle, then the lines are equal.

This theorem belongs in `HilbertInterfaceIV`, not in `Proposition4_18.lean`, because it is a general theorem about the project's tangent predicate.

Its proof is itself fully synthetic.

Suppose l, m are tangent at A . Unpack the two tangent witnesses: there are points $U \in l$ and $V \in m$ such that

$\angle UAK$ and $\angle VAK$

are right.

After swapping the angle arms, both

$\angle KAU$ and $\angle KAV$

are right.

The reusable perpendicular-uniqueness theorem

```
hilbert_two_right_angles_same_first_arm_collinear
```

then shows that

U, A, V

are collinear.

Since A, U determine l and A, V determine m , incidence uniqueness gives

$$l = m.$$

Applied in IV.18:

$$\boxed{\ell = \ell'}.$$

18.10 Why Equality of Carriers Is Not Yet Enough

At this point both T and T' lie on the same tangent carrier through A .

But an angle uses a *ray*, not merely a line.

The line equality alone allows two possibilities:

$$T, T' \text{ lie on the same ray from } A,$$

or

$$A \text{ lies between } T, T'.$$

The second alternative must be excluded.

Suppose

$$T - A - T'.$$

Since A lies on chord AX , the segment TT' would meet the chord at A . This means that T, T' lie on opposite sides of chord AX .

But the angle construction deliberately placed T' on the same side of the chord as T .

Contradiction.

Hence A is not between T, T' , and the collinearity of A, T, T' therefore yields

$$\text{SameRay}(A, T, T').$$

This is the order-theoretic step that turns equality of tangent carriers into equality of tangent rays.

18.11 Replacing T' by T

Same-ray equality gives literal equality of the two angles:

$$\angle XAT = \angle XAT'.$$

The constructed ray already satisfies

$$\angle XAT' \cong \angle XYA.$$

Therefore

$$\boxed{\angle XAT \cong \angle XYA}.$$

This is the IV.18 conclusion.

The final Lean step is conceptually just:

```
have hAngleXAT_XAT' :
  Geo.Angle X A T =
  Geo.Angle X A T' :=
  hilbert_angle_eq_of_sameRay_second
  Geo A X T T' hRayATT'

unfold Geometry.Geo.AngleCongruent
at hXAT'_XYA |-

rw [hAngleXAT_XAT']
exact hXAT'_XYA
```

Thus the last stage of the proof contains no new circle geometry: it is ray transport.

18.12 Lean Formalization

The public layer is small:

```
HilbertForderIV18TangentChord
hilbert_IV18_tangent_chord
```

The proposition-specific proof is essentially the former stage-61 theorem:

```
hilbert_forder_IV18_tangent_chord_stage61
```

The general uniqueness theorem is no longer proposition-local. Its production name is

```
hilbert_circle_tangent_carrier_unique
```

in `HilbertInterfaceIV`.

The second general ingredient is

```
hilbert_two_right_angles_same_first_arm_collinear
```

which expresses the underlying uniqueness of the perpendicular through a point.

The numbered dependency is simply:

```
hilbert_IV17_tangent_converse
```

Everything else is reusable incidence, side-of-line, tangent, perpendicular, and ray infrastructure.

18.13 Relation to IV.17

IV.17 and IV.18 together form the tangent–chord equivalence.

IV.17 proves:

$$\angle XAT \cong \angle XYA \implies AT \text{ tangent.}$$

IV.18 proves:

$$AT \text{ tangent} \implies \angle XAT \cong \angle XYA.$$

The proof of IV.18 is deliberately asymmetric: instead of reproducing the secant analysis of IV.17 in reverse, it constructs a comparison ray and reuses IV.17.

This is a strong modularity property. The difficult secant contradiction is proved exactly once, in IV.17.

IV.18 then needs only:

$$\text{angle construction} + \text{IV.17} + \text{tangent uniqueness.}$$

18.14 Why Tangent Uniqueness Belongs to the Interface

During development the theorem `hilbert_circle_tangent_carrier_unique` first appeared as support for IV.18.

Its statement, however, mentions no feature specific to IV.18. It says only:

$$\text{two tangent carriers at the same circle point are equal.}$$

The production refactor therefore promotes it to `HilbertInterfaceIV`.

This ownership is important later. Any theorem that needs uniqueness of a tangent can use the interface result without importing IV.18 merely to obtain an unrelated helper.

The numbered module therefore contains the tangent–chord theorem, while the generic uniqueness principle remains proposition-independent.

18.15 Hilbert Components Used

18.15.1 Angle construction

Hilbert III.4 is the constructive heart of the proof. It produces the ray AT' realizing the angle $\angle XYA$ on a prescribed side of chord AX .

The prescribed-side clause matters: it is what later ensures T, T' are on the same ray after the tangent carriers are identified.

18.15.2 Plane separation

The proof uses side transport twice.

First:

$$T' \sim T, \quad T \mid Y \implies T' \mid Y.$$

Second, after tangent carrier equality, same-side information excludes

$$T - A - T'.$$

Thus plane separation enters both before and after IV.17.

18.15.3 Perpendicular uniqueness

The tangent predicate is defined through a right angle with the radius. Therefore uniqueness of tangent carriers ultimately reduces to uniqueness of the perpendicular to the radius through A .

18.15.4 Incidence uniqueness

Once the two tangent witnesses are shown collinear with A , uniqueness of the line through two distinct points identifies the two tangent carriers.

18.15.5 Ray equality

The final angle transport is justified only after the points T, T' are shown to lie on the same ray from A .

This distinguishes the formal proof from a looser statement that two tangent lines coincide.

18.16 Foundational Boundary

IV.18 introduces no new tangent axiom and no circle-specific uniqueness axiom.

The proof uses:

- Hilbert angle construction;
- plane separation;
- IV.17;
- the synthetic tangent predicate;
- uniqueness of perpendiculars;
- line uniqueness;
- ray transport.

The uniqueness of the tangent is itself derived inside the existing Hilbert Euclidean-plane framework.

No numerical angle measure and no analytic tangent construction are used.

18.17 Classical and Formal Dependency Structures

Forder's source proof is:

$$\begin{array}{c}
 AT \text{ tangent, } T \mid Y \\
 \downarrow \\
 \text{construct } T' \sim T \text{ with } \angle XAT' \cong \angle XYA \\
 \downarrow \\
 \text{IV.17} \\
 \downarrow \\
 AT' \text{ tangent} \\
 \downarrow \\
 \text{uniqueness of tangent} \\
 \downarrow \\
 AT' = AT \\
 \downarrow \\
 \angle XAT \cong \angle XYA.
 \end{array}$$

The formal proof expands the uniqueness and ray steps:

$$\begin{array}{c}
 \text{Hilbert angle construction} \\
 \downarrow \\
 T' \sim T, \quad \angle XAT' \cong \angle XYA \\
 \downarrow \\
 T' \mid Y \\
 \downarrow \\
 \text{carrier } \ell' = AT' \\
 \downarrow \\
 \text{IV.17} \\
 \downarrow \\
 \ell' \text{ tangent} \\
 \downarrow \\
 \text{hilbert_circle_tangent_carrier_unique} \\
 \downarrow \\
 \ell' = \ell \\
 \downarrow \\
 A, T, T' \text{ collinear} \\
 \downarrow \\
 T - A - T' \text{ excluded by same-side} \\
 \downarrow \\
 \text{SameRay}(A, T, T') \\
 \downarrow \\
 \angle XAT = \angle XAT' \\
 \downarrow \\
 \angle XAT \cong \angle XYA.
 \end{array}$$

At module level:

```

HilbertInterfaceIV
Proposition4_17
  -> Proposition4_18

```

18.18 Diagrammatic Audit

Two figures are sufficient for IV.18.

1. The source tangent–chord configuration.
2. The comparison construction showing the given tangent ray AT and the constructed ray AT' before uniqueness identifies their carriers.

A third figure for tangent uniqueness would largely duplicate the second figure; the logical argument is clearer in the text.

18.19 Conclusion

Forder IV.18 is short because almost all of its mathematical work has already been concentrated in IV.17.

The proof constructs a comparison ray AT' with the required angle, places it on the correct side of chord AX , and invokes IV.17 to prove that its carrier is tangent. A reusable tangent-carrier uniqueness theorem then identifies that carrier with the original tangent.

The formal reconstruction exposes one final distinction invisible in the source shorthand: equality of carriers does not yet give equality of rays. The same-side condition from the angle construction is used once more to exclude opposite orientation, yielding

$$\text{SameRay}(A, T, T').$$

Only then may T' be replaced by T in the angle.

Thus IV.18 is a compact but instructive example of modular proof design: construction, converse theorem, uniqueness, and ray transport combine to recover the direct tangent–chord relation without repeating the secant analysis of IV.17.

Chapter 19

Proposition IV.19

19.1 Introduction

Forer IV.19 is the converse of the same-segment theorem.

In source notation, if C, D lie on the same side of chord AB and

$$\angle ACB \cong \angle ADB,$$

then the four points A, B, C, D are cyclic [1].

The production interface uses a relabelled but equivalent form. The fixed chord is CD ; the points A, B lie on the same side of its carrier; and the equal-angle hypothesis is

$$\angle CAD \cong \angle CBD.$$

The conclusion is

$$A, C, D, B \text{ are concyclic.}$$

This proposition is a genuine converse theorem. The circle containing all four points is not assumed. The proof therefore begins by constructing the circumcircle through three points and then proves that the fourth point must lie on it.

The formal architecture is:

$$\text{circle through } C, D, A \longrightarrow \begin{cases} B \text{ already on the circle,} \\ B \text{ off the circle} \end{cases}$$

and in the second case:

$$\text{line } CB \longrightarrow \begin{cases} \text{secant,} \\ \text{tangent.} \end{cases}$$

The secant branch is itself divided into two ray-order cases. The tangent branch is excluded by IV.18.

The production implementation is

```
CGJteamLab/Proposition4_19.lean
```

and it exposes both a dependency-explicit theorem

```
hilbert_IV19_circle_converse_of_IV16_IV18
```

and the closed public endpoint

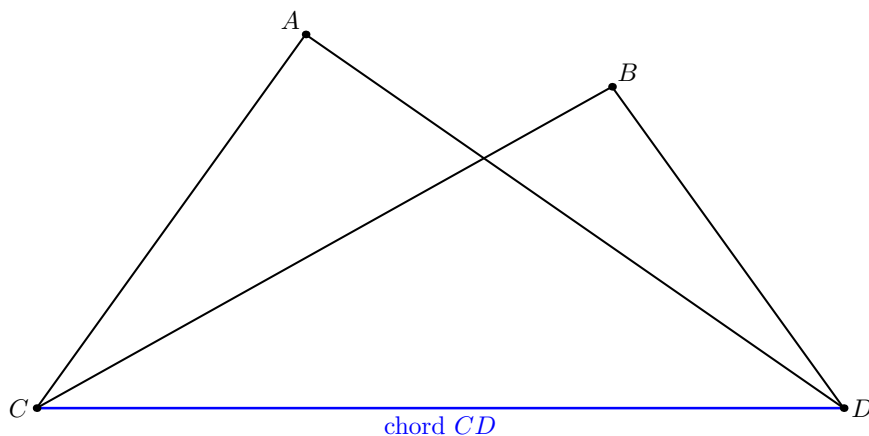
```
hilbert_IV19_circle_converse
```

19.2 Classical Configuration

Let C, D determine the chord carrier. Assume A, B lie on the same side of that line and

$$\angle CAD \cong \angle CBD.$$

A, B same side of CD



$$\angle CAD \cong \angle CBD$$

Figure 19.1: Production naming for Forder IV.19. The points A, B lie on the same side of chord CD , and the equal-angle hypothesis is $\angle CAD \cong \angle CBD$. The theorem concludes that A, C, D, B are concyclic.

The source theorem is usually remembered as the converse of “angles in the same segment are equal.” The formal proof, however, cannot simply invoke a uniqueness theorem for circles from an equality of angle measures. It constructs the relevant circle and excludes all ways in which the fourth point could fail to belong to it.

19.3 Forder’s Statement and Proof

Forder argues as follows.

Because A, C, D are noncollinear, there is a circle through those three points. If B lies on that circle, the theorem is proved.

Suppose B does not lie on it. Then the line through C and B either meets the circle again in a point $E \neq C$, or is tangent to the circle at C .

In the secant case, IV.16 compares the angle at E with the angle at A . According to the order of E relative to B and C , the resulting angles are either congruent or supplementary. Both alternatives contradict elementary triangle angle order unless $E = B$. But $E = B$ would put B on the circle.

In the tangent case, IV.18 gives the tangent–chord angle relation. Combined with the assumed equality of the two inscribed angles, this makes a remote interior angle congruent to an exterior angle of a triangle, again impossible.

Thus the line CB cannot have a second circle point distinct from B and cannot be tangent while B remains off the circle. Therefore B must lie on the constructed circumcircle.

The formal proof preserves this source architecture almost literally, but makes every line-order and side-of-chord distinction explicit.

19.4 Formal Statement

The proposition interface in `HilbertInterfaceIV.lean` is:

```
def HilbertForderIV19CircleConverse
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall C D A B : Geo.Point,
    forall chord : Geo.Line,
      Ne C D ->
      H.OnLine C chord ->
      H.OnLine D chord ->
      HilbertSameSide Geo A B chord ->
      Geo.AngleCongruent
        C A D
        C B D ->
      HilbertConcyclic4 Geo A C D B
```

The hypothesis

```
HilbertSameSide Geo A B chord
```

contains the off-line information for both A and B . In particular, A is off chord CD , so C, D, A form a genuine triangle.

The closed production theorem is:

```
theorem hilbert_IV19_circle_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV19CircleConverse (Geo := Geo)
```

The production module also exports a dependency-explicit variant

```
hilbert_IV19_circle_converse_of_IV16_IV18
```

which exposes the mathematical fact that the proof requires both halves of IV.16 and IV.18.

19.5 Constructing the Circumcircle Through Three Points

Since A lies off the carrier CD , the points

$$C, D, A$$

are noncollinear.

The proof invokes the triangle circumcenter theorem

```
hilbert_triangle_circumcenter_exists_XI
```

to construct a point K satisfying

$$KC \cong KD, \quad KC \cong KA.$$

Using C itself as the reference-radius point, the proof obtains circle membership:

$$C, D, A$$

all lie on the circle centered at K .

The theorem now splits on whether B belongs to this circle.

If it does, the target

$$\text{Concyclic4}(A, C, D, B)$$

is immediate.

All remaining work occurs under the contradiction hypothesis that B is off the constructed circle.

19.6 Preparing the Carrier CB

Assume B is not on the constructed circle.

Because B lies on the same side of chord CD as A , it is off that chord. In particular

$$C \neq B.$$

Let ℓ_{CB} be the carrier through C, B .

The proof also shows

$$D \notin \ell_{CB}.$$

Otherwise C, D, B would be collinear, forcing B onto chord CD , contrary to the same-side hypothesis.

The circle centre is distinct from C , so the general theorem

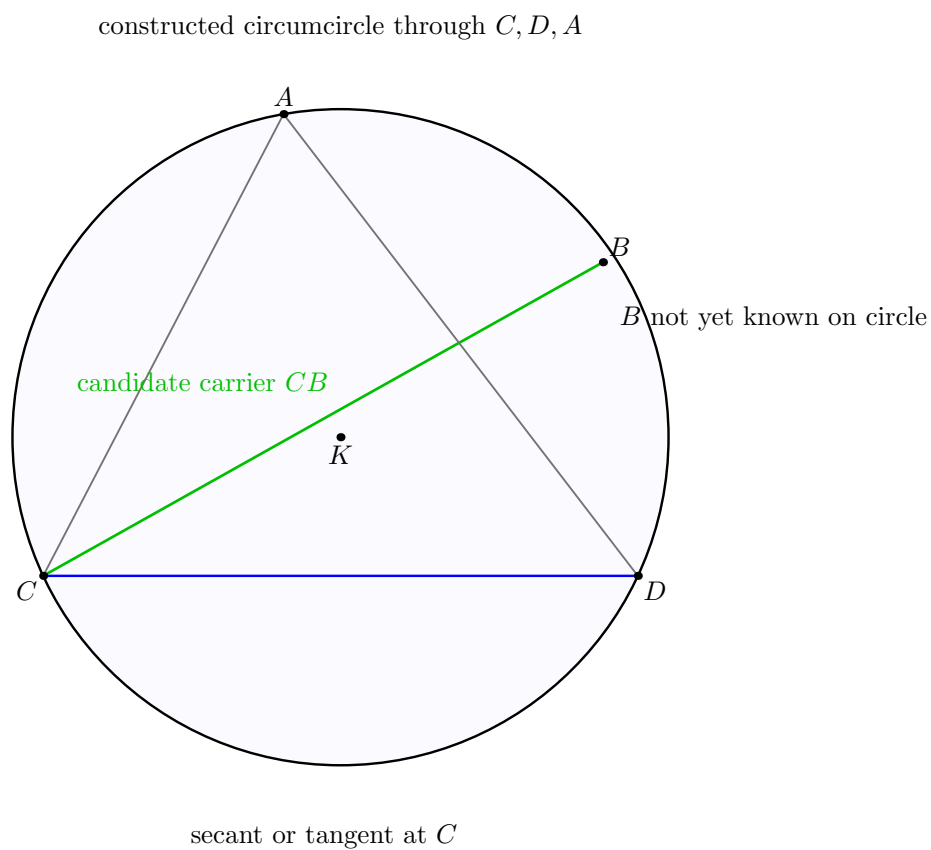


Figure 19.2: The converse proof starts by constructing the circumcircle through C, D, A . The fourth point B is not initially assumed to lie on this circle. If B is off it, the carrier CB is analyzed as secant or tangent.

```
hilbert_circle_line_second_or_tangent
```

applies to the line ℓ_{CB} through circle point C .

It returns exactly two alternatives:

1. there exists a second circle point $E \neq C$ on ℓ_{CB} ;
2. ℓ_{CB} is tangent to the circle at C .

This is the main structural dichotomy of IV.19.

19.7 Secant Branch: Order Normalization

Suppose $E \neq C$ is the second circle point on line CB .

Since B was assumed off the circle,

$$E \neq B.$$

The three points C, B, E are distinct and collinear. The reusable theorem

```
hilbert_secant_ray_order_split
```

reduces all possible linear orders to two geometrically relevant cases:

$$\text{SameRay}(C, B, E)$$

or

$$B - C - E.$$

This semantic production theorem was originally discovered inside the IV.19 development, but it now belongs to `HilbertInterfaceIV`. Its promotion is essential to the acyclic Book IV architecture, because IV.17 also needs the same general ray-order fact.

19.8 Same-Ray Secant Branch

Assume

$$\text{SameRay}(C, B, E).$$

The point E lies on the same circle as C, D, A .

Because A, B lie on the same side of chord CD , and E lies on the same ray CB as B , the relevant IV.16 same-side configuration gives

$$\angle CAD \cong \angle CED.$$

The IV.19 hypothesis is

$$\angle CAD \cong \angle CBD.$$

Hence

$$\angle CED \cong \angle CBD.$$

But B, E lie on the same ray from C . This is exactly the setting of the reusable secant-rigidity theorem

```
hilbert_equal_angles_same_secant_unique
```

which says that if two points on the same ray from a fixed origin subtend the same angle against a point off the carrier, then the two points coincide.

Therefore

$$E = B.$$

This contradicts $E \neq B$, which came from the assumption that B is off the circle.

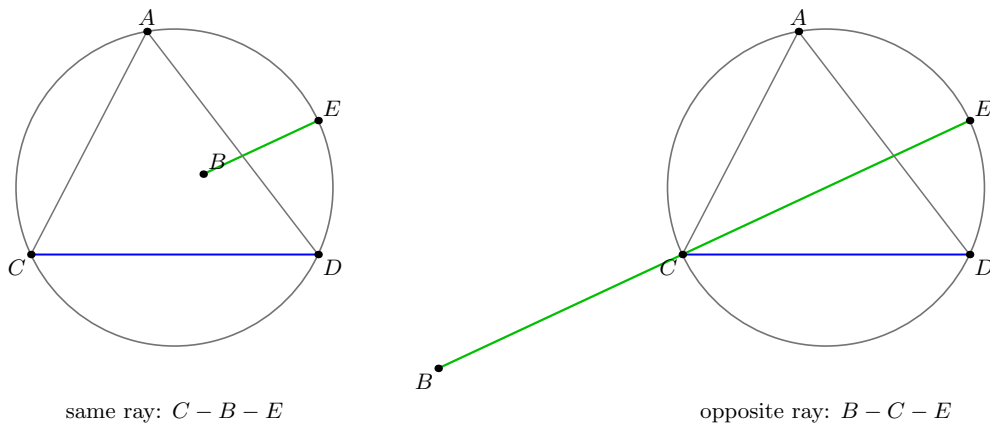


Figure 19.3: The two secant orders in IV.19. Left: E lies on the same ray CB as B , and equal-angle secant rigidity forces $E = B$. Right: $B - C - E$; then B, E lie on opposite sides of the chord through C, D , and IV.16 opposite-side leads to an exterior-angle contradiction.

The proposition-specific helper closing this branch is historically the same-ray IV.19 secant lemma; its general uniqueness ingredient now lives in the interface.

19.9 Why Same-Secant Equality Forces Coincidence

The theorem

```
hilbert_equal_angles_same_secant_unique
```

deserves separate comment.

Its input is:

$$O, B, B'$$

on one ray and a point C off that carrier, together with

$$\angle OBC \cong \angle OB'C.$$

If $B \neq B'$, one of the two points lies strictly between the origin and the other. The assumed angle congruence then identifies an exterior angle of the triangle determined by the nearer point with a remote interior angle.

Euclid's exterior-angle theorem rules this out.

Thus

$$B = B'.$$

This is a general synthetic rigidity fact, not something peculiar to circles, which is why the production refactor correctly places it in `HilbertInterfaceIV`.

19.10 Opposite-Ray Secant Branch

Assume now

$$B - C - E.$$

Because C lies on the chord carrier CD , the segment BE crosses the chord at C . Hence B, E lie on opposite sides of chord CD .

Since A, B are on the same side of the chord,

$$A, E$$

are on opposite sides.

The reusable production helper

```
hilbert_oppositeRay_gives_oppositeSide
```

packages exactly this order-to-half-plane conversion.

Apply IV.16 opposite-side to the circle points C, D, A, E . It produces a supplement of the angle at E congruent to

$$\angle CAD.$$

Using the IV.19 hypothesis

$$\angle CAD \cong \angle CBD$$

and the collinearity/ray relation created by $B - C - E$, the proof transports the angle at B into the corresponding remote interior angle of triangle EDB .

The result is that a remote interior angle of triangle EDB is congruent to an exterior angle at E .

Hilbert's exterior-angle theorem forbids this.

Therefore the opposite-ray secant branch is impossible.

In the original staged proof this was the theorem

```
hilbert_forder_IV19_secant_oppositeRay_impossible_stage5
```

The production module retains the proposition-specific contradiction while using semantic interface names for the general order lemmas.

19.11 Tangent Branch

The only remaining alternative from the circle-line dichotomy is that the line CB is tangent to the constructed circle at C .

This branch is also impossible.

Extend BA through A to a point Z :

$$B - A - Z$$

in production naming, equivalently the source extension of the line through the noncircle point and the relevant vertex.

The important side relation is that the extension point Z lies on the opposite side of chord CD from B , hence also on the opposite side from A , because A, B are same-side.

Now IV.18 applies to the tangent at C .

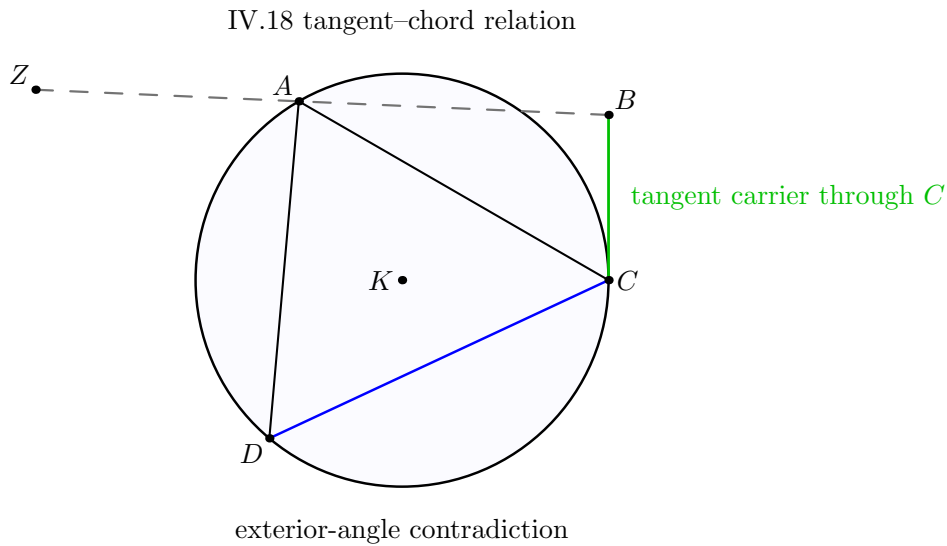


Figure 19.4: The tangent alternative in IV.19. If the carrier through the candidate point were tangent to the constructed circle, IV.18 would turn the tangent–chord angle into the same inscribed angle appearing in the hypothesis. After transport, an exterior angle of the relevant triangle would be congruent to a remote interior angle.

The tangent–chord theorem gives an angle congruence between the tangent angle at the circle point and the inscribed angle standing on the same chord.

Combining it with the IV.19 hypothesis identifies that tangent angle with the angle at the noncircle point.

But the extension point makes the tangent angle an exterior angle of a proper triangle, while the other angle is a remote interior angle.

This contradicts the exterior-angle theorem.

Hence the tangent branch is impossible.

The proposition-specific helper is

```
hilbert_forder_IV19_tangent_impossible_stage5
```

and its only numbered circle input is IV.18.

19.12 Completion of the Converse

The complete proof is now exhaustive.

Construct the circumcircle through C, D, A .

If B lies on it, return concyclicity.

Otherwise consider line CB .

If it is secant, let E be the second circle point.

- same ray: equal-angle secant rigidity forces $E = B$, contradicting that B is off the circle;
- opposite ray: IV.16 opposite-side creates an impossible exterior-angle congruence.

If the line is tangent, IV.18 creates another impossible exterior-angle congruence.

Every way for B to remain off the circle has therefore been eliminated.

Hence

A, C, D, B are concyclic.

The staged complete theorem was

```
hilbert_forder_IV19_circle_converse_stage6
```

The production module exports the same mathematics under the public names

```
hilbert_IV19_circle_converse_of_IV16_IV18
hilbert_IV19_circle_converse
```

19.13 Dependency-Explicit and Closed Theorems

IV.19 is unusual in this block because the production module exposes two useful endpoints.

The dependency-explicit theorem

```
hilbert_IV19_circle_converse_of_IV16_IV18
```

makes the proof parameters visible: it consumes the same-side and opposite-side forms of IV.16 together with IV.18.

The closed theorem

```
hilbert_IV19_circle_converse
```

supplies the production results

```
hilbert_IV16_same_side
hilbert_IV16_opposite_side
hilbert_IV18_tangent_chord
```

and returns the proposition interface with no theorem parameters.

This dual API is useful downstream. The explicit form records the exact logical reduction; the closed form is the convenient theorem imported by later numbered propositions.

19.14 Lean Formalization

The public API is:

```
HilbertForderIV19CircleConverse
hilbert_IV19_circle_converse_of_IV16_IV18
hilbert_IV19_circle_converse
```

The proposition-specific proof architecture contains the secant and tangent contradiction lemmas.

The general reusable ingredients have production semantic names in `HilbertInterfaceIV`:

```
hilbert_circle_line_second_or_tangent
hilbert_equal_angles_same_secant_unique
hilbert_secant_ray_order_split
hilbert_oppositeRay_gives_oppositeSide
```

This ownership is important. These theorems were discovered during the IV.19 development, but none of their statements is specific to the conclusion of IV.19.

They therefore belong to the reusable interface, while `Proposition4_19.lean` owns only the proof that the secant and tangent alternatives contradict the IV.19 hypothesis.

19.15 Relation to IV.16

Both halves of IV.16 occur.

In the same-ray secant branch, the relevant circle points are on the same side of the chord, so IV.16 same-side gives a direct angle congruence.

In the opposite-ray branch, the line order forces the second circle point onto the opposite side of the chord, so IV.16 opposite-side gives a supplement relation.

Thus the two-way public split of IV.16 again aligns exactly with the geometry of a later proof.

19.16 Relation to IV.18

IV.18 is used only in the tangent branch.

This is conceptually clean:

secant branches $\longleftrightarrow$ IV.16,

whereas

tangent branch $\longleftrightarrow$ IV.18.

The circle-line dichotomy therefore routes the proof directly to the theorem appropriate for each geometric type.

19.17 Circumcircle Existence and the Converse Direction

The converse nature of IV.19 makes circumcircle existence foundationally visible.

The forward same-segment theorem IV.16 begins with an already given circle. IV.19 does not.

It must first construct a circle through the three noncollinear points C, D, A .

Only after that construction can the fourth point B be tested against the circle.

This is an important logical distinction:

equal angles $\nrightarrow$ “there is already a circle”.

The proof must use the established circumcenter existence theorem to create the candidate circle.

19.18 Hilbert Components Used

19.18.1 Circumcenter existence

The triangle CDA is proper because A is off chord CD . The Euclidean-plane circumcenter theorem supplies the centre K .

19.18.2 Circle-line dichotomy

A line through a noncentral circle point is either secant again or tangent. This theorem supplies the exhaustive top-level classification once B is assumed off the candidate circle.

19.18.3 Ray order and plane separation

The secant point E must be classified relative to ray CB . The order determines whether the relevant pair of points is same-side or opposite-side with respect to chord CD .

19.18.4 Angle rigidity on a secant

The same-ray branch closes through `hilbert_equal_angles_same_secant_unique`. Its proof is ultimately an exterior-angle contradiction.

19.18.5 IV.16 and IV.18

IV.16 controls the secant branches; IV.18 controls the tangent branch.

19.18.6 Exterior-angle comparison

Every impossible branch eventually reduces to the same geometric principle: an exterior angle of a proper triangle cannot be congruent to the appropriate remote interior angle.

This repeated terminal contradiction is the synthetic substitute for the usual informal uniqueness intuition.

19.19 Foundational Boundary

IV.19 introduces no new circle axiom.

It does use the existing theorem guaranteeing a circumcircle for a proper triangle. Once that circle exists, the rest of the proof uses:

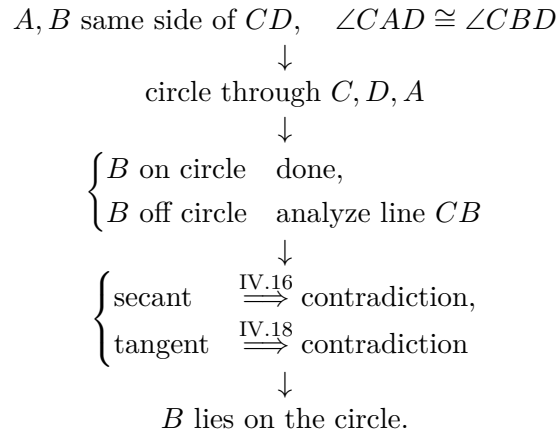
- incidence and line uniqueness;
- ray order;
- plane separation;
- the secant/tangent dichotomy;
- IV.16 same-side and opposite-side;
- IV.18;
- synthetic exterior-angle comparison.

No coordinate equation of a circle, no determinant criterion for concyclicity, and no numerical angle measure appear.

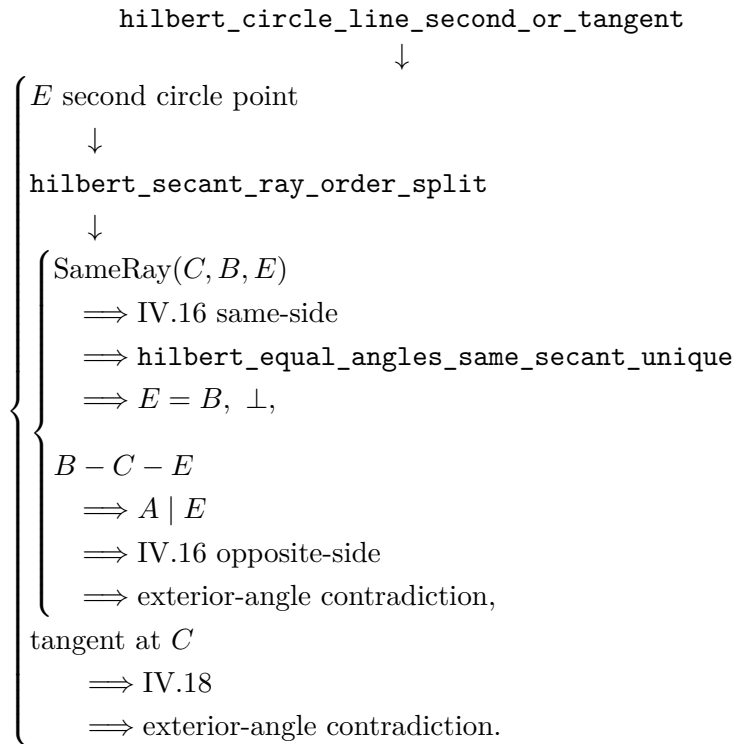
The converse is therefore proved entirely within the synthetic Hilbert framework.

19.20 Classical and Formal Dependency Structures

Forder's proof can be summarized as:



The formal proof expands the secant line order:



The production import graph is:

```

HilbertInterfaceIV
Proposition4_16
Proposition4_18
  -> Proposition4_19

```

There is no direct dependency on IV.17 because IV.18 already encapsulates the tangent-chord theorem needed here.

19.21 Architectural Significance

IV.19 is where the semantic extraction of general helpers from the old monolithic development becomes especially important.

The following results were first needed while proving IV.19:

```
hilbert_circle_line_second_or_tangent
hilbert_equal_angles_same_secant_unique
hilbert_secant_ray_order_split
hilbert_oppositeRay_gives_oppositeSide
```

But none is mathematically “Proposition IV.19”.

The production refactor therefore places them in `HilbertInterfaceIV`.

That move has two consequences:

1. IV.17 can reuse the secant-order split without depending forward on IV.19;
2. IV.19 itself becomes a numbered theorem module containing only its proposition-specific converse argument.

This is one of the clearest examples in the Book IV block of semantic module ownership replacing historical proof-development ownership.

19.22 Diagrammatic Audit

Four figures are appropriate.

1. The source same-segment converse configuration.
2. The constructed circumcircle through three points with B not yet known to lie on it.
3. A two-panel figure for the two secant ray orders.
4. The tangent contradiction controlled by IV.18.

These figures mirror the actual logical branching and are preferable to a single crowded picture containing all auxiliary points simultaneously.

19.23 Conclusion

Forder IV.19 is the first full circle-converse theorem in this sequence.

The proof does not infer concyclicity from an angle equation by a black-box criterion. It constructs the circumcircle through three points and then eliminates every possible way for the fourth point to stay off that circle.

If the line through the fourth point is secant, IV.16 and secant rigidity eliminate both possible ray orders. If it is tangent, IV.18 and the exterior-angle theorem give the contradiction.

The proposition also illustrates the production architecture particularly well. General secant/tangent and ray-order facts live in `HilbertInterfaceIV`; the numbered module owns only the circle converse itself and its secant/tangent contradiction branches.

The result is a source-faithful synthetic reconstruction of the same-segment converse, with the hidden existence, order, and uniqueness arguments made fully explicit.

Chapter 20

Proposition IV.20

20.1 Introduction

Forder IV.20 is the converse theorem for the opposite-segment configuration.

In Forder’s source notation, if C, D lie on opposite sides of chord AB and the angles

$$\angle ACB, \quad \angle ADB$$

are supplementary, then A, C, B, D form a cyclic quadrilateral [1].

The production interface uses a relabelled synthetic form. The fixed chord is CD , the points A, B lie on opposite sides of its carrier, and an explicit point X witnesses that

$$\angle XAD$$

is the supplement of

$$\angle CAD.$$

If

$$\angle XAD \cong \angle CBD,$$

then

$$A, C, D, B$$

are concyclic.

The implementation lives in

```
CGJteamLab/Proposition4_20.lean
```

and exposes two public endpoints:

```
hilbert_IV20_circle_converse  
hilbert_crossing_rays_transfer
```

These endpoints have different roles. The first is Forder IV.20 itself. The second is the downstream crossing-rays theorem for which the completed circle block was needed in the proportion development.

This distinction is central to the production architecture. The numbered circle theorem is one layer; the crossing-rays transfer built on the completed circle theory is another.

20.2 Forder's Statement

Forder states:

If C, D are on opposite sides of AB , and $\angle ACB, \angle ADB$ are supplementary, then $ACBD$ is a cyclic quadrilateral.

His printed proof is very short.

Choose a point E on the ray determined by B, C and on the circle through A, B, D . IV.16 says that

$$\angle AEB$$

and

$$\angle ADB$$

are supplementary. By hypothesis,

$$\angle ACB$$

and

$$\angle ADB$$

are also supplementary. Hence

$$\angle ACB \cong \angle AEB.$$

IV.19 then places E on the circle through A, B, C . Since E already lies on the circle through A, B, D , the two circles coincide, and the desired four points are cyclic.

Forder immediately adds the corollary that if

$$AB \perp BC, \quad AD \perp DC,$$

then A, B, C, D are cyclic.

The present production block formalizes IV.20 itself. The printed IV.20.1 corollary is not a separate public endpoint in the current Book IV production sequence.

20.3 Production Configuration

The production theorem uses the chord CD as the reference line.

Assume:

$$A \mid B$$

with respect to CD .

Let X satisfy

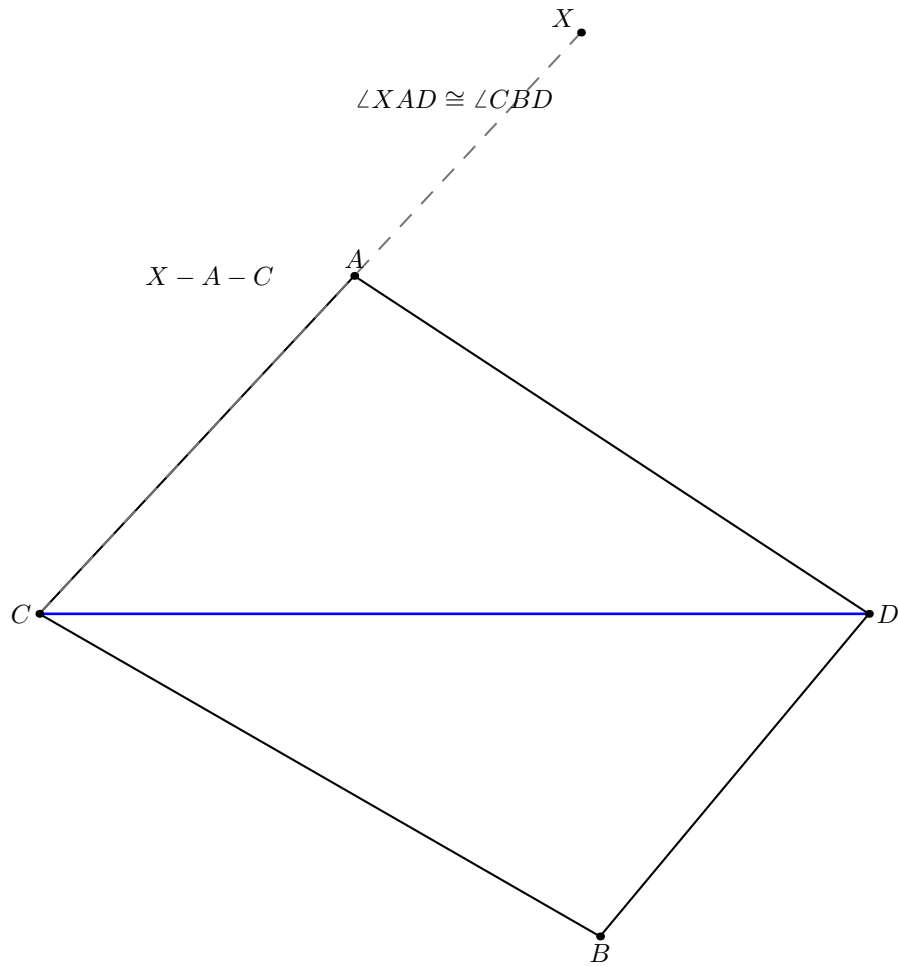
$$X - A - C.$$

Then the ray AD is unchanged and the angle

$$\angle XAD$$

is the adjacent supplement of

$$\angle CAD.$$



A, B opposite sides of CD

Figure 20.1: Production form of Forder IV.20. The points A, B lie on opposite sides of chord CD . The point X extends CA through A , so $\angle XAD$ is the supplement of $\angle CAD$. The hypothesis $\angle XAD \cong \angle CBD$ expresses that $\angle CAD$ and $\angle CBD$ are supplementary.

The hypothesis is

$$\angle XAD \cong \angle CBD.$$

The conclusion is

$$\text{Concyclic4}(A, C, D, B).$$

20.4 Formal Statement

The interface definition is:

```
def HilbertForderIV20CircleConverse
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall C D A B X : Geo.Point,
    forall chord : Geo.Line,
      Ne C D ->
        H.OnLine C chord ->
        H.OnLine D chord ->
        HilbertOppositeSide Geo A B chord ->
        BookZeroSupplement Geo
          X A D
          D C ->
        Geo.AngleCongruent
          X A D
          C B D ->
        HilbertConcyclic4 Geo A C D B
```

The public theorem is:

```
theorem hilbert_IV20_circle_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV20CircleConverse (Geo := Geo)
```

The `BookZeroSupplement` hypothesis is essential. No numerical equation

$$\angle CAD + \angle CBD = \pi$$

appears in the formal statement. Supplemantarity is represented by an actual extension point and a same-ray condition.

20.5 Source Proof and Production Proof

The source and production proofs establish the same geometric theorem, but their proof organizations are not literally identical.

Forder starts from a circle through A, B, D , chooses a second circle point on the ray through B, C , uses IV.16, then IV.19, and finally circle uniqueness.

The production theorem uses the converse engine already developed for IV.19:

1. construct the circumcircle through C, D, A ;
2. if B lies on it, finish;
3. otherwise analyze the line CB as secant or tangent;
4. eliminate both secant ray orders;
5. eliminate the tangent alternative.

This organization is longer than Forder's printed proof, but it has a significant formal advantage: the same circle-line and ray-order infrastructure used in IV.19 can be reused without introducing a second circle-intersection existence construction specialized to IV.20.

Thus the formal reconstruction is source-faithful at the level of the synthetic geometry and dependencies, while the proof normalization is adapted to the reusable production API.

20.6 Constructing the Candidate Circumcircle

Since A, B lie on opposite sides of chord CD , both are off the chord. Therefore C, D, A are noncollinear.

The proof invokes

```
hilbert_triangle_circumcenter_exists_XI
```

to construct a centre K with

$$KC \cong KD, \quad KC \cong KA.$$

This gives a candidate circle containing

$$C, D, A.$$

If B already belongs to the circle, the target concyclicity follows immediately.

Hence all subsequent work takes place under

$$B \notin \text{circle}(K).$$

20.7 The Carrier CB : Secant or Tangent

Since B is off chord CD ,

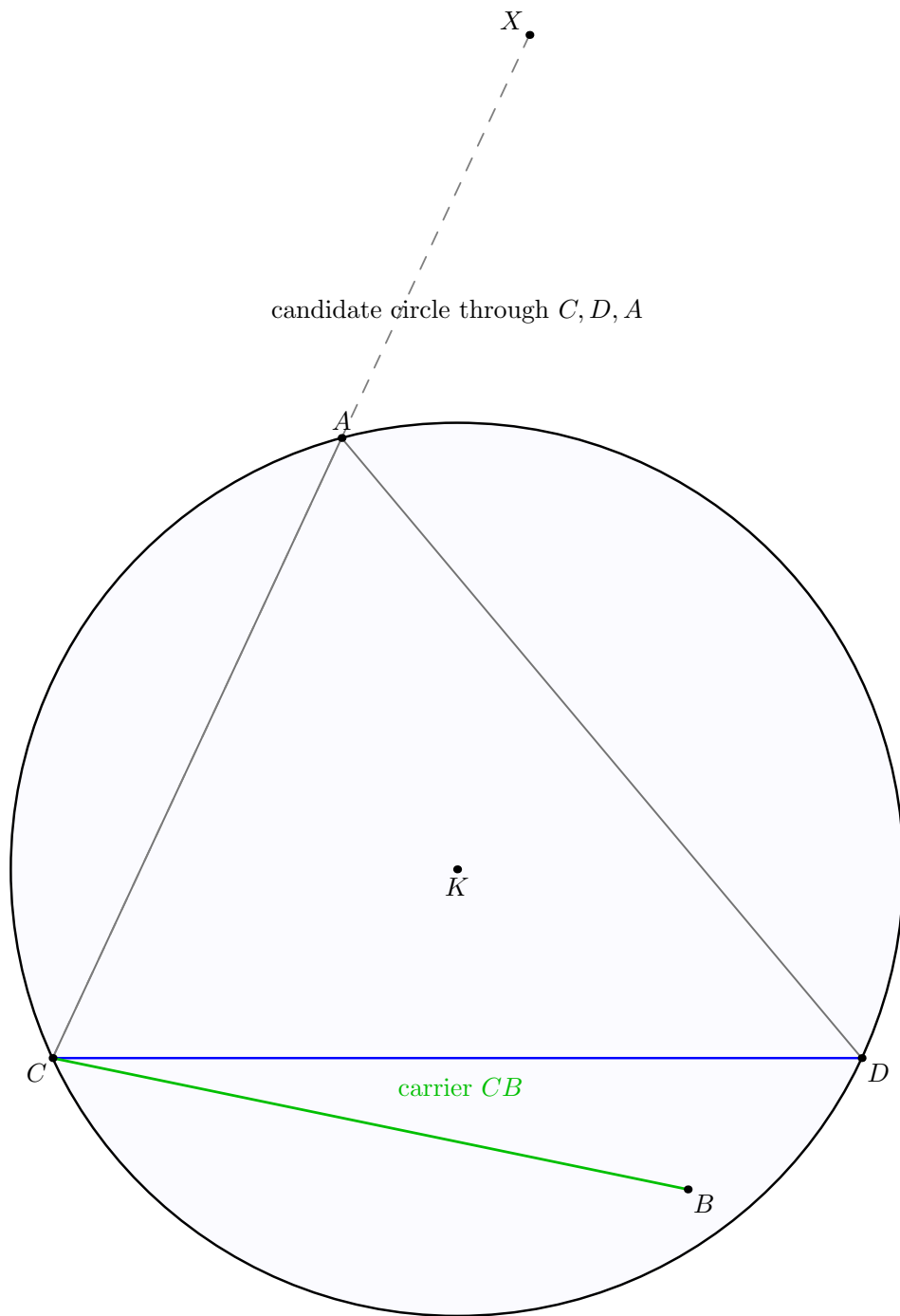
$$C \neq B.$$

Let ℓ_{CB} be the line through C, B .

The point D does not lie on ℓ_{CB} , because that would make C, D, B collinear and place B on the chord carrier.

The centre differs from C , so the reusable theorem

```
hilbert_circle_line_second_or_tangent
```



B off circle: secant or tangent

Figure 20.2: The production converse begins with the circumcircle through C, D, A . The point B is initially outside the logical circle data. If B is not on the candidate circle, the carrier CB is classified as secant or tangent.

applies.

It returns:

$$\begin{cases} \text{a second circle point } E \neq C \text{ on } \ell_{CB}, \\ \text{or} \\ \ell_{CB} \text{ tangent at } C. \end{cases}$$

This is the same structural dichotomy used in IV.19.

20.8 Secant Branch: Ray-Order Split

Assume there is a second circle point $E \neq C$ on the carrier CB .

Since B is assumed off the circle,

$$E \neq B.$$

The general order theorem

```
hilbert_secant_ray_order_split
```

reduces the line order to:

$$\text{SameRay}(C, B, E)$$

or

$$B - C - E.$$

The two cases use different halves of IV.16.

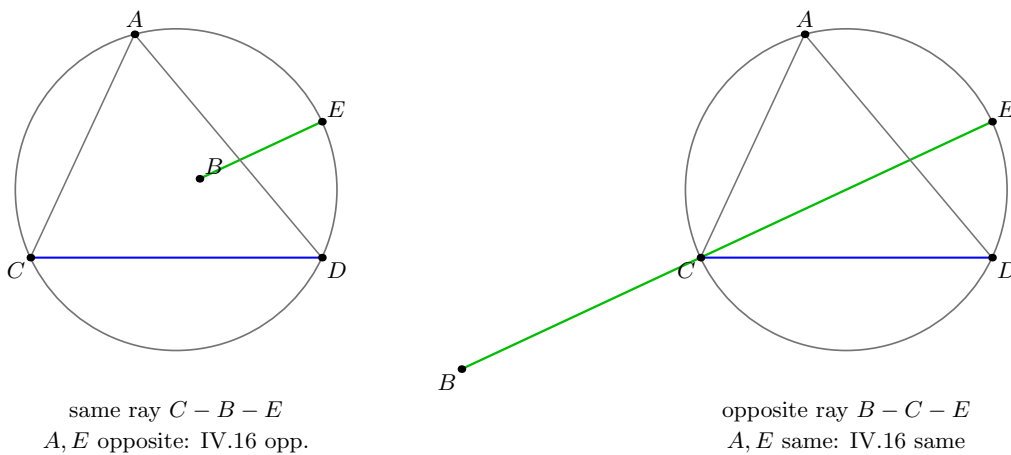


Figure 20.3: The two secant orders in IV.20. In the same-ray case B, E lie on the same side of chord CD , so A, E are opposite. In the opposite-ray case $B - C - E$, the points B, E are opposite across the chord; since A, B are opposite, A, E are on the same side.

20.9 Same-Ray Secant Branch

Assume

$$\text{SameRay}(C, B, E).$$

Because B, E lie on the same ray from the chord point C , they lie on the same side of chord CD .

The theorem hypothesis says

$$A \mid B.$$

Therefore

$$A \mid E.$$

Apply IV.16 opposite-side to the circle points

$$C, D, A, E.$$

It produces a point Y such that the angle

$$\angle YED$$

is a supplement of

$$\angle CED$$

and

$$\angle CAD \cong \angle YED.$$

The input at A already says that

$$\angle XAD$$

is a supplement of

$$\angle CAD.$$

After orienting both supplement witnesses consistently, Book Zero theorem 43 transports supplements of congruent angles and yields

$$\angle XAD \cong \angle CED.$$

The IV.20 angle hypothesis is

$$\angle XAD \cong \angle CBD.$$

Hence

$$\angle CED \cong \angle CBD.$$

Now B, E lie on the same ray from C . The reusable theorem

```
hilbert_equal_angles_same_secant_unique
```

forces

$$E = B.$$

But $E \neq B$ because B was assumed off the circle.

Contradiction.

The proposition-specific helper is

```
hilbert_forder_IV20_secant_sameRay_stage7
```

and its essential numbered input is IV.16 opposite-side.

20.10 Opposite-Ray Secant Branch

Assume

$$B - C - E.$$

The open segment BE meets the chord carrier CD at C , so

$$B \mid E$$

relative to the chord.

But the theorem hypothesis also says

$$A \mid B.$$

Two points opposite to the same point across a line lie on the same side. Hence

$$A \sim E.$$

The general production theorem underlying this parity step is

```
hilbert_two_oppositeSides_sameSide
```

which was promoted from the old stage-labelled development into `HilbertInterfaceIV`.

Apply IV.16 same-side to A, E :

$$\angle CAD \cong \angle CED.$$

The production proof now compares supplements.

The input witness says:

$$\angle XAD$$

is a supplement of

$$\angle CAD.$$

Extend CE through E to a point Y . Then

$$\angle DEY$$

is a supplement of

$$\angle CED.$$

Since

$$\angle CAD \cong \angle CED,$$

Book Zero supplement transport gives

$$\angle XAD \cong \angle DEY.$$

Using the IV.20 hypothesis,

$$\angle XAD \cong \angle CBD,$$

we obtain

$$\angle DEY \cong \angle CBD.$$

Finally $B - C - E$ identifies the ray BC with the ray BE , so

$$\angle CBD = \angle EBD.$$

Therefore

$$\angle DEY \cong \angle EBD.$$

But $\angle DEY$ is an exterior angle of triangle EBD , while $\angle EBD$ is a remote interior angle.

The exterior-angle theorem gives a contradiction.

Thus the opposite-ray secant branch is impossible.

The proposition-specific theorem is

```
hilbert_forder_IV20_secant_oppositeRay_impossible_stage7
```

20.11 Tangent Branch

Assume now that the line CB is tangent to the candidate circle at C .

IV.18 gives the tangent-chord relation:

$$\angle DCB \cong \angle DAC.$$

The IV.20 supplement witness says that

$$\angle XAD$$

is supplementary to

$$\angle DAC.$$

Extend CB through B to a point Z :

$$C - B - Z.$$

Then

$$\angle DBZ$$

is supplementary to

$$\angle CBD.$$

The theorem hypothesis gives

$$\angle XAD \cong \angle CBD.$$

Transporting supplements therefore yields

$$\angle DAC \cong \angle DBZ.$$

Combining with IV.18:

$$\angle DCB \cong \angle DBZ.$$

After reversing the arms at C ,

$$\angle BCD \cong \angle DBZ.$$

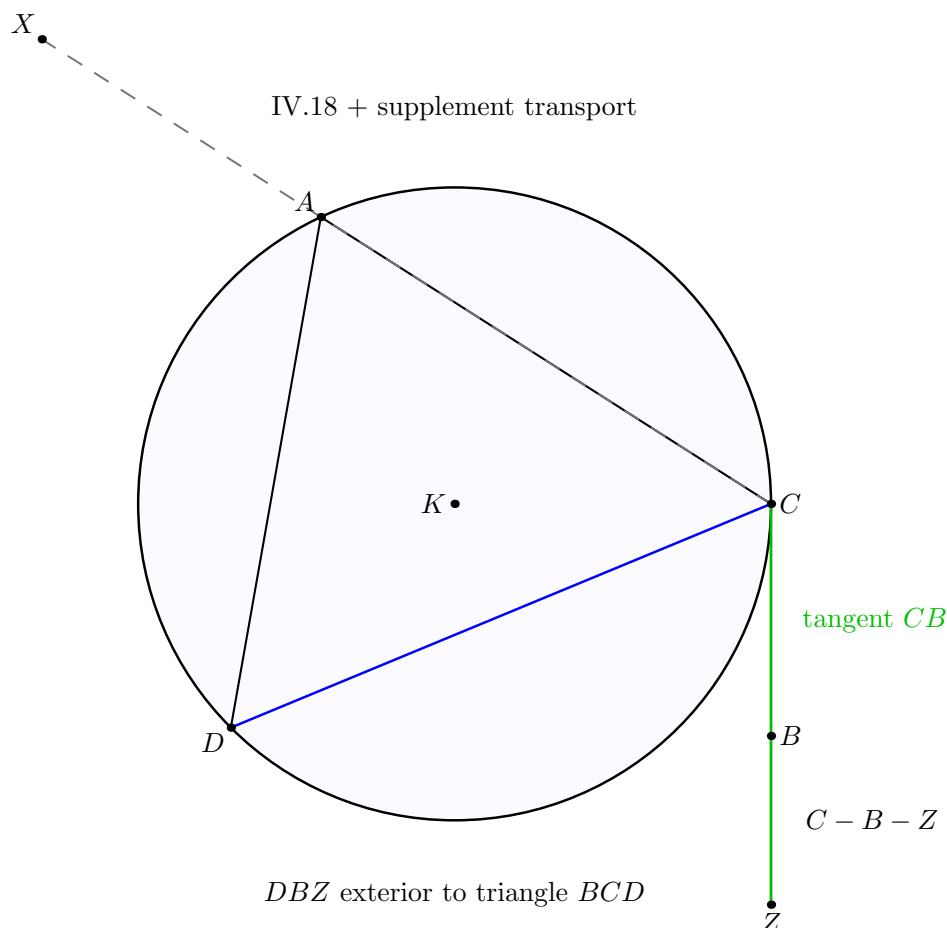


Figure 20.4: Tangent branch of IV.20. IV.18 identifies the tangent-chord angle at C with $\angle DAC$. Supplement transport then identifies the exterior angle DBZ of triangle BCD with the remote interior angle BCD , which is impossible.

But DBZ is an exterior angle of triangle BCD .

It cannot be congruent to the remote interior angle BCD .

Hence the tangent alternative is impossible.

This is the private helper

```
hilbert_forder_IV20_tangent_impossible_stage7
```

and its essential numbered input is IV.18.

20.12 Completion of Forder IV.20

The final IV.20 proof is now exhaustive.

Construct the circle through C, D, A .

If B lies on it, finish.

If not, classify the carrier CB :

- secant, same ray: IV.16 opposite-side + supplement transport + same-secant uniqueness force $E = B$;
- secant, opposite ray: IV.16 same-side + supplement transport create an exterior-angle contradiction;
- tangent: IV.18 + supplement transport create an exterior-angle contradiction.

Thus no off-circle position for B is possible.

Therefore

 A, C, D, B are concyclic.

The private assembly theorem is

```
hilbert_forder_IV20_circle_converse_stage8
```

and the public wrapper is

```
theorem hilbert_IV20_circle_converse
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV20CircleConverse (Geo := Geo) :=
  hilbert_forder_IV20_circle_converse_stage8
  Geo
  (hilbert_IV16_same_side Geo)
  (hilbert_IV16_opposite_side Geo)
  (hilbert_IV18_tangent_chord Geo)
```

This is an important dependency fact:

IV.20 itself uses IV.16 and IV.18, not IV.19.

The module imports IV.19 for the separate crossing-rays endpoint documented below.

20.13 IV.20 versus Forder's Short Proof

The production proof is intentionally more uniform than Forder's printed argument.

Forder proves IV.20 by choosing a circle point on a specified ray, invoking IV.16, then IV.19, and finally uniqueness of the circle.

The production implementation instead reuses the same “candidate circumcircle + secant/tangent exclusion” architecture as IV.19.

This choice has three formal advantages.

First, all circle-line alternatives are handled by the already verified `hilbert_circle_line_second_or_tangent` theorem.

Second, both secant order cases use the semantic ray-order API already extracted to the interface.

Third, the terminal contradictions are expressed entirely by the established IV.16/IV.18 and exterior-angle machinery, without needing a new theorem about two circles coinciding after sharing three points.

The documentation therefore records both the source proof and the actual production proof rather than silently identifying them.

20.14 The Crossing-Rays Endpoint

The same production module exports

```
theorem hilbert_crossing_rays_transfer
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertCrossingRaysTransfer (Geo := Geo)
```

The proposition interface is:

```
def HilbertCrossingRaysTransfer
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo] : Prop :=
  forall O A C B D : Geo.Point,
    Not (PrimCollinear Geo A O B) ->
    HilbertSameRay Geo O A C ->
    HilbertSameRay Geo O B D ->
    Geo.AngleCongruent
      O A D
      O B C ->
    Geo.AngleCongruent
      O D C
      O A B
```

Thus:

$$A, C$$

lie on one ray from O ,

$$B, D$$

lie on a second ray from O , and the crossing configuration is nondegenerate.

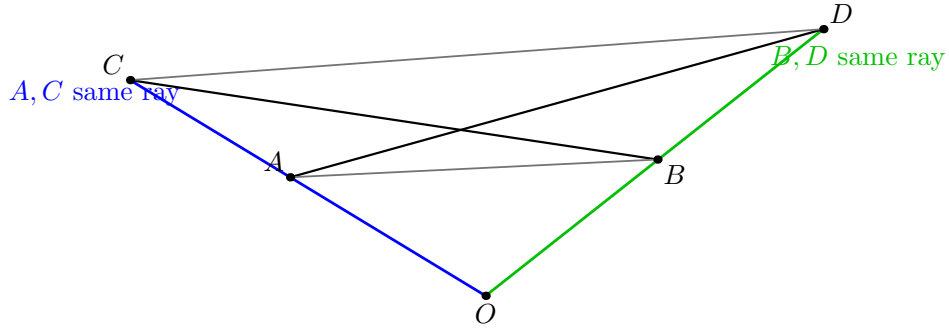
From

$$\angle OAD \cong \angle OBC$$

one proves

$$\angle ODC \cong \angle OAB.$$

given: $\angle OAD \cong \angle OBC$



target: $\angle ODC \cong \angle OAB$

Figure 20.5: Crossing-rays transfer. The points A, C lie on one ray from O , while B, D lie on the other. One cross-angle congruence, $\angle OAD \cong \angle OBC$, transfers to the complementary congruence $\angle ODC \cong \angle OAB$.

This theorem is not Forder IV.20 itself. It is the downstream result for which the IV.19–IV.20 circle converses provide the concyclicity mechanism.

20.15 Crossing Chords Become Concyclic

For the proper crossing configuration, the proof first establishes

$$A, C, D, B$$

concyclic.

The local theorem is

```
hilbert_crossing_chords_concyclic_stage9
```

It obtains dependency-explicit versions of IV.19 and IV.20 and passes them to the reusable interface theorem

```
hilbert_crossing_chords_concyclic_of_forder
```

The latter already contains the classification of the crossing configuration into:

- same-side chord cases, handled by IV.19;
- opposite-side/supplement cases, handled by IV.20;
- degenerate configurations, handled separately.

This is exactly where IV.19 becomes a genuine dependency of the *module-level endpoint*, even though it is not a dependency of `hilbert_IV20_circle_converse` itself.

20.16 Degenerate Crossing Cases

The public crossing-rays theorem handles two degenerate possibilities explicitly before entering the circle argument.

20.16.1 $B = D$

If

$$B = D,$$

the desired conclusion reduces to symmetry of the given angle congruence.

No circle theorem is required.

20.16.2 $A = C$, with $B \neq D$

If

$$A = C,$$

the circle construction is again unnecessary.

The proof compares triangles

$$OAD \quad \text{and} \quad OBA.$$

Because A, C lie on the same ray and B, D lie on the same ray, the angles at O are congruent.

The second pair is exactly the theorem hypothesis.

The reusable theorem

```
hilbert_AA_third_angle
```

then supplies the required third-angle congruence.

Thus the full crossing theorem does not force a circle into a degenerate configuration where one is not needed.

20.17 Proper Crossing Case

Assume now:

$$A \neq C, \quad B \neq D.$$

The crossing-chords theorem described above gives

$$\text{Concyclic4}(A, C, D, B).$$

The final angle transfer is then provided by the reusable theorem

```
hilbert_crossing_chords_angle_nondegenerate
```

This theorem converts the concyclic crossing configuration and the given cross-angle congruence into

$$\angle ODC \cong \angle OAB.$$

Hence the proper case proceeds as:

$$\text{IV.19} + \text{IV.20} \implies \text{crossing-chord concyclicity} \implies \text{crossing-angle transfer}.$$

The complete private theorem is

```
hilbert_crossing_rays_transfer_stage11
```

and its closed wrappers finally expose

```
hilbert_crossing_rays_transfer
```

20.18 Why the Transfer Theorem Lives in Proposition4_20

The proposition `HilbertCrossingRaysTransfer` itself is declared in `HilbertInterfaceIV` because downstream modules must be able to state their contracts without importing the completed circle block.

Its proof, however, cannot live there.

The proper crossing proof needs the completed numbered circle converses: IV.19 and IV.20. Moving the proof into the interface would therefore force the interface to import later numbered propositions and would destroy the acyclic production architecture.

The correct direction is:

```
HilbertInterfaceIV → Proposition4_19 → Proposition4_20 → hilbert_crossing_rays_transfer.
```

The proposition interface is early; the theorem proving it is late.

This separation between *statement ownership* and *proof ownership* is intentional.

20.19 Downstream Role in V.22

The crossing-rays theorem is the circle-geometric kernel needed by the angle-coded proportion calculus.

The downstream production dependency is

Book IV circle theory $\implies$ crossing-rays transfer $\implies$ alternando $\implies$ Hilbert combination rule / Euclid V.22.

Accordingly, the production proportion module imports `Proposition4_20.lean`, because that is the narrow numbered module exporting

```
hilbert_crossing_rays_transfer
```

The proportion theory does not need to import the old monolithic Book IV development.

20.20 Downstream Role in XI.27

The V.22 result is then consumed by the XI.27 scaling argument.

The dependency route is:

Forder IV.12–IV.20 → crossing-rays transfer → V.22 → XI.27

This is why the Book IV development became production-critical even though its subject matter is planar circle geometry and the eventual application is three-dimensional Euclidean geometry.

The circle block supplies the planar angle-transfer mechanism needed by the proportion layer; the proportion layer then supports the spatial construction.

20.21 Lean Formalization

The public layer of `Proposition4_20.lean` contains:

```
hilbert_IV20_circle_converse
hilbert_crossing_rays_transfer
```

The proposition-specific IV.20 layer contains:

```
hilbert_forder_IV20_secant_sameRay_stage7
hilbert_forder_IV20_secant_oppositeRay_impossible_stage7
hilbert_forder_IV20_tangent_impossible_stage7
hilbert_forder_IV20_circle_converse_stage8
```

The crossing endpoint adds private composition theorems:

```
hilbert_crossing_chords_concyclic_stage9
hilbert_crossing_rays_transfer_stage11
hilbert_crossing_rays_transfer_of_IV18_stage55
hilbert_crossing_rays_transfer_complete_stage62
```

The reusable general infrastructure remains in `HilbertInterfaceIV`:

```
hilbert_circle_line_second_or_tangent
hilbert_equal_angles_same_secant_unique
hilbert_secant_ray_order_split
hilbert_two_oppositeSides_sameSide
hilbert_crossing_chords_concyclic_of_forder
```

```

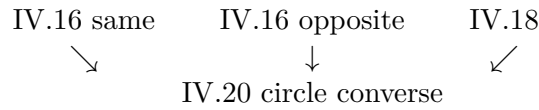
hilbert_crossing_chords_angle_nondegenerate
hilbert_crossing_angle_at_0
hilbert_AA_third_angle

```

The stage-labelled names above are private implementation details. The public API uses semantic production names.

20.22 Dependency Structure of IV.20 Proper

For the numbered theorem itself:

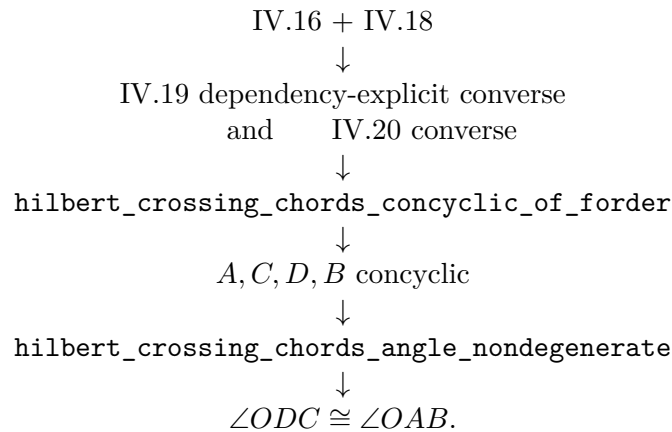


At module-import level, `Proposition4_20.lean` also imports IV.19, but that dependency belongs to the crossing-rays endpoint, not to the numbered IV.20 theorem.

This distinction should be preserved whenever the dependency graph is reported.

20.23 Dependency Structure of Crossing-Rays Transfer

The proper crossing branch uses:



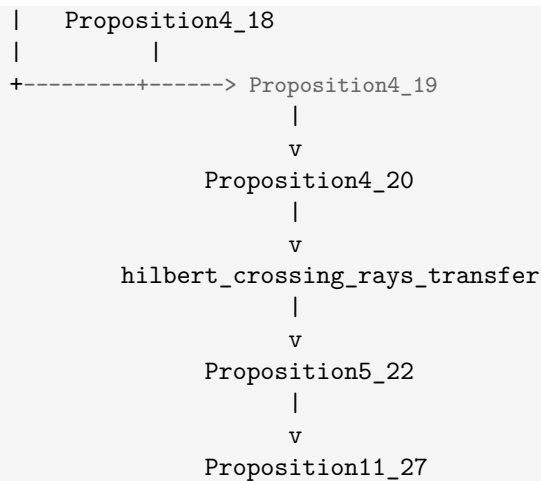
Together with the two degenerate cases, this yields the public transfer theorem.

The full module dependency graph at the endpoint is therefore:

```

HilbertInterfaceIV
|
+--> Proposition4_16
|   |
|   v
|   Proposition4_17
|   |
|   v

```



The actual direct imports of `Proposition4_20.lean` are:

```

import CGJteamLab.HilbertInterfaceIV
import CGJteamLab.Proposition4_16
import CGJteamLab.Proposition4_18
import CGJteamLab.Proposition4_19

```

20.24 Foundational Boundary

Neither IV.20 nor crossing-rays transfer introduces a new axiom.

IV.20 uses:

- circumcenter existence for a proper triangle;
- the secant/tangent dichotomy;
- plane separation;
- ray order;
- IV.16;
- IV.18;
- Book Zero supplement transport;
- exterior-angle comparison.

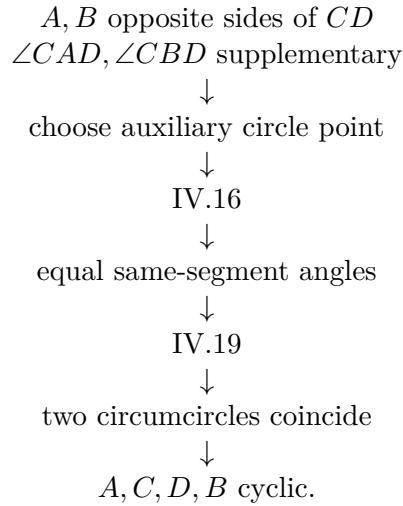
The crossing-rays theorem then uses:

- the completed IV.19 and IV.20 converse machinery;
- the reusable classification of crossing chords;
- AA third-angle transport in a degenerate branch;
- the nondegenerate crossing-chord angle theorem.

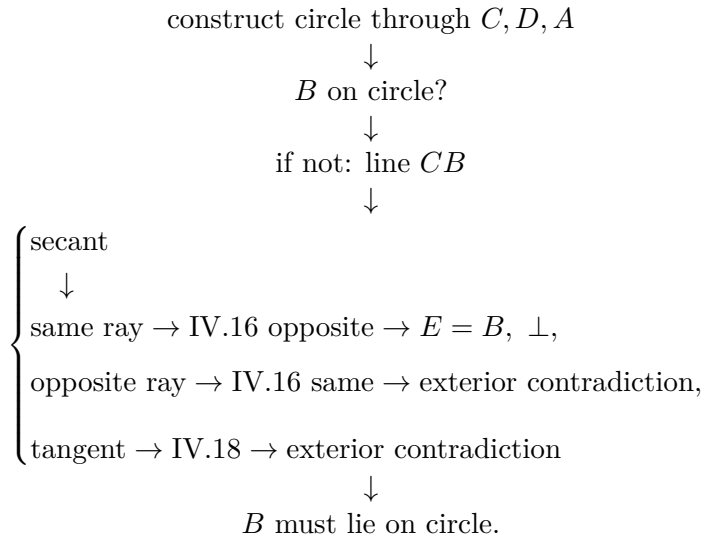
No coordinate slopes, directed-angle arithmetic, determinant concyclicity test, or analytic circle equations are introduced.

20.25 Classical and Formal Dependency Diagrams

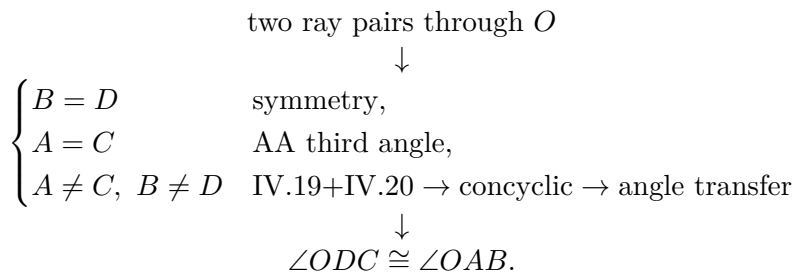
For order IV.20 itself has the source-level shape:



The production proof is:



The crossing endpoint continues:



20.26 Diagrammatic Audit

Five figures are justified for IV.20 and its endpoint.

1. The synthetic production form of IV.20 with the supplement witness.
2. The candidate circumcircle through C, D, A .
3. A two-panel comparison of the two secant orders.
4. The tangent contradiction using IV.18.
5. The crossing-rays transfer configuration used downstream.

The crossing figure belongs here because the public theorem is exported from the same production module and represents the actual endpoint of the Book IV development.

20.27 Conclusion

Forder IV.20 closes the pair of circle converses begun by IV.19.

The source proof is short: construct an auxiliary circle point, use IV.16, apply IV.19, and identify the circles.

The production proof adopts the uniform converse engine developed for IV.19. It constructs the circumcircle through three points and eliminates every secant or tangent alternative for the fourth point. The same-ray secant branch uses IV.16 opposite-side; the opposite-ray branch uses IV.16 same-side; the tangent branch uses IV.18.

The module then goes beyond the numbered proposition. Together with IV.19, the completed circle converses prove concyclicity in the proper crossing-rays configuration, and that concyclicity yields the public `hilbert_crossing_rays_transfer` theorem.

This transfer is the actual production bridge from Forder Book IV to the proportion theory:

$$\text{circle geometry} \rightarrow \text{crossing-ray angle transfer} \rightarrow \text{V.22} \rightarrow \text{XI.27}.$$

With IV.20 and the transfer theorem documented, the Forder IV.12–IV.20 production block is mathematically and architecturally complete.

Appendix A

HilbertInterfaceIV: Reusable Circle Infrastructure

A.1 Role of the Interface

`HilbertInterfaceIV.lean` contains the public proposition predicates and the reusable lemmas shared by several numbered proofs. It imports no `Proposition4_NN` module.

This direction is deliberate. A general geometric fact belongs to the interface when later propositions can reuse it independently of the proposition in which it happened to be discovered.

A.2 Public Proposition Predicates

```
HilbertForderIV12CentralHalf
HilbertForderIV12_1SameSegment
HilbertForderIV13DiameterRightAngle
HilbertForderIV15CyclicSupplement
HilbertForderIV16SameSide
HilbertForderIV16OppositeSide
HilbertForderIV17TangentConverse
HilbertForderIV18TangentChord
HilbertForderIV19CircleConverse
HilbertForderIV20CircleConverse
HilbertCrossingRaysTransfer
```

IV.14 is the exception at the interface level: its production theorem is exported directly with its geometric arguments and supplement witness.

A.3 Selected Reusable Theorems

Among the semantic production names in the interface are:

```
hilbert_circle_line_second_or_tangent
hilbert_equal_angles_same_secant_unique
```

```

hilbert_secant_ray_order_split
hilbert_oppositeRay_gives_oppositeSide
hilbert_two_oppositeSides_sameSide
hilbert_crossing_chords_angle_nondegenerate
hilbert_AA_third_angle
hilbert_right_triangle_third_angle_congruent
hilbert_circle_diameter_chord_intersection_inside
hilbert_triangle_angle_not_supplement
hilbert_two_right_angles_same_first_arm_collinear
hilbert_circle_tangent_carrier_unique

```

The interface also contains the general circle/order/angle support extracted during the IV.12 proof: circle-centre nondegeneracy, antipodes, diameter-midpoint facts, chord-interior order, same-side order, angle bisectors, and angle-decomposition lemmas.

A.4 The Cycle Removed by Semantic Ownership

During development, one generic secant-order lemma carried an IV.19 stage name because it had first been proved while working on IV.19. IV.17 also needed that lemma, creating an apparent dependency of IV.17 on IV.19.

The production split replaces that development provenance by semantic ownership. The theorem is now

```

hilbert_secant_ray_order_split

```

in `HilbertInterfaceIV`. The resulting numbered chain is acyclic: IV.16 feeds IV.17, IV.17 feeds IV.18, and IV.18 together with IV.16 feeds IV.19.

A.5 Production Validation

The split was checked incrementally by compiling `HilbertInterfaceIV.lean` and each numbered module `Proposition4_12.lean` through `Proposition4_20.lean`. The downstream modules `Proposition5_22.lean` and `Proposition11_27.lean` were then rebuilt against the new import graph, followed by a successful full `lake build` reported at the production checkpoint.

Bibliography

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.
- [2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.